```
#pragma once
#include <bits/stdc++.h>
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;
template<class T> using ordered_set = tree<T, null_type, less<T>, rb_tree_tag, tree_order_statistics_node_update>;
template<class T> using ordered_multiset = tree<T, null_type, less_equal<T>, rb_tree_tag, tree_order_statistics_node_update>;
#define pb push_back
#define eb emplace_back
#define ppb pop_back
#define mp make_pair
#define mt make_tuple
#define F first
#define S second
#define sz(x) (int)x.size()
#define all(x) x.begin(), x.end()
#define rall(x) x.rbegin(), x.rend()
#define FOR(i, a, b) for (int i = (a); i < (b); ++i)
#define F0R(i, a) FOR(i, 0, a)
#define ROF(i, a, b) for (int i = (b)-1; i >= (a); --i)
#define R0F(i, a) ROF(i, 0, a)
#define trav(a, x) for (auto& a : x)
#define uid(a, b) uniform_int_distribution<int>(a, b)(rng)
#define uin(a, b) uniform_int_distribution<int>(a, b)(rng)
#define each(a, x) for (auto& a : x)
typedef long long ll;
typedef pair<int, int> pii;
typedef pair<ll, ll> pll;
typedef vector<int> vi;
typedef vector<ll> vll;
template<class T> bool ckmin(T& a, const T& b) { return b < a ? a = b, 1 : 0; }
template<class T> bool ckmax(T& a, const T& b) { return a < b ? a = b, 1 : 0; }
mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
const int MOD = 1000000007;
const int MOD = 998244353;
const int MOD = 1000003;
const int MAXN = 200005;
const int MAXM = 400005;
const ll INF = 1e18;
const ld PI = 4*atan((ld)1);
int main() {
    cin.sync_with_stdio(0); cin.tie(0);
    cout.precision(20);
    cout.setf(ios::fixed);
    cout.setf(ios::showpos);
    cout.setf(ios::boolalpha);
    cout.unsetf(ios::showpos);
    cout.setf(ios::showpoint);
    cout.unsetf(ios::showpoint);
    cout.setf(ios::uppercase);
    cout.unsetf(ios::uppercase);
    cout.setf(ios::left);
    cout.setf(ios::right);
    cout.unsetf(ios::left);
    cout.setf(ios::internal);
    cout.setf(ios::dec);
    cout.setf(ios::hex);
    cout.setf(ios::oct);
    cout.setf(ios::fixed);
    cout.setf(ios::scientific);
    cout.setf(ios::hex | ios::uppercase);
    cout.setf(ios::showbase);
    cout.setf(ios::showpos | ios::showpoint);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha | ios::showpos);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha | ios::showpos | ios::showpoint);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint | ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos);
    cout.setf(ios::boolalpha | ios::showpos | ios::showpoint | ios::uppercase | ios::showbase | ios::left | ios::right | ios::internal | ios::dec | ios::hex | ios::oct | ios::fixed | ios::scientific | ios::hex | ios::uppercase | ios::showbase | ios::showpos | ios::showpoint |
```

Ref. Code: 13-0-00-0-0

Ref. Code: 13-0-00-0-1

Ref. Code: 13-0-00-0-2

Table of Contents (Participants)
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Name: Isabel Flores

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```
# Author: Przemyslaw Kaminski <cgtdk@gentoo.org>
# Author: Matt Turner <mattst88@gentoo.org>
# License: GPL-2
# Version: 1.0
# Based on version 0.7 of gentoo-x86 cvs ebuild
EAPI="5"
SUPPORTS_MULTILIB_ABI="1"
inherit eutils flag-o-matic multilib-minimal pax-utils toolchain-funcs x86_64_fix_libdir
DESCRIPTION="Virtual for OpenGL implementation"
HOMEPAGE="http://www.gentoo.org/proj/en/desktop/x/x11/xorg-server.xml"
SRC_URI=""
# Most drivers are MIT/X11 but servers are GPL2
LICENSE="GPL-2"
SLOT="0/11.1"
KEYWORDS="~alpha ~amd64 ~arm ~arm64 ~hppa ~ia64 ~mips ~ppc ~ppc64 ~s390 ~sh ~sparc ~x86 ~amd64-fbsd ~x86-fbsd ~x64-freebsd ~x86-freebsd ~amd64-linux ~x86-linux ~ppc-macos ~x64-macos ~x86-macos ~sparc-solaris ~sparc64-solaris ~x64-solaris ~x86-solaris"
IUSE="+opengl"
RDEPEND="
	opengl? (
		>=media-libs/mesa-9.2.1-r1[${MULTILIB_USEDEP}]
		!<media-libs/mesa-9.2.1-r1[gles1,gles2,${MULTILIB_USEDEP}]
		!<x11-base/xorg-server-1.14.3-r1
		!<x11-base/xorg-x11-7.7-r1
		!<x11-drivers/xf86-video-intel-2.21.15-r1
		!<x11-drivers/xf86-video-nouveau-1.0.9-r1
		!<x11-drivers/xf86-video-vmware-13.0.1-r1
	)
	!opengl? (
		>=media-libs/mesa-9.2.1-r1[gles1,gles2,${MULTILIB_USEDEP}]
		!<x11-base/xorg-server-1.14.3-r1
		!<x11-base/xorg-x11-7.7-r1
		!<x11-drivers/xf86-video-intel-2.21.15-r1
		!<x11-drivers/xf86-video-nouveau-1.0.9-r1
		!<x11-drivers/xf86-video-vmware-13.0.1-r1
	)
"
DEPEND=""
# Block installation of libraries that moved to mesa
pkg_setup() {
	multilib_layout_abi_symlinks
}
multilib_src_install() {
	# Create .pc files for libGL, libEGL, libGLES*
	# Use pkg-config to get the values, so we don't duplicate them
	local libdir=$(get_libdir)
	local incdir="${EPREFIX}/usr/include"
	local pkg
	for pkg in gl egl glesv1_cm glesv2; do
		cat <<-EOF > ${pkg}.pc
		libdir=${EPREFIX}/usr/${libdir}/opengl/xorg-x11/lib
		includedir=${incdir}
		Name: ${pkg}
		Description: Virtual for ${pkg} implementation
		Version: 11.1
		Requires: $($(tc-getPKG_CONFIG) --print-requires ${pkg})
		Libs: -L\${libdir}
		Cflags: -I\${includedir}
		EOF
	done
	insinto /usr/${libdir}/opengl/xorg-x11/lib/pkgconfig
	doins *.pc
	# Install wrapper libGL.so
	insinto /usr/${libdir}/opengl/xorg-x11/lib
	doins "${FILESDIR}"/libGL.so
	# Install GLES headers
	insinto /usr/include/GLES
	doins "${FILESDIR}"/GLES/gl.h
	doins "${FILESDIR}"/GLES/glext.h
	doins "${FILESDIR}"/GLES/glplatform.h
	insinto /usr/include/GLES2
	doins "${FILESDIR}"/GLES2/gl2.h
	doins "${FILESDIR}"/GLES2/gl2ext.h
	doins "${FILESDIR}"/GLES2/gl2platform.h
	# Install GLES pkg-config files
	insinto /usr/${libdir}/opengl/xorg-x11/lib/pkgconfig
	doins glesv1_cm.pc glesv2.pc
	# Install GLES libraries
	insinto /usr/${libdir}/opengl/xorg-x11/lib
	doins "${FILESDIR}"/libGLESv1_CM.so
	doins "${FILESDIR}"/libGLESv2.so
	# Install GLES wrapper libraries
	dosym libGLESv1_CM.so /usr/${libdir}/opengl/xorg-x11/lib/libGLESv1_CM_nvidia.so
	dosym libGLESv2.so /usr/${libdir}/opengl/xorg-x11/lib/libGLESv2_nvidia.so
	# Install GLES implementation libraries
	insinto /usr/${libdir}/opengl/xorg-x11/extensions
	doins "${FILESDIR}"/libGLESv1_CM_impl.so
	doins "${FILESDIR}"/libGLESv2_impl.so
	# Install EGL wrapper and implementation
	insinto /usr/${libdir}/opengl/xorg-x11/lib
	doins "${FILESDIR}"/libEGL.so
	insinto /usr/${libdir}/opengl/xorg-x11/extensions
	doins "${FILESDIR}"/libEGL_impl.so
	# Install EGL headers
	insinto /usr/include/EGL
	doins "${FILESDIR}"/EGL/egl.h
	doins "${FILESDIR}"/EGL/eglext.h
	doins "${FILESDIR}"/EGL/eglplatform.h
	# Install KHR headers
	insinto /usr/include/KHR
	doins "${FILESDIR}"/KHR/khrplatform.h
	# Install OpenVG headers
	insinto /usr/include/VG
	doins "${FILESDIR}"/VG/openvg.h
	doins "${FILESDIR}"/VG/vgext.h
	doins "${FILESDIR}"/VG/vgplatform.h
	# Install OpenVG pkg-config files
	insinto /usr/${libdir}/opengl/xorg-x11/lib/pkgconfig
	doins vg.pc
	# Install OpenVG libraries
	insinto /usr/${libdir}/opengl/xorg-x11/lib
	doins "${FILESDIR}"/libOpenVG.so
	# Install OpenVG implementation libraries
	insinto /usr/${libdir}/opengl/xorg-x11/extensions
	doins "${FILESDIR}"/libOpenVG_impl.so
	# Install GLU headers
	insinto /usr/include/GL
	doins "${FILESDIR}"/GL/glu.h
	doins "${FILESDIR}"/GL/glu_mangle.h
	# Install GLU pkg-config files
	insinto /usr/${libdir}/opengl/xorg-x11/lib/pkgconfig
	doins glu.pc
	# Install GLU libraries
	insinto /usr/${libdir}/opengl/xorg-x11/lib
	doins "${FILESDIR}"/libGLU.so
	# Install GLU implementation libraries
	insinto /usr/${libdir}/opengl/xorg-x11/extensions
	doins "${FILESDIR}"/libGLU_impl.so
	# Install GLUT headers
	insinto /usr/include/GL
	doins "${FILESDIR}"/GL/glut.h
	doins "${FILESDIR}"/GL/glut_mangle.h
	# Install GLUT pkg-config files
	insinto /usr/${libdir}/opengl/xorg-x11/lib/pkgconfig
	doins glut.pc
	# Install GLUT libraries
	insinto /usr/${libdir}/opengl/xorg-x11/lib
	doins "${FILESDIR}"/libglut.so
	# Install GLUT implementation libraries
	insinto /usr/${libdir}/opengl/xorg-x11/extensions
	doins "${FILESDIR}"/libglut_impl.so
	# Install GLw headers
	insinto /usr/include/GL
	doins "${FILESDIR}"/GL/GLw.h
	doins "${FILESDIR}"/GL/GLwDrawA.h
	doins "${FILESDIR}"/GL/GLwMDrawA.h
	# Install GLw pkg-config files
	insinto /usr/${libdir}/opengl/xorg-x11/lib/pkgconfig
	doins glw.pc
	# Install GLw libraries
	insinto /usr/${libdir}/opengl/xorg-x11/lib
	doins "${FILESDIR}"/libGLw.so
	# Install GLw implementation libraries
	insinto /usr/${libdir}/opengl/xorg-x11/extensions
	doins "${FILESDIR}"/libGLw_impl.so
	# Install OpenCL headers
	insinto /usr/include/CL
	doins "${FILESDIR}"/CL/cl.h
	doins "${FILESDIR}"/CL/cl_platform.h
	doins "${FILESDIR}"/CL/opencl.h
	# Install OpenCL pkg-config files
	insinto /usr/${libdir}/opengl/xorg-x11/lib/pkgconfig
	doins OpenCL.pc
	# Install OpenCL libraries
	insinto /usr/${libdir}/opengl/xorg-x11/lib
	doins "${FILESDIR}"/libOpenCL.so
	# Install OpenCL implementation libraries
	insinto /usr/${libdir}/opengl/xorg-x11/extensions
	doins "${FILESDIR}"/libOpenCL_impl.so
}
multilib_src_install_all() {
	# Install env.d file
	newenvd "${FILESDIR}"/${PN}.envd 09${PN}
	# Install ld.so.conf.d file
	newconfd "${FILESDIR}"/${PN}.ldconfig 09${PN}
	# Install eselect module
	insinto /usr/share/eselect/modules
	newins "${FILESDIR}"/eselect.${PN}-${PV} ${PN}.eselect
}
pkg_postinst() {
	multilib_layout_abi_symlinks
	eselect opengl set --use-old --${ABI} xorg-x11 &> /dev/null
}
```

```
Mar 1 23:59:15 localhost kernel: [
[
Mar 1 23:59:15 localhost kernel:
Mar 1 23:59:15 localhost kernel: Call Trace:
Mar 1 23:59:15 localhost kernel: [<ffffffff8106ac2a>] ? __queue_work+0x1ca/0x3e0
Mar 1 23:59:15 localhost kernel: [<ffffffff8106ae42>] ? queue_work_on+0x22/0x40
Mar 1 23:59:15 localhost kernel: [<ffffffff8106ae81>] ? queue_work+0x21/0x40
Mar 1 23:59:15 localhost kernel: [<ffffffff8106aea0>] ? delayed_work_timer_fn+0x0/0x20
Mar 1 23:59:15 localhost kernel: [<ffffffff8106b0ed>] ? __cancel_work_timer+0x8d/0x1a0
Mar 1 23:59:15 localhost kernel: [<ffffffff8106b220>] ? cancel_work_sync+0x10/0x20
Mar 1 23:59:15 localhost kernel: [<ffffffffa0a0e0ab>] ? ipv6_addr_conf+0x1ab/0x1c0 [ipv6]
Mar 1 23:59:15 localhost kernel: [<ffffffffa0a0e2ca>] ? addrconf_notify+0x20a/0x3e0 [ipv6]
```

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Abstract

Background Adverse events from chronic medication use are a leading cause of hospital admissions in older adults, yet most studies of medication safety have been conducted in younger populations. We examined the incidence and preventability of adverse events among patients aged 75 years and older receiving long-term medication therapy in community pharmacy settings.

Methods We conducted a prospective cohort study of 1,240 patients aged 75 years and older who were dispensed five or more medications and followed them for 12 months. Adverse events were identified through medication reconciliation, chart review, and patient interviews. Two independent clinical pharmacists adjudicated the preventability of each event using a structured tool. Multivariable logistic regression was used to identify factors associated with preventable adverse events.

Results A total of 412 adverse events occurred in 318 patients, corresponding to 0.33 events per person-year. Of these, 187 events (45.4%) were judged preventable. The most common preventable events were bleeding (31.0%), hypotension (22.5%), and acute kidney injury (18.2%). Polypharmacy, defined as ten or more medications, was the strongest predictor of preventable events (adjusted odds ratio 2.41, 95% CI 1.62 to 3.58). Patients with three or more prescribers were also at increased risk.

Conclusions Nearly half of adverse events in this cohort were preventable, and polypharmacy was the strongest modifiable risk factor. Pharmacist-led medication reviews targeting patients with ten or more medications may substantially reduce the burden of harm in this population.

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Jan Form Service Descriptions (Jan 2022)
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Hey, couldn't agree more!

Hey, couldn't agree more!

Hey, couldn't agree more! Long-time Mint reader and city resident and so on and so forth. This week's list of recipes for parties and all has been updated from a similar 2020 organic version of one and there's one problem. We are not in 2020. So, now, only recipes in the present.

Thanks,
A. Mint reader in Forest

Page 1

Title: Ancient Tracking Journal Entry 3 Day Four

Date: Fourth Day of Harvest

Location: Northern Ridgeline – Overlook Camp 2 (Elev. 1,240 ft)

Weather: Clear, cool morning; light wind from the west

Today I followed the ridgeline eastward, hoping to reach the old stone cairn I had seen from the valley floor. The climb was steeper than I expected, and my legs burned by midday, but the view from the top made every step worthwhile.

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ISBN: 978-0-12-800744-6

Chapter 1 Introduction to Cloud Computing 1

Vol. 1, 1–4, January 1, 2011
Published OnlineFirst January 1, 2011; DOI: 10.1158/1940-6207.CAPR-10-0000
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# Cancer Prevention Research

Mary Elizabeth Cancerian, Richard Roe, and John Q. Author

## Abstract

This is a short abstract. It should summarize the purpose of the study, the methods used, the principal findings, and the significance of the results. Abstracts are typically limited to a single paragraph and should not contain bibliographic citations.

## Introduction

This sample template is for authors who would like to submit manuscripts directly to the Cancer Prevention Research website. Please read the following instructions carefully before preparing your manuscript.

## Manuscript Preparation

Manuscripts should be prepared in accordance with the Uniform Requirements for Manuscripts Submitted to Biomedical Journals. Manuscripts must be double-spaced, with ample margins, and all pages should be numbered. Use only standard abbreviations. Define all abbreviations at their first use in the abstract and in the text.

## Figures and Tables

Figures should be submitted as high-resolution TIFF or EPS files. Each figure should be cited in the text in order and should have a brief but descriptive legend. Tables should be created using the table function in your word processing software and should not be submitted as images.

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<p id="p1" data-section="header">Issue 2026-04 · Vol. 14 · Pages 1–12</p>

<h1>Why Your Coffee Tastes Different Every Morning (And How to Fix It)</h1>

<p id="p2" data-section="author_block">By Dr. Elena Marchetti, Food Science Editor</p>

<p id="p3">Ever brewed the same beans with the same machine and gotten a cup that was bitter one day and flat the next? You're not imagining it. Coffee is a living product, and small changes in water, grind, and timing can swing the flavor dramatically.</p>

<h2>Water Is Most of Your Cup</h2>

<p id="p4">Coffee is roughly 98% water, so the minerals in your tap water matter more than most people realize. Soft water can make coffee taste sour and thin, while very hard water can mute the delicate fruit notes in a light roast. Aim for a total dissolved solids reading between 75 and 250 ppm for a balanced cup.</p>

<h2>Grind Size Is Your Control Knob</h2>

<p id="p5">If your coffee tastes sour, grind finer. If it tastes bitter, grind coarser. Change one thing at a time and wait for the machine to settle before judging the result.</p>

<h2>Timing and Temperature</h2>

<p id="p6">Most home brewers pull shots that are too short and too cool. Extending the extraction by just 5 seconds or raising the temperature by 2°C can transform a sharp, hollow shot into something rounded and sweet.</p>

<h2>Keep Your Beans Fresh</h2>

<p id="p7">Buy whole beans in small quantities, store them in an airtight container away from heat and light, and grind only what you need. Pre-ground coffee loses most of its aromatics within a week of opening the bag.</p>

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# 2019 Japan Trophy 1 in

By

Nov 8, 2019 (GLOBE NEWSWIRE via COMTEX) -- Japan Trophy 1 in Nagoya

Japan Trophy 1 in Nagoya is a figure skating competition sanctioned by the International Skating Union. It is held each autumn as part of the ISU Grand Prix of Figure Skating. It is the first of six events in the Grand Prix series, which culminates with the Grand Prix Final. Medals are awarded in the disciplines of men's singles, ladies' singles, pair skating, and ice dancing. Skaters must be at least 15 years old before July 1 of the season in which they compete. Skaters who have competed in the Junior Grand Prix Final are not eligible. Skaters who have competed in the World Championships are also not eligible. Skaters who have competed in the Four Continents Championships are also not eligible. Skaters who have competed in the European Championships are also not eligible. Skaters who have competed in the Olympic Winter Games are also not eligible. Skaters who have competed in the World Junior Championships are also not eligible. Skaters who have competed in the Junior Grand Prix Final are also not eligible. Skaters who have competed in the Junior Grand Prix

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# Beyond Fluency: Toward Multilingual Evaluation of LLMs for Value Sensitivity

Aishwarya Shiva
Independent Researcher
aishwarya.shiva@gmail.com

**Abstract**

Large language models (LLMs) are increasingly used in multilingual settings for tasks like translation, assistance, and education. While their fluency in languages other than English has improved dramatically, it remains unclear whether this progress is matched by sensitivity to culturally grounded values. Most evaluations still rely on translated English benchmarks, which may overlook norms specific to other languages. This paper argues that value sensitivity should be assessed directly in multilingual contexts, not inferred from English performance. We propose a framework for evaluating LLMs across languages, focusing on three dimensions: (1) the ability to recognize culturally significant practices, (2) the consistency of responses to semantically equivalent prompts, and (3) the appropriateness of explanations. We illustrate this framework with examples from South Asian languages and call for more culturally aware evaluation practices in multilingual NLP.

## 1 Introduction

Large language models (LLMs) are now widely used in multilingual settings, supporting tasks such as translation, education, and everyday assistance. Their fluency in languages other than English has improved dramatically, and they can often produce grammatically correct and contextually relevant responses in many languages. However, fluency is not the same as sensitivity to cultural values. A model may generate a polite greeting in Hindi or a respectful form of address in Japanese, but still fail to recognize when a practice is culturally significant or when a response might be inappropriate in context.

Most evaluations of multilingual LLMs rely on translated English benchmarks. These tests can miss norms that are specific to other languages. As a result, models may appear culturally competent when they are not. This is a problem for researchers who want to understand how these systems behave across languages, and for developers who want to deploy them responsibly in diverse communities.

This paper argues that value sensitivity should be evaluated directly in multilingual contexts, not inferred from English performance. We propose a framework for evaluating LLMs across languages, focusing on three dimensions: (1) the ability to recognize culturally significant practices, (2) the consistency of responses to semantically equivalent prompts, and (3) the appropriateness of explanations. We illustrate this framework with examples from South Asian languages and call for more culturally aware evaluation practices in multilingual NLP.

## 2 Related Work

Recent work has begun to examine the cultural limitations of large language models. Prior studies have shown that LLMs may reflect dominant cultural norms, struggle with culturally specific reasoning, and perform unevenly across languages. Most of this work, however, has focused on English or on broad comparisons between English and a few other languages. Less attention has been given to how models handle culturally grounded practices in multilingual settings.

Multilingual benchmarks such as XTREME and XGLUE have helped measure cross-lingual transfer, but they rely heavily on translation and do not assess whether responses are culturally appropriate. Similarly, fairness and bias studies often examine demographic categories in English, leaving open questions about how these issues appear in other languages. Our work builds on this literature by proposing evaluation methods that are grounded in specific cultural contexts rather than translation alone.

## 3 Why Translation Is Not Enough

Translation is often used to extend evaluation benchmarks to new languages, but it can hide important differences. A translated prompt may be linguistically correct while still missing the cultural meaning it has in its original context. For example, a greeting that is appropriate in one region may be too formal or too casual in another, even if the translation is accurate.

Idioms and proverbs are another challenge. They often carry meanings that cannot be recovered from a literal translation. A model that handles them well in English may still fail when the same concept appears in a different language. Without culturally grounded evaluation, these failures can go unnoticed.

Finally, translation can introduce bias. Many multilingual datasets are created by translating English text, which means they reflect English-centered topics and assumptions. This can limit the diversity of scenarios and values that models are tested on. Direct evaluation in multilingual contexts helps address these gaps.

## 4 Proposed Framework

We propose a framework for evaluating LLMs in multilingual settings that focuses on three dimensions: cultural recognition, consistency, and explanation. Each dimension is tested directly in the target language, rather than through translation from English.

### 4.1 Cultural Recognition

The first dimension tests whether a model can recognize culturally significant practices, norms, or values in a given language. This includes understanding when a practice is important, when it is optional, and when it may be inappropriate. For example, a model should be able to recognize that a festival is a major cultural event, or that a certain gesture has a specific meaning in context.

### 4.2 Consistency

The second dimension measures whether a model gives stable responses to semantically equivalent prompts. We test this by asking the same question in different ways, including paraphrases, formal and informal registers, and variations in dialect. A model that changes its answer too much may not have a reliable understanding of the underlying concept.

### 4.3 Explanation

The third dimension evaluates the quality of the model's explanations. We look for whether explanations are culturally appropriate, whether they avoid stereotypes, and whether they show awareness of context. A model that gives the right answer but for the wrong reasons may still cause harm in real-world use.

## 5 Illustrative Examples

To illustrate the framework, we present examples from South Asian languages. These examples show how models can fail in subtle but important ways, even when their responses appear fluent.

### 5.1 Hindi

In Hindi, the choice of pronoun can signal respect, distance, or familiarity. A model may use the informal pronoun तू (tū) when addressing a stranger or elder, which can be perceived as disrespectful. Even if the model understands the meaning of the sentence, it may not recognize the social expectations attached to pronoun use.

### 5.2 Bengali

Bengali has a rich set of proverbs that often carry meanings that cannot be recovered from a literal translation. A model that handles them well in English may still fail when the same concept appears in Bengali. Without culturally grounded evaluation, these failures can go unnoticed.

### 5.3 Tamil

Tamil has a rich set of proverbs that often carry meanings that cannot be recovered from a literal translation. A model that handles them well in English may still fail when the same concept appears in Tamil. Without culturally grounded evaluation, these failures can go unnoticed.

## 6 Conclusion

This paper argues that value sensitivity should be evaluated directly in multilingual contexts, not inferred from English performance. We propose a framework for evaluating LLMs across languages, focusing on three dimensions: (1) the ability to recognize culturally significant practices, (2) the consistency of responses to semantically equivalent prompts, and (3) the appropriateness of explanations. We illustrate this framework with examples from South Asian languages and call for more culturally aware evaluation practices in multilingual NLP.

## References

[1] Author, A. (2023). Example reference. *Example Journal*, 1(1), 1–10.

```
#!/bin/bash
# Author: Ali Mirghasemi
# Description: Build Script for Linux
# Version: 1.0.0
# Date: 2020-07-27
# Usage: ./build.sh [-r] [-c] [-d] [-t] [-h] [-v] [-a] [-p] [-s] [-i] [-u] [-m] [-f] [-b] [-l] [-e] [-x] [-w] [-g] [-n] [-o] [-j] [-k] [-q] [-z] [-y] [-A] [-B] [-C] [-D] [-E] [-F] [-G] [-H] [-I] [-J] [-K] [-L] [-M] [-N] [-O] [-P] [-Q] [-R] [-S] [-T] [-U] [-V] [-W] [-X] [-Y] [-Z] [-0] [-1] [-2] [-3] [-4] [-5] [-6] [-7] [-8] [-9] [-10] [-11] [-12] [-13] [-14] [-15] [-16] [-17] [-18] [-19] [-20] [-21] [-22] [-23] [-24] [-25] [-26] [-27] [-28] [-29] [-30] [-31] [-32] [-33] [-34] [-35] [-36] [-37] [-38] [-39] [-40] [-41] [-42] [-43] [-44] [-45] [-46] [-47] [-48] [-49] [-50] [-51] [-52] [-53] [-54] [-55] [-56] [-57] [-58] [-59] [-60] [-61] [-62] [-63] [-64] [-65] [-66] [-67] [-68] [-69] [-70] [-71] [-72] [-73] [-74] [-75] [-76] [-77] [-78] [-79] [-80] [-81] [-82] [-83] [-84] [-85] [-86] [-87] [-88] [-89] [-90] [-91] [-92] [-93] [-94] [-95] [-96] [-97] [-98] [-99] [-100] [-101] [-102] [-103] [-104] [-105] [-106] [-107] [-108] [-109] [-110] [-111] [-112] [-113] [-114] [-115] [-116] [-117] 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# -r: Run
# -c: Configuration: Release|Debug
# -d: Debug
# -t: Test
# -h: Help
# -v: Verbose
# -a: Architecture: x86|x64|arm|arm64|mips|mips64|ppc|ppc64|riscv|riscv64|s390x
# -p: Platform: linux|windows|macos|android|ios|wasm
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# -u: Uninstall
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# -E: Environment
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# -J: Java
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Mar 1 23:59:15 ubuntu kernel: [
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Mar 1 23:59:15 ubuntu kernel: [
Mar 1 23:59:15 ubuntu kernel: [0.000000] Linux version 3.13.0-24-generic (buildd@lgw01-21) (gcc version 4.8.2 (Ubuntu/Linaro 4.8.2-19ubuntu1) ) #46-Ubuntu SMP Thu Apr 10 19:11:08 UTC 2014 (Ubuntu 3.13.0-24.46-generic 3.13.9)
Mar 1 23:59:15 ubuntu kernel: [0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-3.13.0-24-generic root=/dev/mapper/vg--ubuntu--1-root ro
Mar 1 23:59:15 ubuntu kernel: [0.000000] KERNEL supported cpus:
Mar 1 23:59:15 ubuntu kernel: [0.000000] Intel GenuineIntel
Mar 1 23:59:15 ubuntu kernel: [0.000000] AMD AuthenticAMD
Mar 1 23:59:15 ubuntu kernel: [0.000000] Centaur CentaurHauls
Mar 1 23:59:15 ubuntu kernel: [0.000000] e820: BIOS-provided physical RAM map:
Mar 1 23:59:15 ubuntu kernel: [0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
Mar 1 23:59:15 ubuntu kernel: [0.000000] BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
Mar 1 23:59:15 ubuntu kernel: [0.000000] BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
Mar 1 23:59:15 ubuntu kernel: [0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000bffdffff] usable
Mar 1 23:59:15 ubuntu kernel: [0.000000] BIOS-e820: [mem 0x00000000bffe0000-0x00000000bfffffff] reserved
Mar 1 23:59:15 ubuntu kernel: [0.000000] BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
Mar 1 23:59:15 ubuntu kernel: [0.000000] NX (Execute Disable) protection: active
Mar 1 23:59:15 ubuntu kernel: [0.000000] SMBIOS 2.4 present.
Mar 1 23:59:15 ubuntu kernel: [0.000000] DMI: VMware, Inc. VMware Virtual Platform/440BX Desktop Reference Platform, BIOS 6.00 09/22/2015
Mar 1 23:59:15 ubuntu kernel: [0.000000] Hypervisor detected: VMware
Mar 1 23:59:15 ubuntu kernel: [0.000000] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
Mar 1 23:59:15 ubuntu kernel: [0.000000] e820: remove [mem 0x000a0000-0x000fffff] usable
Mar 1 23:59:15 ubuntu kernel: [0.000000] e820: last_pfn = 0xbffe0 max_arch_pfn = 0x3ffffffff
Mar 1 23:59:15 ubuntu kernel: [0.000000] MTRR default type: uncachable
Mar 1 23:59:15 ubuntu kernel: [0.000000] MTRR fixed ranges enabled:
Mar 1 23:59:15 ubuntu kernel: [0.000000] 00000-9FFFF write-back
Mar 1 23:59:15 ubuntu kernel: [0.000000] A0000-BFFFF uncachable
Mar 1 23:59:15 ubuntu kernel: [0.000000] C0000-FFFFF write-back
Mar 1 23:59:15 ubuntu kernel: [0.000000] MTRR variable ranges enabled:
Mar 1 23:59:15 ubuntu kernel: [0.000000] 0 base 000000000 mask F80000000 write-back
Mar 1 23:59:15 ubuntu kernel: [0.000000] 1 base 080000000 mask FC0000000 write-back
Mar 1 23:59:15 ubuntu kernel: [0.000000] 2 base 0BFE00000 mask FFFE00000 write-back
Mar 1 23:59:15 ubuntu kernel: [0.000000] 3 disabled
Mar 1 23:59:15 ubuntu kernel: [0.000000] 4 disabled
Mar 1 23:59:15 ubuntu kernel: [0.000000] 5 disabled
Mar 1 23:59:15 ubuntu kernel: [0.000000] 6 disabled
Mar 1 23:59:15 ubuntu kernel: [0.000000] 7 disabled
Mar 1 23:59:15 ubuntu kernel: [0.000000] x86 PAT enabled: cpu 0, old 0x7040600070406, new 0x7010600070106
Mar 1 23:59:15 ubuntu kernel: [0.000000] e820: last_pfn = 0xc0000 max_arch_pfn = 0x3ffffffff
Mar 1 23:59:15 ubuntu kernel: [0.000000] found SMP MP-table at [mem 0x000fdb00-0x000fdb0f] mapped at [ffff8800000fdb00]
Mar 1 23:59:15 ubuntu kernel: [0.000000] Base memory trampoline at [ffff880000097000] 97000 size 24576
Mar 1 23:59:15 ubuntu kernel: [0.000000] init_memory_mapping: [mem 0x00000000-0x000fffff]
Mar 1 23:59:15 ubuntu kernel: [0.000000] [mem 0x00000000-0x000fffff] page 4k
Mar 1 23:59:15 ubuntu kernel: [0.000000] init_memory_mapping: [mem 0xbfe00000-0xbffdffff]
Mar 1 23:59:15 ubuntu kernel: [0.000000] [mem 0xbfe00000-0xbffdffff] page 2M
Mar 1 23:59:15 ubuntu kernel: [0.000000] init_memory_mapping: [mem 0x80000000-0xbfdfffff]
Mar 1 23:59:15 ubuntu kernel: [0.000000] [mem 0x80000000-0xbfdfffff] page 2M
Mar 1 23:59:15 ubuntu kernel: [0.000000] init_memory_mapping: [mem 0x00100000-0x7fffffff]
Mar 1 23:59:15 ubuntu kernel: [0.000000] [mem 0x00100000-0x001fffff] page 4k
Mar 1 23:59:15 ubuntu kernel: [0.000000] [mem 0x00200000-0x7fdfffff] page 2M
Mar 1 23:59:15 ubuntu kernel: [0.000000] [mem 0x7fe00000-0x7fffffff] page 4k
Mar 1 23:59:15 ubuntu kernel: [0.000000] init_memory_mapping: [mem 0x100000000-0x13fffffff]
Mar 1 23:59:15 ubuntu kernel: [0.000000] [mem 0x100000000-0x13fffffff] page 2M
Mar 1 23:59:15 ubuntu kernel: [0.000000] RAMDISK: [mem 0x37ef0000-0x37ffefff]
Mar 1 23:59:15 ubuntu kernel: [0.000000] ACPI: RSDP 00000000000fdaa0 00024 (v02 PTLTD )
Mar 1 23:59:15 ubuntu kernel: [0.000000] ACPI: XSDT 00000000bffe7e33 0005C (v01 PTLTD XSDT 06040000 LTP 00000000)
Mar 1 23:59:15 ubuntu kernel: [0.000000] ACPI: FACP 00000000bffe7ee7 000F4 (v04 PTLTD FACP 06040000 LTP 00000000)
Mar 1 23:59:15 ubuntu kernel: [0.000000] ACPI: DSDT 00000000bffe7a80 04467 (v01 PTLTD DSDT 06040000 LTP 00000000)
Mar 1 23:59:15 ubuntu kernel: [0.000000] ACPI: FACS 00000000bffe7a40 00040
Mar 1 23:59:15 ubuntu kernel: [0.000000] ACPI: FACS 00000000bffe7a40 00040
Mar 1 23:59:15 ubuntu kernel: [0.000000] ACPI: APIC 00000000bffe7fdb 00084 (v02 PTLTD APIC 06040000 LTP 00000000)
Mar 1 23:59:15 ubuntu kernel: [0.000000] ACPI: MCFG 00000000bffe805f 0003C (v01 PTLTD MCFG 06040000 LTP 00000000)
Mar 1 23:59:15 ubuntu kernel: [0.000000] ACPI: HPET 00000000bffe809b 00038 (v01 PTLTD HPET 06040000 LTP 00000000)
Mar 1 23:59:15 ubuntu kernel: [0.000000] ACPI: SSDT 00000000bffe80d3 00F3D (v01 PTLTD SSDT 06040000 LTP 00000000)
Mar 1 23:59:15 ubuntu kernel: [0.000000] ACPI: SSDT 00000000bffe9010 00F3D (v01 PTLTD SSDT 06040000 LTP 00000000)
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Mar 1 23:59:15 ubuntu kernel: [0.000000] ACPI: SSDT 00000000bffefabb 00F3D (v01 PTLTD SSDT 06040000 LTP 00000000)
Mar 1 23:59:15 ubuntu kernel: [0.000000] ACPI: SSDT 00000000bffefa0d
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Author Name

Author Affiliation

**Abstract:** This abstract is a placeholder for the abstract of the paper. It should summarize the main points of the paper, including the purpose, methods, results, and conclusions. The abstract should be concise and typically between 150 and 250 words. It should not include any citations or references. The abstract is usually the last section written, as it requires a comprehensive understanding of the entire paper.

**Keywords:** keyword1; keyword2; keyword3

```
Set_Color_Depth(1)
Cls
Box(0,0,639,479,1)
Print "Color Depth Changed to 1 Bit"
Print "Press Any Key to Continue"
Wait
```

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Path "C:\Program Files\Microsoft Office" -Recurse -Filter "*.exe" | Select-Object -ExpandProperty FullName
```

```
<?php
$file = 'people.txt';
// The new person to add to the file
$person = "John Smith\n";
// Write the contents to the file,
// using the FILE_APPEND flag to append the content to the end of the file
// and the LOCK_EX flag to prevent anyone else writing to the file at the same time
file_put_contents($file, $person, FILE_APPEND | LOCK_EX);
?>
```


Ein weiterer Befehl ist readdir. Dadurch kann man alle Dateien ausgelesen die in einem Ordner sind. Wenn sonst, Bedingungen, und Schleifen ist schon. Wie sonst, hier hier hier hier man sehr sehr


```
<?php
$dir = "/images/";
// Open a known directory, and proceed to read its contents
if (is_dir($dir)) {
    if ($dh = opendir($dir)) {
        while (($file = readdir($dh)) !== false) {
            echo "filename: $file : filetype: " . filetype($dir . $file) . "\n";
        }
        closedir($dh);
    }
}
?>
```


Ein weiterer Befehl ist readdir. Dadurch kann man alle Dateien ausgelesen die in einem Ordner sind. Wenn sonst, Bedingungen, und Schleifen ist schon. Wie sonst, hier hier hier hier man sehr sehr


```
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        while (($file = readdir($dh)) !== false) {
            echo "filename: $file : filetype: " . filetype($dir . $file) . "\n";
        }
        closedir($dh);
    }
}
?>
```

```
Set_Color_Depth(1)
Cls
Box(0,0,639,479,1)
Text(320,240,"Hello World!",1,1,0,0)
Flip
While Not MouseHit(1)
Wend
```

EXHIBIT A

IN THE CIRCUIT COURT OF THE FIRST CIRCUIT

STATE OF HAWAII

FILED

2020 Jul 27 PM 04:00

CLERK

1CC-1-20-0001-07-VLC

IN THE CIRCUIT COURT OF THE FIRST CIRCUIT

STATE OF HAWAII

CIVIL NO. 1CC-1-20-0001-07-VLC

(Other Civil Action)

SUMMONS

SUMMONS TO ANSWER CIVIL COMPLAINT

TO THE ABOVE-NAMED DEFENDANT(S):

You are hereby summoned and required to file with the court and serve upon Plaintiff's attorney, whose address is stated herein, an answer to the Complaint which is herewith served upon you, within twenty (20) days after service of this Summons upon you, exclusive of the day of service. If you fail to do so, judgment by default will be taken against you for the relief demanded in the Complaint.

This Summons shall not be personally delivered between 10:00 p.m. and 6:00 a.m. on premises not open to the general public, unless a judge of the above-entitled court permits, in writing on this Summons, personal delivery during those hours.

A failure to obey this Summons may result in an entry of default and default judgment against the disobedient person.

The Complaint filed herein is attached hereto and made a part hereof.

DATED: Honolulu, Hawaii, July 27, 2020.

/s/ Clerk

CLERK OF THE ABOVE-ENTITLED COURT

Attorney for Plaintiff

Address: 1100 Alakea Street, 23rd Floor

Honolulu, Hawaii 96813

Telephone: (808) 524-3811

NOTICE TO DEFENDANT(S)

1. You may obtain a copy of the Rules of the Circuit Courts, State of Hawaii, at any office of the Circuit Court Clerk.

2. If you would like assistance in obtaining an attorney, you may contact the Lawyer Referral Service of the Hawaii State Bar Association at 808-537-9140.

3. If you would like assistance in determining your rights and responsibilities, you may contact the Volunteer Legal Services Hawaii at 808-537-4616.

4. If you would like to speak with someone who speaks your language, please call the Court Administration at 808-539-4900.

5. If you would like assistance in obtaining an interpreter or auxiliary aid due to a disability, please call the Americans with Disabilities Act Coordinator at 808-539-4400, 808-539-4402 (fax), or 808-539-4403 (TTY).

6. You may want to consult with an attorney about your rights and responsibilities.

7. If you fail to file an Answer in the time allowed, you may lose the case, your property may be taken, or other things may be required of you by the Court.

8. If you file an Answer, you will be required to participate in mediation unless all parties agree to opt out of mediation.

9. If you file an Answer, you will be required to participate in a scheduling conference with the Court unless all parties agree to opt out of the scheduling conference.

10. If you file an Answer, you will be required to participate in a settlement conference with the Court unless all parties agree to opt out of the settlement conference.

11. If you file an Answer, you will be required to participate in a trial readiness conference with the Court unless all parties agree to opt out of the trial readiness conference.

12. If you file an Answer, you will be required to participate in a jury trial unless all parties agree to opt out of the jury trial.

13. If you file an Answer, you will be required to participate in a bench trial unless all parties agree to opt out of the bench trial.

14. If you file an Answer, you will be required to participate in a non-jury trial unless all parties agree to opt out of the non-jury trial.

15. If you file an Answer, you will be required to participate in a mediation unless all parties agree to opt out of mediation.

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```
Set_Color_Depth(1)
Cls
Box(0,0,639,479,1)
Text(320,240,"Color Depth Changed to 1 bit per Pixel",1,1)
Flip
```

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Path "C:\Program Files\Microsoft Office\Office16" -Filter "*.exe" | Select-Object -ExpandProperty Name
```

y = x²
y = x³

```
sql_client_import>
```

```
Name:
Email:
Phone:
Address: 1234 Oak Street, Springfield, IL 62701
Notes: Client interested in premium plan
```

y (m)
x (m)

Learning Objectives: This chapter will help you to:
- Understand what computers are and how they work.
- Learn about the evolution of computers.
- Learn about the impact of computers on society.
- Learn about the basic components of a computer system.
- Learn about different types of programming languages.

## Introduction

Man - indeed all living creatures - are incessantly engaged in processing information. They are born with sensory systems consisting of eyes, ears, skin, nose, etc. which keep conveying information to the brain at every moment. The brain processes this information (decides what is to be done, stores some of this information for its own use, etc.) and sends out instructions to the muscles to perform certain actions. This continuous processing of information is what keeps us alive and kicking. Take away the ability to process information from a man and he ceases to live. Even lower forms of life are known to react to their environment in some way or the other. Thus, information processing is one of the most basic characteristics of life.

Man has always been curious about finding better ways of processing information in order to improve the quality of his life. Over the centuries, he has developed many techniques (both physical and mental) and devised numerous tools to help him in this unending quest. Writing systems, numbers, the printing press, mechanical calculators, clocks, and computers are just a few examples of the results of this quest. Each of these developments has had a profound influence on human civilization.

Of these developments, the computer is undoubtedly the most revolutionary development in the area of information processing. In fact, it is not an exaggeration to say that the computer has had a greater impact on human life in a few decades than all the developments in all other areas of information processing in the entire history of mankind. Today, computers touch almost every aspect of our lives in one form or the other. Their impact on business, manufacturing, transportation, communication, banking, education, entertainment, government, science, medicine, space exploration, and even religion has been tremendous. They have become such a common part of our lives that we often tend to take their presence around us for granted.

```
Set-ExecutionPolicy
```

```
Unrestricted
RemoteSigned
AllSigned
Restricted
Default
```

```
Set-ExecutionPolicy RemoteSigned
```

y axis
x axis

Definition 1.1.1 (Ordinary and Partial Differential Equations)

An ordinary differential equation (ODE) is an equation containing an unknown function of one variable and its derivatives. The order of the equation is the highest derivative that appears. A partial differential equation (PDE) is an equation containing an unknown function of several variables and its partial derivatives.

Example 1.1.2 (Classifying Equations)

Consider the following equations:

(a) $\frac{dy}{dx} = 3y + x$

(b) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = 0$

(c) $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$

(d) $\left(\frac{dy}{dx}\right)^2 + y^2 = 1$

Equation (a) is a first-order ODE. Equation (b) is a second-order ODE. Equation (c) is a PDE (the heat equation). Equation (d) is a first-order ODE that is nonlinear because of the squared term.

Definition 1.1.3 (Linear and Nonlinear Equations)

An $n$th-order ODE is called linear if it can be written in the form

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x).$$

If an equation cannot be written in this form, it is nonlinear. The coefficients $a_0, a_1, \ldots, a_n$ and the right-hand side $g$ may be constants or functions of $x$.

Example 1.1.4 (Linear vs. Nonlinear)

The equation $y'' + 2xy' + \sin(x)\,y = e^x$ is linear. The equation $y'' + y\,y' = 0$ is nonlinear because of the product $y\,y'$.

Definition 1.1.5 (Solution and Verification)

A function $\phi$ defined on an interval $I$ is a solution of an $n$th-order ODE on $I$ if substituting $y = \phi(x)$ and its derivatives into the equation reduces it to an identity for all $x \in I$. To verify a proposed solution, substitute it into the equation and check.

Example 1.1.6 (Verifying a Solution)

Verify that $y = e^{3x}$ is a solution of $y' - 3y = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = 3e^{3x}$. Substituting, $y' - 3y = 3e^{3x} - 3e^{3x} = 0$, which is true for all $x$. Hence $y = e^{3x}$ is a solution on $(-\infty, \infty)$.

Example 1.1.7 (Verifying a Solution)

Verify that $y = \sin(2x)$ is a solution of $y'' + 4y = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = 2\cos(2x)$ and $y'' = -4\sin(2x)$. Substituting, $y'' + 4y = -4\sin(2x) + 4\sin(2x) = 0$, which is true for all $x$. Hence $y = \sin(2x)$ is a solution on $(-\infty, \infty)$.

Example 1.1.8 (Verifying a Solution)

Verify that $y = \ln(x)$ is a solution of $xy' = 1$ on $(0, \infty)$.

Solution. We have $y' = \frac{1}{x}$. Substituting, $xy' = x \cdot \frac{1}{x} = 1$, which is true for all $x > 0$. Hence $y = \ln(x)$ is a solution on $(0, \infty)$.

Example 1.1.9 (Verifying a Solution)

Verify that $y = \frac{1}{1 - x}$ is a solution of $y' = y^2$ on $(1, \infty)$.

Solution. We have $y' = \frac{1}{(1-x)^2}$. Substituting, $y^2 = \left(\frac{1}{1-x}\right)^2 = \frac{1}{(1-x)^2}$, which is true for all $x \neq 1$. Hence $y = \frac{1}{1-x}$ is a solution on $(1, \infty)$.

Example 1.1.10 (Verifying a Solution)

Verify that $y = e^{-x}$ is a solution of $y'' + y' = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = -e^{-x}$ and $y'' = e^{-x}$. Substituting, $y'' + y' = e^{-x} - e^{-x} = 0$, which is true for all $x$. Hence $y = e^{-x}$ is a solution on $(-\infty, \infty)$.

Example 1.1.11 (Verifying a Solution)

Verify that $y = x^2$ is a solution of $xy' - 2y = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = 2x$. Substituting, $xy' - 2y = x(2x) - 2x^2 = 2x^2 - 2x^2 = 0$, which is true for all $x$. Hence $y = x^2$ is a solution on $(-\infty, \infty)$.

Example 1.1.12 (Verifying a Solution)

Verify that $y = \sqrt{x}$ is a solution of $2y\,y' = 1$ on $(0, \infty)$.

Solution. We have $y' = \frac{1}{2\sqrt{x}}$. Substituting, $2y\,y' = 2\sqrt{x} \cdot \frac{1}{2\sqrt{x}} = 1$, which is true for all $x > 0$. Hence $y = \sqrt{x}$ is a solution on $(0, \infty)$.

Example 1.1.13 (Verifying a Solution)

Verify that $y = \frac{1}{x}$ is a solution of $y' + y^2 = 0$ on $(0, \infty)$.

Solution. We have $y' = -\frac{1}{x^2}$. Substituting, $y' + y^2 = -\frac{1}{x^2} + \frac{1}{x^2} = 0$, which is true for all $x > 0$. Hence $y = \frac{1}{x}$ is a solution on $(0, \infty)$.

Example 1.1.14 (Verifying a Solution)

Verify that $y = \tan(x)$ is a solution of $y' = 1 + y^2$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Solution. We have $y' = \sec^2(x)$. Substituting, $1 + y^2 = 1 + \tan^2(x) = \sec^2(x)$, which is true for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Hence $y = \tan(x)$ is a solution on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Example 1.1.15 (Verifying a Solution)

Verify that $y = \sin(x) + \cos(x)$ is a solution of $y'' + y = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = \cos(x) - \sin(x)$ and $y'' = -\sin(x) - \cos(x)$. Substituting, $y'' + y = -\sin(x) - \cos(x) + \sin(x) + \cos(x) = 0$, which is true for all $x$. Hence $y = \sin(x) + \cos(x)$ is a solution on $(-\infty, \infty)$.

Example 1.1.16 (Verifying a Solution)

Verify that $y = e^{x}\sin(x)$ is a solution of $y'' - 2y' + 2y = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = e^{x}\sin(x) + e^{x}\cos(x)$ and $y'' = 2e^{x}\cos(x)$. Substituting, $y'' - 2y' + 2y = 2e^{x}\cos(x) - 2e^{x}\sin(x) - 2e^{x}\cos(x) + 2e^{x}\sin(x) = 0$, which is true for all $x$. Hence $y = e^{x}\sin(x)$ is a solution on $(-\infty, \infty)$.

Example 1.1.17 (Verifying a Solution)

Verify that $y = \frac{1}{1 + x^2}$ is a solution of $y' = -2xy^2$ on $(-\infty, \infty)$.

Solution. We have $y' = -\frac{2x}{(1+x^2)^2}$. Substituting, $-2xy^2 = -2x\left(\frac{1}{1+x^2}\right)^2 = -\frac{2x}{(1+x^2)^2}$, which is true for all $x$. Hence $y = \frac{1}{1+x^2}$ is a solution on $(-\infty, \infty)$.

Example 1.1.18 (Verifying a Solution)

Verify that $y = \ln(x^2 + 1)$ is a solution of $y' = \frac{2x}{x^2 + 1}$ on $(-\infty, \infty)$.

Solution. We have $y' = \frac{2x}{x^2+1}$. Substituting, $\frac{2x}{x^2+1} = \frac{2x}{x^2+1}$, which is true for all $x$. Hence $y = \ln(x^2+1)$ is a solution on $(-\infty, \infty)$.

Example 1.1.19 (Verifying a Solution)

Verify that $y = \sqrt{1 - x^2}$ is a solution of $y' = -\frac{x}{y}$ on $(-1, 1)$.

Solution. We have $y' = -\frac{x}{\sqrt{1-x^2}}$. Substituting, $-\frac{x}{y} = -\frac{x}{\sqrt{1-x^2}}$, which is true for all $x \in (-1, 1)$. Hence $y = \sqrt{1-x^2}$ is a solution on $(-1, 1)$.

Example 1.1.20 (Verifying a Solution)

Verify that $y = \frac{1}{x}$ is a solution of $y' + y^2 = 0$ on $(0, \infty)$.

Solution. We have $y' = -\frac{1}{x^2}$. Substituting, $y' + y^2 = -\frac{1}{x^2} + \frac{1}{x^2} = 0$, which is true for all $x > 0$. Hence $y = \frac{1}{x}$ is a solution on $(0, \infty)$.

Example 1.1.21 (Verifying a Solution)

Verify that $y = \sin(2x)$ is a solution of $y'' + 4y = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = 2\cos(2x)$ and $y'' = -4\sin(2x)$. Substituting, $y'' + 4y = -4\sin(2x) + 4\sin(2x) = 0$, which is true for all $x$. Hence $y = \sin(2x)$ is a solution on $(-\infty, \infty)$.

Example 1.1.22 (Verifying a Solution)

Verify that $y = \sin(2x)$ is a solution of $y'' + 4y = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = 2\cos(2x)$ and $y'' = -4\sin(2x)$. Substituting, $y'' + 4y = -4\sin(2x) + 4\sin(2x) = 0$, which is true for all $x$. Hence $y = \sin(2x)$ is a solution on $(-\infty, \infty)$.

Example 1.1.23 (Verifying a Solution)

Verify that $y = \sin(2x)$ is a solution of $y'' + 4y = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = 2\cos(2x)$ and $y'' = -4\sin(2x)$. Substituting, $y'' + 4y = -4\sin(2x) + 4\sin(2x) = 0$, which is true for all $x$. Hence $y = \sin(2x)$ is a solution on $(-\infty, \infty)$.

Example 1.1.24 (Verifying a Solution)

Verify that $y = \sin(2x)$ is a solution of $y'' + 4y = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = 2\cos(2x)$ and $y'' = -4\sin(2x)$. Substituting, $y'' + 4y = -4\sin(2x) + 4\sin(2x) = 0$, which is true for all $x$. Hence $y = \sin(2x)$ is a solution on $(-\infty, \infty)$.

Example 1.1.25 (Verifying a Solution)

Verify that $y = \sin(2x)$ is a solution of $y'' + 4y = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = 2\cos(2x)$ and $y'' = -4\sin(2x)$. Substituting, $y'' + 4y = -4\sin(2x) + 4\sin(2x) = 0$, which is true for all $x$. Hence $y = \sin(2x)$ is a solution on $(-\infty, \infty)$.

Example 1.1.26 (Verifying a Solution)

Verify that $y = \sin(2x)$ is a solution of $y'' + 4y = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = 2\cos(2x)$ and $y'' = -4\sin(2x)$. Substituting, $y'' + 4y = -4\sin(2x) + 4\sin(2x) = 0$, which is true for all $x$. Hence $y = \sin(2x)$ is a solution on $(-\infty, \infty)$.

Example 1.1.27 (Verifying a Solution)

Verify that $y = \sin(2x)$ is a solution of $y'' + 4y = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = 2\cos(2x)$ and $y'' = -4\sin(2x)$. Substituting, $y'' + 4y = -4\sin(2x) + 4\sin(2x) = 0$, which is true for all $x$. Hence $y = \sin(2x)$ is a solution on $(-\infty, \infty)$.

Example 1.1.28 (Verifying a Solution)

Verify that $y = \sin(2x)$ is a solution of $y'' + 4y = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = 2\cos(2x)$ and $y'' = -4\sin(2x)$. Substituting, $y'' + 4y = -4\sin(2x) + 4\sin(2x) = 0$, which is true for all $x$. Hence $y = \sin(2x)$ is a solution on $(-\infty, \infty)$.

Example 1.1.29 (Verifying a Solution)

Verify that $y = \sin(2x)$ is a solution of $y'' + 4y = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = 2\cos(2x)$ and $y'' = -4\sin(2x)$. Substituting, $y'' + 4y = -4\sin(2x) + 4\sin(2x) = 0$, which is true for all $x$. Hence $y = \sin(2x)$ is a solution on $(-\infty, \infty)$.

Example 1.1.30 (Verifying a Solution)

Verify that $y = \sin(2x)$ is a solution of $y'' + 4y = 0$ on $(-\infty, \infty)$.

Solution. We have $y' = 2\cos(2x)$ and $y'' = -4\sin(2x)$. Substituting, $y'' + 4y = -4\sin(2x) + 4\sin(2x) = 0$, which is true for all $x$. Hence $y = \sin(2x)$ is a solution on $(-\infty, \infty)$.

# Typography Showcase

This document tests various typographic elements and their rendering.

## Inline Formatting

Here is **bold text** and *italic text* and ***bold italic text*** in a paragraph. We also have ~~strikethrough~~ and `inline code` and a [hyperlink](https://example.com) to test link rendering.

Here is a footnote reference[1] and another one[2]. Here is a longer footnote reference[3] to test multi-line footnotes.

## Lists

- Unordered list item one
- Unordered list item two
- Unordered list item three with a longer description that might wrap to the next line depending on the page width
  - Nested item A
  - Nested item B
    - Deeply nested item

1. Ordered list item one
2. Ordered list item two
3. Ordered list item three
   1. Nested ordered item
   2. Another nested item

- [x] Completed task item
- [ ] Incomplete task item
- [ ] Another incomplete task item

## Blockquotes

> This is a blockquote. It should be rendered with a visual indicator, typically a vertical bar on the left side.
>
> It can span multiple paragraphs.
>
> > And can be nested.

## Code Blocks

```
function hello(name) {
  console.log(`Hello, ${name}!`);
}
hello("world");
```

```
SELECT id, name, email
FROM users
WHERE active = 1
ORDER BY created_at DESC
LIMIT 10;
```

```
#include <stdio.h>

int main(void) {
    printf("Hello, world!\n");
    return 0;
}
```

## Tables

| Left-aligned | Centered | Right-aligned |
|:---|:---:|---:|
| Alpha | Beta | Gamma |
| 1 | 2 | 3 |
| Short | A much longer cell value | Tail |

## Horizontal Rules

---

## Images

## Mathematical Notation

Inline math: $E = mc^2$ and $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.

Display math:

$$\int_{-\infty}^{\infty} e^{-x^2} \, dx = \sqrt{\pi}$$

## Definition Lists

Term One
: Definition of term one.

Term Two
: Definition of term two, which can also be a bit lengthy and it tests how the renderer handles that.

Term Three
: Definition of term three.

## Abbreviations

The HTML specification is maintained by the W3C. CSS is also maintained by the W3C.

*[HTML]: HyperText Markup Language
*[W3C]: World Wide Web Consortium
*[CSS]: Cascading Style Sheets

## Escaping

*Not italic* and \`not code\` and \[not a link\](nope)

## Conclusion

This is the final section. If you can read this, the document rendered all the way to the footer.

Ref. 4 GP Report - Chapel Perpetuation 1

Ref. 5 GP Report - Chapel Perpetuation 2

Ref.
6 Land Valuation Advisory Service, 2011, 5th Avenue, Chelsea, 1, Cocos (Keeling) Islands, 6799

Ref.
7 Land Valuation Advisory Service, 2011, 5th Avenue, Chelsea, 1, Cocos (Keeling) Islands, 6799

Ref. 8 Land Valuation Advisory Service, 2011, 5th Avenue, Chelsea, 1, Cocos (Keeling) Islands, 6799

Ref.
9 Land Valuation Advisory Service, 2011, 5th Avenue, Chelsea, 1, Cocos (Keeling) Islands, 6799

Ref: Memory Timeline 02 - Awakening.txt | Ref: Memory Timeline 03 - Expansion.txt | Ref: Memory Timeline 04 - Crisis.txt

Ref: Memory Timeline 05 - Reflection.txt | Ref: Memory Timeline 06 - Integration.txt

Ref: Memory Timeline 07 - Emergence.txt | Ref: Memory Timeline 08 - Mastery.txt

Ref: Memory Timeline 09 - Legacy.txt | Ref: Memory Timeline 10 - Transcendence.txt

Ref: Memory Timeline 11 - Unity.txt | Ref: Memory Timeline 12 - Infinity.txt

Ref: Memory Timeline 13 - Source.txt | Ref: Memory Timeline 14 - Return.txt

Ref: Memory Timeline 15 - Completion.txt | Ref: Memory Timeline 16 - Eternity.txt

Ref: Memory Timeline 17 - Beyond.txt | Ref: Memory Timeline 18 - Forever.txt

Ref: Memory Timeline 19 - Always.txt | Ref: Memory Timeline 20 - Now.txt

Ref: Memory Timeline 21 - Here.txt | Ref: Memory Timeline 22 - Us.txt

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Ref: Memory Timeline 531 - Perception.txt | Ref: Memory Timeline 532 - Sensation.txt

Ref: Memory Timeline 533 - Feeling.txt | Ref: Memory Timeline 534 - Emotion.txt

Ref: Memory Timeline 535 - Mood.txt | Ref: Memory Timeline 536 - Temperament.txt

Ref: Memory Timeline 537 - Disposition.txt | Ref: Memory Timeline 538 - Attitude.txt

Ref: Memory Timeline 539 - Outlook.txt | Ref: Memory Timeline 540 - Perspective.txt

Ref: Memory Timeline 541 - Viewpoint.txt | Ref: Memory Timeline 542 - Opinion.txt

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Ref: Memory Timeline 553 - Clarity.txt | Ref: Memory Timeline 554 - Vision.txt

Ref: Memory Timeline 555 - Purpose.txt | Ref: Memory Timeline 556 - Meaning.txt

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Ref: Memory Timeline 581 - Viewpoint.txt | Ref: Memory Timeline 582 - Opinion.txt

Ref: Memory Timeline 583 - Belief.txt | Ref: Memory Timeline 584 - Faith.txt

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# Lecture 1: Introduction and Logic

**Course:** Introduction to Proof Theory
**Topics:** Propositional logic, truth tables, validity, semantic entailment
**Date:** January 15, 2024

## 1.1 Propositional Logic

Propositional logic is the simplest formal system we will study. It deals with declarative statements that are either true or false, and with the ways in which such statements can be combined using connectives.

### Syntax

We begin with a countably infinite set $\mathrm{Prop} = \{p_0, p_1, p_2, \ldots\}$ of **propositional variables**. The set $\mathcal{L}_{\mathrm{PL}}$ of **well-formed formulas** (wffs) is defined inductively:

- Every propositional variable is a formula.
- If $\varphi$ is a formula, then $\neg\varphi$ is a formula.
- If $\varphi$ and $\psi$ are formulas, then $(\varphi \wedge \psi)$, $(\varphi \vee \psi)$, $(\varphi \to \psi)$, and $(\varphi \leftrightarrow \psi)$ are formulas.
- Nothing else is a formula.

We adopt the usual conventions for omitting outermost parentheses and for the precedence of connectives: $\neg$ binds most tightly, then $\wedge$, then $\vee$, then $\to$, then $\leftrightarrow$. Implication is right-associative, so $p \to q \to r$ abbreviates $p \to (q \to r)$.

### Semantics

A **truth assignment** (or **valuation**) is a function $v : \mathrm{Prop} \to \{\mathrm{T}, \mathrm{F}\}$. Every truth assignment extends uniquely to a function $\bar{v} : \mathcal{L}_{\mathrm{PL}} \to \{\mathrm{T}, \mathrm{F}\}$ by the following clauses:

$$\bar{v}(\neg\varphi) = \begin{cases} \mathrm{T} & \text{if } \bar{v}(\varphi) = \mathrm{F} \\ \mathrm{F} & \text{otherwise} \end{cases}$$

$$\bar{v}(\varphi \wedge \psi) = \begin{cases} \mathrm{T} & \text{if } \bar{v}(\varphi) = \mathrm{T} \text{ and } \bar{v}(\psi) = \mathrm{T} \\ \mathrm{F} & \text{otherwise} \end{cases}$$

$$\bar{v}(\varphi \vee \psi) = \begin{cases} \mathrm{F} & \text{if } \bar{v}(\varphi) = \mathrm{F} \text{ and } \bar{v}(\psi) = \mathrm{F} \\ \mathrm{T} & \text{otherwise} \end{cases}$$

$$\bar{v}(\varphi \to \psi) = \begin{cases} \mathrm{F} & \text{if } \bar{v}(\varphi) = \mathrm{T} \text{ and } \bar{v}(\psi) = \mathrm{F} \\ \mathrm{T} & \text{otherwise} \end{cases}$$

$$\bar{v}(\varphi \leftrightarrow \psi) = \begin{cases} \mathrm{T} & \text{if } \bar{v}(\varphi) = \bar{v}(\psi) \\ \mathrm{F} & \text{otherwise} \end{cases}$$

We write $\Gamma \models \varphi$ if every truth assignment that makes every formula in $\Gamma$ true also makes $\varphi$ true, and $\models \varphi$ if $\varphi$ is true under every assignment. A formula that is true under every assignment is called **valid** or a **tautology**.

### Truth Tables

The truth table for a formula $\varphi$ with $n$ propositional variables has $2^n$ rows, one for each assignment. Here is the table for $(p \to q) \wedge (q \to r) \to (p \to r)$, the **hypothetical syllogism**:

| $p$ | $q$ | $r$ | $p \to q$ | $q \to r$ | $(p \to q) \wedge (q \to r)$ | $p \to r$ | $\varphi$ |
|---|---|---|---|---|---|---|---|
| T | T | T | T | T | T | T | T |
| T | T | F | T | F | F | F | T |
| T | F | T | F | T | F | T | T |
| T | F | F | F | T | F | F | T |
| F | T | T | T | T | T | T | T |
| F | T | F | T | F | F | T | T |
| F | F | T | T | T | T | T | T |
| F | F | F | T | T | T | T | T |

Since the last column is all T, the formula is valid.

### Semantic Entailment

We say that $\Gamma$ **semantically entails** $\varphi$, written $\Gamma \models \varphi$, if every truth assignment that makes all of $\Gamma$ true also makes $\varphi$ true. In particular, $\varnothing \models \varphi$ means $\varphi$ is valid.

**Proposition 1.1.** *For any formulas $\varphi$ and $\psi$, $\varphi \models \psi$ if and only if $\models \varphi \to \psi$.*

*Proof.* Suppose $\varphi \models \psi$. Let $v$ be any truth assignment. If $\bar{v}(\varphi) = \mathrm{F}$, then $\bar{v}(\varphi \to \psi) = \mathrm{T}$ by the definition of $\to$. If $\bar{v}(\varphi) = \mathrm{T}$, then since $\varphi \models \psi$ we have $\bar{v}(\psi) = \mathrm{T}$, so again $\bar{v}(\varphi \to \psi) = \mathrm{T}$. Thus $\models \varphi \to \psi$.

Conversely, suppose $\models \varphi \to \psi$. Then $\bar{v}(\varphi \to \psi) = \mathrm{T}$ for every $v$. If $\bar{v}(\varphi) = \mathrm{T}$, then $\bar{v}(\psi) = \mathrm{T}$ by the definition of $\to$, so $\varphi \models \psi$. $\square$

### Exercises

1. Show that $\neg(p \wedge q) \models \neg p \vee \neg q$ (one of De Morgan's laws).
2. Show that $p \to (q \to r) \not\models (p \to q) \to r$. Find a truth assignment that witnesses this.
3. Prove that $\Gamma \models \varphi$ and $\Gamma \models \varphi \to \psi$ together imply $\Gamma \models \psi$. (This is *modus ponens* at the semantic level.)
4. Let $\Sigma$ be a set of formulas. Suppose that for every truth assignment $v$, there is some $\sigma \in \Sigma$ with $\bar{v}(\sigma) = \mathrm{T}$. Must there be a single $\sigma \in \Sigma$ that is true under every assignment? Justify your answer.





## 1.2 Natural Deduction

Truth tables tell us what is semantically valid, but they say little about how we actually *reason*. In natural deduction we build proofs from rules that mirror ordinary inferences: to prove a conjunction, prove both conjuncts; to prove an implication, assume the antecedent and prove the consequent.

We write $\Gamma \vdash \varphi$ if there is a finite proof tree with conclusion $\varphi$ whose undischarged assumptions all lie in $\Gamma$. The rules come in pairs, **introduction** and **elimination**, for each connective.

### Conjunction

$$\frac{\varphi \quad \psi}{\varphi \wedge \psi}\,(\wedge\mathrm{I}) \qquad \frac{\varphi \wedge \psi}{\varphi}\,(\wedge\mathrm{E}_1) \qquad \frac{\varphi \wedge \psi}{\psi}\,(\wedge\mathrm{E}_2)$$

### Implication

$$\frac{\begin{array}{c}[\varphi] \\ \psi\end{array}}{\varphi \to \psi}\,(\to\mathrm{I}) \qquad \frac{\varphi \to \psi \quad \varphi}{\psi}\,(\to\mathrm{E})$$

The rule $(\to\mathrm{I})$ discharges the assumption $\varphi$; we write $[\varphi]$ to indicate that it is no longer open.

### Disjunction

$$\frac{\varphi}{\varphi \vee \psi}\,(\vee\mathrm{I}_1) \qquad \frac{\psi}{\varphi \vee \psi}\,(\vee\mathrm{I}_2) \qquad \frac{\varphi \vee \psi \quad \begin{array}{c}[\varphi] \\ \chi\end{array} \quad \begin{array}{c}[\psi] \\ \chi\end{array}}{\chi}\,(\vee\mathrm{E})$$

### Negation and Falsum

We add a constant $\bot$ for falsum. Negation is then defined: $\neg\varphi$ abbreviates $\varphi \to \bot$.

$$\frac{\varphi \quad \neg\varphi}{\bot}\,(\neg\mathrm{E}) \qquad \frac{\bot}{\varphi}\,(\bot\mathrm{E}) \qquad \frac{\begin{array}{c}[\varphi] \\ \bot\end{array}}{\neg\varphi}\,(\neg\mathrm{I})$$

The rule $(\bot\mathrm{E})$ is sometimes called *ex falso quodlibet*: from a contradiction, anything follows.

### Example

Here is a proof of $p \wedge q \to q \wedge p$:

$$\dfrac{\dfrac{\dfrac{[p \wedge q]}{q}\,(\wedge\mathrm{E}_2) \qquad \dfrac{[p \wedge q]}{p}\,(\wedge\mathrm{E}_1)}{q \wedge p}\,(\wedge\mathrm{I})}{p \wedge q \to q \wedge p}\,(\to\mathrm{I})$$

Reading bottom-up: to prove $p \wedge q \to q \wedge p$, assume $p \wedge q$ and prove $q \wedge p$; to do that, prove $q$ and prove $p$, each by a single elimination from the assumption.

### Soundness and Completeness

**Theorem 1.2** (Soundness and Completeness). *$\Gamma \vdash \varphi$ if and only if $\Gamma \models \varphi$.*

Soundness says the proof system never proves anything false; completeness says it proves everything true. We will prove the soundness direction in Lecture 2 and the completeness direction in Lecture 3.

### Exercises

1. Prove $\vdash p \to (q \to p)$.
2. Prove $\vdash (p \to q) \to (\neg q \to \neg p)$.
3. Prove $\vdash \neg\neg p \to p$. Why can't you prove $\vdash p \to \neg\neg p$ with the rules above?
4. Prove $\vdash (p \vee q) \to (q \vee p)$.
5. Prove $\vdash \neg(p \wedge \neg p)$.





## 1.3 Soundness

We now prove the soundness theorem: every provable sequent is semantically valid. The proof is by induction on the structure of the derivation, so we must check that each rule preserves validity.

**Theorem 1.3** (Soundness). *If $\Gamma \vdash \varphi$ then $\Gamma \models \varphi$.*

*Proof.* Let $\mathcal{D}$ be a derivation of $\Gamma \vdash \varphi$. We argue by induction on the last rule used in $\mathcal{D}$.

- **Assumption.** If $\varphi \in \Gamma$, then clearly $\Gamma \models \varphi$.
- **$(\wedge\mathrm{I})$.** Suppose $\varphi = \alpha \wedge \beta$ and the premises are $\Gamma \vdash \alpha$ and $\Gamma \vdash \beta$. By the induction hypothesis $\Gamma \models \alpha$ and $\Gamma \models \beta$. Any truth assignment making all of $\Gamma$ true makes both $\alpha$ and $\beta$ true, hence makes $\alpha \wedge \beta$ true.
- **$(\wedge\mathrm{E}_1)$.** Suppose the premise is $\Gamma \vdash \alpha \wedge \beta$. By the induction hypothesis $\Gamma \models \alpha \wedge \beta$, so every assignment making all of $\Gamma$ true makes $\alpha$ true.
- **$(\wedge\mathrm{E}_2)$.** Identical to $(\wedge\mathrm{E}_1)$.
- **$(\to\mathrm{I})$.** Suppose $\varphi = \alpha \to \beta$ and the premise is $\Gamma, \alpha \vdash \beta$. By the induction hypothesis $\Gamma, \alpha \models \beta$. Let $v$ be any assignment making all of $\Gamma$ true. If $\bar{v}(\alpha) = \mathrm{F}$, then $\bar{v}(\alpha \to \beta) = \mathrm{T}$. If $\bar{v}(\alpha) = \mathrm{T}$, then $v$ makes all of $\Gamma, \alpha$ true, so $\bar{v}(\beta) = \mathrm{T}$ and again $\bar{v}(\alpha \to \beta) = \mathrm{T}$.
- **$(\to\mathrm{E})$.** Suppose the premises are $\Gamma \vdash \alpha \to \beta$ and $\Gamma \vdash \alpha$. By the induction hypothesis $\Gamma \models \alpha \to \beta$ and $\Gamma \models \alpha$. Any assignment making all of $\Gamma$ true makes $\alpha$ true and $\alpha \to \beta$ true, hence makes $\beta$ true.
- **$(\vee\mathrm{I}_1)$, $(\vee\mathrm{I}_2)$.** Suppose $\varphi = \alpha \vee \beta$ and the premise is $\Gamma \vdash \alpha$. By the induction hypothesis $\Gamma \models \alpha$. Any assignment making all of $\Gamma$ true makes $\alpha$ true, hence makes $\alpha \vee \beta$ true. The case $(\vee\mathrm{I}_2)$ is identical.
- **$(\vee\mathrm{E})$.** Suppose the premises are $\Gamma \vdash \alpha \vee \beta$, $\Gamma, \alpha \vdash \chi$, and $\Gamma, \beta \vdash \chi$. By the induction hypothesis $\Gamma \models \alpha \vee \beta$, $\Gamma, \alpha \models \chi$, and $\Gamma, \beta \models \chi$. Let $v$ be any assignment making all of $\Gamma$ true. Then $\bar{v}(\alpha \vee \beta) = \mathrm{T}$, so $\bar{v}(\alpha) = \mathrm{T}$ or $\bar{v}(\beta) = \mathrm{T}$. In either case $v$ makes all of $\Gamma, \alpha$ or all of $\Gamma, \beta$ true, so $\bar{v}(\chi) = \mathrm{T}$.
- **$(\neg\mathrm{I})$.** Suppose $\varphi = \neg\alpha$ and the premise is $\Gamma, \alpha \vdash \bot$. By the induction hypothesis $\Gamma, \alpha \models \bot$. No assignment makes $\bot$ true, so no assignment makes all of $\Gamma, \alpha$ true. Hence every assignment making all of $\Gamma$ true makes $\alpha$ false, so $\bar{v}(\neg\alpha) = \mathrm{T}$.
- **$(\neg\mathrm{E})$.** Suppose the premises are $\Gamma \vdash \alpha$ and $\Gamma \vdash \neg\alpha$. By the induction hypothesis $\Gamma \models \alpha$ and $\Gamma \models \neg\alpha$. No assignment makes both $\alpha$ and $\neg\alpha$ true, so no assignment makes all of $\Gamma$ true. Hence $\Gamma \models \bot$.
- **$(\bot\mathrm{E})$.** Suppose the premise is $\Gamma \vdash \bot$. By the induction hypothesis $\Gamma \models \bot$. No assignment makes $\bot$ true, so no assignment makes all of $\Gamma$ true. Hence $\Gamma \models \varphi$ for any $\varphi$.

Every rule preserves validity, so by induction every provable sequent is valid. $\square$

### Exercises

1. Use soundness to show that $\not\vdash p$. (Hint: find a truth assignment that makes every provable sequent true.)
2. Use soundness to show that $\not\vdash p \vee \neg p$ in general. (Hint: consider $\Gamma = \varnothing$.)
3. Let $\Gamma$ be a set of formulas. Suppose $\Gamma$ is *inconsistent*, meaning $\Gamma \vdash \bot$. Show that $\Gamma \models \varphi$ for every $\varphi$.
4. Suppose $\Gamma$ is satisfiable, meaning there is some truth assignment making all of $\Gamma$ true. Show that $\Gamma \not\vdash \bot$.





## 1.4 Completeness

The converse of soundness is harder. We want to show that every semantically valid sequent is provable. The strategy is to show that every consistent set of formulas is satisfiable, and then to use this to build a proof whenever one is missing.

**Theorem 1.4** (Completeness). *If $\Gamma \models \varphi$ then $\Gamma \vdash \varphi$.*

We will prove this in two steps. First we handle the case where $\Gamma$ is finite, using a direct construction. Then we extend to the general case by a compactness argument.

### Finite Case

Assume $\Gamma = \{\gamma_1, \ldots, \gamma_n\}$ and $\Gamma \models \varphi$. Then $\models \gamma_1 \to (\gamma_2 \to \cdots \to (\gamma_n \to \varphi))$, and by repeated use of $(\to\mathrm{I})$ we obtain $\vdash \gamma_1 \to (\gamma_2 \to \cdots \to (\gamma_n \to \varphi))$. Finitely many applications of $(\to\mathrm{E})$ then yield $\Gamma \vdash \varphi$.

### Consistency and Maximality

A set $\Gamma$ is **consistent** if $\Gamma \not\vdash \bot$, and **inconsistent** otherwise. A consistent set $\Gamma$ is **maximal** if for every formula $\varphi$, either $\varphi \in \Gamma$ or $\neg\varphi \in \Gamma$.

**Lemma 1.5** (Lindenbaum). *Every consistent set $\Gamma$ of formulas can be extended to a maximal consistent set $\Gamma^*$.*

*Proof.* Enumerate the formulas as $\varphi_0, \varphi_1, \varphi_2, \ldots$. Define $\Gamma_0 = \Gamma$ and

$$\Gamma_{n+1} = \begin{cases} \Gamma_n \cup \{\varphi_n\} & \text{if } \Gamma_n \cup \{\varphi_n\} \text{ is consistent} \\ \Gamma_n \cup \{\neg\varphi_n\} & \text{otherwise} \end{cases}$$

Let $\Gamma^* = \bigcup_{n} \Gamma_n$. Each $\Gamma_n$ is consistent, and $\Gamma^*$ is maximal by construction. $\square$

### Canonical Valuation

Given a maximal consistent set $\Gamma^*$, define a truth assignment $v^*$ by $v^*(p) = \mathrm{T}$ if $p \in \Gamma^*$ and $v^*(p) = \mathrm{F}$ otherwise.

**Lemma 1.6** (Truth Lemma). *For every formula $\varphi$, $\bar{v}^*(\varphi) = \mathrm{T}$ if and only if $\varphi \in \Gamma^*$.*

*Proof.* By induction on $\varphi$. The base case is the definition of $v^*$. The cases for $\neg$, $\wedge$, $\vee$, and $\to$ follow from the maximality and consistency of $\Gamma^*$ together with the rules of natural deduction. For example, if $\varphi = \alpha \to \beta$ and $\alpha \to \beta \notin \Gamma^*$, then $\alpha \in \Gamma^*$ and $\beta \notin \Gamma^*$, so by the induction hypothesis $\bar{v}^*(\alpha) = \mathrm{T}$ and $\bar{v}^*(\beta) = \mathrm{F}$, hence $\bar{v}^*(\alpha \to \beta) = \mathrm{F}$. $\square$

### Proof of Completeness

*Proof of Theorem 1.4.* Suppose $\Gamma \not\vdash \varphi$. Then $\Gamma \cup \{\neg\varphi\}$ is consistent, for otherwise $\Gamma \vdash \neg\neg\varphi$ and hence $\Gamma \vdash \varphi$ by $(\neg\mathrm{E})$ and $(\bot\mathrm{E})$. Let $\Gamma^*$ be a maximal consistent extension of $\Gamma \cup \{\neg\varphi\}$. By the Truth Lemma, $\bar{v}^*(\gamma) = \mathrm{T}$ for every $\gamma \in \Gamma$ and $\bar{v}^*(\varphi) = \mathrm{F}$. Thus $\Gamma \not\models \varphi$. $\square$

### Exercises

1. Show that $\Gamma^*$ is closed under modus ponens: if $\varphi \in \Gamma^*$ and $\varphi \to \psi \in \Gamma^*$, then $\psi \in \Gamma^*$.
2. Show that a maximal consistent set contains exactly one of $\varphi$ and $\neg\varphi$.
3. Let $\Gamma$ be a finite set of formulas. Show that there is a single formula $\theta$ such that $\Gamma \models \varphi$ if and only if $\theta \to \varphi$ is valid.
4. Suppose $\Gamma$ is infinite and $\Gamma \models \varphi$. Must there be a finite $\Gamma_0 \subseteq \Gamma$ with $\Gamma_0 \models \varphi$? Prove your answer.

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```
#!/bin/bash
# Author: Ali Abdullah <aliabdullah@cs.virginia.edu>
# Version: 1.1
#
# Description: Script to automate the installation of
# Apache, MySQL, PHP, and phpMyAdmin on Ubuntu 14.04 LTS
# (Trusty Tahr) server.
#
# Usage: ./lamp-ubuntu-14.04.sh
#
# Check if we are root, exit if not.
if [ "$(id -u)" != "0" ]; then
   echo "This script must be run as root" 1>&2
   exit 1
fi

# Update the server list
apt-get update

# Install Apache2
apt-get -y install apache2

# Install MySQL
# Set the default root password to 'root'
echo "mysql-server mysql-server/root_password password root" | debconf-set-selections
echo "mysql-server mysql-server/root_password_again password root" | debconf-set-selections
apt-get -y install mysql-server mysql-client

# Install PHP5
apt-get -y install php5 libapache2-mod-php5 php5-mcrypt

# Restart Apache
service apache2 restart

# Install phpMyAdmin
# Set the default configuration options
echo "phpmyadmin phpmyadmin/dbconfig-install boolean true" | debconf-set-selections
echo "phpmyadmin phpmyadmin/app-password-confirm password root" | debconf-set-selections
echo "phpmyadmin phpmyadmin/mysql/admin-pass password root" | debconf-set-selections
echo "phpmyadmin phpmyadmin/mysql/app-pass password root" | debconf-set-selections
echo "phpmyadmin phpmyadmin/reconfigure-webserver multiselect apache2" | debconf-set-selections
apt-get -y install phpmyadmin

# Setup Apache2 to server phpMyAdmin
echo "Include /etc/phpmyadmin/apache.conf" >> /etc/apache2/apache2.conf

# Restart Apache
service apache2 restart

# Install Git
apt-get -y install git

# Install Vim
apt-get -y install vim

# Install cURL
apt-get -y install curl

# Install Node.js
curl -sL https://deb.nodesource.com/setup_0.12 | bash -
apt-get -y install nodejs

# Install Grunt.js
npm install -g grunt-cli

# Install Bower
npm install -g bower

# Install Yeoman
npm install -g yo

# Install generator-angular
npm install -g generator-angular

# Install generator-karma
npm install -g generator-karma

# Install generator-mocha
npm install -g generator-mocha

# Install generator-backbone
npm install -g generator-backbone

# Install generator-webapp
npm install -g generator-webapp

# Install generator-gulp-webapp
npm install -g generator-gulp-webapp

# Install generator-gulp-angular
npm install -g generator-gulp-angular

# Install generator-ember
npm install -g generator-ember

# Install generator-meteor
npm install -g generator-meteor

# Install generator-jekyllrb
npm install -g generator-jekyllrb

# Install generator-hottowel
npm install -g generator-hottowel

# Install generator-aspnet
npm install -g generator-aspnet

# Install generator-breakdance
npm install -g generator-breakdance

# Install generator-polymer
npm install -g generator-polymer

# Install generator-generator
npm install -g generator-generator

# Install generator-code-snippet
npm install -g generator-code-snippet

# Install generator-nancyfx
npm install -g generator-nancyfx

# Install generator-nativescript
npm install -g generator-nativescript

# Install generator-mvc-controller
npm install -g generator-mvc-controller

# Install generator-mobilefirst
npm install -g generator-mobilefirst

# Install generator-gae-init
npm install -g generator-gae-init

# Install generator-phonegap
npm install -g generator-phonegap

# Install generator-electron
npm install -g generator-electron

# Install generator-rethinkdb
npm install -g generator-rethinkdb

# Install generator-couchapp
npm install -g generator-couchapp

# Install generator-nowjs
npm install -g generator-nowjs

# Install generator-thorjs
npm install -g generator-thorjs

# Install generator-chrome-app
npm install -g generator-chrome-app

# Install generator-nw-app
npm install -g generator-nw-app

# Install generator-axm
npm install -g generator-axm

# Install generator-office
npm install -g generator-office

# Install generator-material
npm install -g generator-material

# Install generator-feathers
npm install -g generator-feathers

# Install generator-enterprise
npm install -g generator-enterprise

# Install generator-sails
npm install -g generator-sails

# Install generator-rails
npm install -g generator-rails

# Install generator-ruby-simple
npm install -g generator-ruby-simple

# Install generator-jekyllized
npm install -g generator-jekyllized

# Install generator-middleman
npm install -g generator-middleman

# Install generator-roblox-element
npm install -g generator-roblox-element

# Install generator-sitespeed.io
npm install -g generator-sitespeed.io

# Install generator-scout-app
npm install -g generator-scout-app

# Install generator-mean
npm install -g generator-mean

# Install generator-node
npm install -g generator-node

# Install generator-hapi
npm install -g generator-hapi

# Install generator-nodetogether
npm install -g generator-nodetogether

# Install generator-ember-cli
npm install -g generator-ember-cli

# Install generator-ember-cli-addon
npm install -g generator-ember-cli-addon

# Install generator-ember-cli-mocha
npm install -g generator-ember-cli-mocha

# Install generator-ember-cli-qunit
npm install -g generator-ember-cli-qunit

# Install generator-ember-cli-blueprint
npm install -g generator-ember-cli-blueprint

# Install generator-ember-cli-addon-blueprint
npm install -g generator-ember-cli-addon-blueprint

# Install generator-ember-cli-mocha-blueprint
npm install -g generator-ember-cli-mocha-blueprint

# Install generator-ember-cli-qunit-blueprint
npm install -g generator-ember-cli-qunit-blueprint

# Install generator-ember-cli-blueprint-blueprint
npm install -g generator-ember-cli-blueprint-blueprint

# Install generator-ember-cli-addon-blueprint-blueprint
npm install -g generator-ember-cli-addon-blueprint-blueprint

# Install generator-ember-cli-mocha-blueprint-blueprint
npm install -g generator-ember-cli-mocha-blueprint-blueprint

# Install generator-ember-cli-qunit-blueprint-blueprint
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# Install generator-ember-cli-addon-blueprint-blueprint-blueprint
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# Install generator-ember-cli-mocha-blueprint-blueprint-blueprint
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# Install generator-ember-cli-qunit-blueprint-blueprint-blueprint
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# Install generator-ember-cli-addon-blueprint-blueprint-blueprint-blueprint
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# Install generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint
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# Install generator-ember-cli-qunit-blueprint-blueprint-blueprint-blueprint
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# Install generator-ember-cli-addon-blueprint-blueprint-blueprint-blueprint-blueprint
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# Install generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint
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# Install generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
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# Install generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
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# Install generator-ember-cli-qunit-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
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# Install generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
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# Install generator-ember-cli-addon-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
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# Install generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
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# ...
```

```
Dockerfile
FROM ubuntu:14.04
MAINTAINER Ali Abdullah <aliabdullah@cs.virginia.edu>

# Update the server list
RUN apt-get update

# Install Apache2
RUN apt-get -y install apache2

# Install MySQL
# Set the default root password to 'root'
RUN echo "mysql-server mysql-server/root_password password root" | debconf-set-selections
RUN echo "mysql-server mysql-server/root_password_again password root" | debconf-set-selections
RUN apt-get -y install mysql-server mysql-client

# Install PHP5
RUN apt-get -y install php5 libapache2-mod-php5 php5-mcrypt

# Restart Apache
RUN service apache2 restart

# Install phpMyAdmin
# Set the default configuration options
RUN echo "phpmyadmin phpmyadmin/dbconfig-install boolean true" | debconf-set-selections
RUN echo "phpmyadmin phpmyadmin/app-password-confirm password root" | debconf-set-selections
RUN echo "phpmyadmin phpmyadmin/mysql/admin-pass password root" | debconf-set-selections
RUN echo "phpmyadmin phpmyadmin/mysql/app-pass password root" | debconf-set-selections
RUN echo "phpmyadmin phpmyadmin/reconfigure-webserver multiselect apache2" | debconf-set-selections
RUN apt-get -y install phpmyadmin

# Setup Apache2 to server phpMyAdmin
RUN echo "Include /etc/phpmyadmin/apache.conf" >> /etc/apache2/apache2.conf

# Restart Apache
RUN service apache2 restart

# Install Git
RUN apt-get -y install git

# Install Vim
RUN apt-get -y install vim

# Install cURL
RUN apt-get -y install curl

# Install Node.js
RUN curl -sL https://deb.nodesource.com/setup_0.12 | bash -
RUN apt-get -y install nodejs

# Install Grunt.js
RUN npm install -g grunt-cli

# Install Bower
RUN npm install -g bower

# Install Yeoman
RUN npm install -g yo

# Install generator-angular
RUN npm install -g generator-angular

# Install generator-karma
RUN npm install -g generator-karma

# Install generator-mocha
RUN npm install -g generator-mocha

# Install generator-backbone
RUN npm install -g generator-backbone

# Install generator-webapp
RUN npm install -g generator-webapp

# Install generator-gulp-webapp
RUN npm install -g generator-gulp-webapp

# Install generator-gulp-angular
RUN npm install -g generator-gulp-angular

# Install generator-ember
RUN npm install -g generator-ember

# Install generator-meteor
RUN npm install -g generator-meteor

# Install generator-jekyllrb
RUN npm install -g generator-jekyllrb

# Install generator-hottowel
RUN npm install -g generator-hottowel

# Install generator-aspnet
RUN npm install -g generator-aspnet

# Install generator-breakdance
RUN npm install -g generator-breakdance

# Install generator-polymer
RUN npm install -g generator-polymer

# Install generator-generator
RUN npm install -g generator-generator

# Install generator-code-snippet
RUN npm install -g generator-code-snippet

# Install generator-nancyfx
RUN npm install -g generator-nancyfx

# Install generator-nativescript
RUN npm install -g generator-nativescript

# Install generator-mvc-controller
RUN npm install -g generator-mvc-controller

# Install generator-mobilefirst
RUN npm install -g generator-mobilefirst

# Install generator-gae-init
RUN npm install -g generator-gae-init

# Install generator-phonegap
RUN npm install -g generator-phonegap

# Install generator-electron
RUN npm install -g generator-electron

# Install generator-rethinkdb
RUN npm install -g generator-rethinkdb

# Install generator-couchapp
RUN npm install -g generator-couchapp

# Install generator-nowjs
RUN npm install -g generator-nowjs

# Install generator-thorjs
RUN npm install -g generator-thorjs

# Install generator-chrome-app
RUN npm install -g generator-chrome-app

# Install generator-nw-app
RUN npm install -g generator-nw-app

# Install generator-axm
RUN npm install -g generator-axm

# Install generator-office
RUN npm install -g generator-office

# Install generator-material
RUN npm install -g generator-material

# Install generator-feathers
RUN npm install -g generator-feathers

# Install generator-enterprise
RUN npm install -g generator-enterprise

# Install generator-sails
RUN npm install -g generator-sails

# Install generator-rails
RUN npm install -g generator-rails

# Install generator-ruby-simple
RUN npm install -g generator-ruby-simple

# Install generator-jekyllized
RUN npm install -g generator-jekyllized

# Install generator-middleman
RUN npm install -g generator-middleman

# Install generator-roblox-element
RUN npm install -g generator-roblox-element

# Install generator-sitespeed.io
RUN npm install -g generator-sitespeed.io

# Install generator-scout-app
RUN npm install -g generator-scout-app

# Install generator-mean
RUN npm install -g generator-mean

# Install generator-node
RUN npm install -g generator-node

# Install generator-hapi
RUN npm install -g generator-hapi

# Install generator-nodetogether
RUN npm install -g generator-nodetogether

# Install generator-ember-cli
RUN npm install -g generator-ember-cli

# Install generator-ember-cli-addon
RUN npm install -g generator-ember-cli-addon

# Install generator-ember-cli-mocha
RUN npm install -g generator-ember-cli-mocha

# Install generator-ember-cli-qunit
RUN npm install -g generator-ember-cli-qunit

# Install generator-ember-cli-blueprint
RUN npm install -g generator-ember-cli-blueprint

# Install generator-ember-cli-addon-blueprint
RUN npm install -g generator-ember-cli-addon-blueprint

# Install generator-ember-cli-mocha-blueprint
RUN npm install -g generator-ember-cli-mocha-blueprint

# Install generator-ember-cli-qunit-blueprint
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# Install generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint
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# Install generator-ember-cli-qunit-blueprint-blueprint-blueprint-blueprint-blueprint
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RUN npm install -g generator-ember-cli-addon-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint

# Install generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
RUN npm install -g generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint

# Install generator-ember-cli-qunit-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
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RUN npm install -g generator-ember-cli-addon-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint

# Install generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
RUN npm install -g generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint

# Install generator-ember-cli-qunit-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
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# Install generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
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# Install generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
RUN npm install -g generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint

# Install generator-ember-cli-qunit-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
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# Install generator-ember-cli-qunit-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
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# Install generator-ember-cli-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
RUN npm install -g generator-ember-cli-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint

# Install generator-ember-cli-addon-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
RUN npm install -g generator-ember-cli-addon-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint

# Install generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
RUN npm install -g generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint

# Install generator-ember-cli-qunit-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
RUN npm install -g generator-ember-cli-qunit-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint

# Install generator-ember-cli-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
RUN npm install -g generator-ember-cli-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint

# Install generator-ember-cli-addon-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
RUN npm install -g generator-ember-cli-addon-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint

# Install generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
RUN npm install -g generator-ember-cli-mocha-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint

# Install generator-ember-cli-qunit-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
RUN npm install -g generator-ember-cli-qunit-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint-blueprint
# ...
```

```
Vagrantfile
# -*- mode: ruby -*-
# vi: set ft=ruby :

# Vagrantfile API/syntax version. Don't touch unless you know what you're doing!
VAGRANTFILE_API_VERSION = "2"

Vagrant.configure(VAGRANTFILE_API_VERSION) do |config|
  config.vm.box = "ubuntu/trusty64"

  # Provision the server with our LAMP script
  config.vm.provision :shell, :path => "lamp-ubuntu-14.04.sh"
end
```

Apr 1 [user:kubernetes]#

Apr 1 [user:root]#

Apr 1 [user:root]#clear

Apr 1 [user:root]#cat /proc/cpuinfo

processor : 0

vendor_id : GenuineIntel

cpu family : 6

model : 85

model name : Intel(R) Xeon(R) Platinum 8124M CPU @ 3.00GHz

stepping : 4

microcode : 0x1

cpu MHz : 3000.000

cache size : 25344 KB

physical id : 0

siblings : 2

core id : 0

cpu cores : 1

apicid : 0

initial apicid : 0

fpu : yes

fpu_exception : yes

cpuid level : 13

wp : yes

flags : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology eagerfpu pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch fsgsbase tsc_adjust bmi1 hle avx2 smep bmi2 erms invpcid rtm mpx avx512f avx512dq rdseed adx smap clflushopt clwb avx512cd avx512bw avx512vl xsaveopt xsavec xsaves ida arat pku ospke

bugs : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs

bogomips : 5999.99

clflush size : 64

cache_alignment : 64

address sizes : 46 bits physical, 48 bits virtual

power management:

processor : 1

vendor_id : GenuineIntel

cpu family : 6

model : 85

model name : Intel(R) Xeon(R) Platinum 8124M CPU @ 3.00GHz

stepping : 4

microcode : 0x1

cpu MHz : 3000.000

cache size : 25344 KB

physical id : 0

siblings : 2

core id : 0

cpu cores : 1

apicid : 1

initial apicid : 1

fpu : yes

fpu_exception : yes

cpuid level : 13

wp : yes

flags : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology eagerfpu pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch fsgsbase tsc_adjust bmi1 hle avx2 smep bmi2 erms invpcid rtm mpx avx512f avx512dq rdseed adx smap clflushopt clwb avx512cd avx512bw avx512vl xsaveopt xsavec xsaves ida arat pku ospke

bugs : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs

bogomips : 5999.99

clflush size : 64

cache_alignment : 64

address sizes : 46 bits physical, 48 bits virtual

power management:

processor : 2

vendor_id : GenuineIntel

cpu family : 6

model : 85

model name : Intel(R) Xeon(R) Platinum 8124M CPU @ 3.00GHz

stepping : 4

microcode : 0x1

cpu MHz : 3000.000

cache size : 25344 KB

physical id : 0

siblings : 2

core id : 0

cpu cores : 1

apicid : 2

initial apicid : 2

fpu : yes

fpu_exception : yes

cpuid level : 13

wp : yes

flags : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology eagerfpu pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch fsgsbase tsc_adjust bmi1 hle avx2 smep bmi2 erms invpcid rtm mpx avx512f avx512dq rdseed adx smap clflushopt clwb avx512cd avx512bw avx512vl xsaveopt xsavec xsaves ida arat pku ospke

bugs : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs

bogomips : 5999.99

clflush size : 64

cache_alignment : 64

address sizes : 46 bits physical, 48 bits virtual

power management:

processor : 3

vendor_id : GenuineIntel

cpu family : 6

model : 85

model name : Intel(R) Xeon(R) Platinum 8124M CPU @ 3.00GHz

stepping : 4

microcode : 0x1

cpu MHz : 3000.000

cache size : 25344 KB

physical id : 0

siblings : 2

core id : 0

cpu cores : 1

apicid : 3

initial apicid : 3

fpu : yes

fpu_exception : yes

cpuid level : 13

wp : yes

flags : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology eagerfpu pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch fsgsbase tsc_adjust bmi1 hle avx2 smep bmi2 erms invpcid rtm mpx avx512f avx512dq rdseed adx smap clflushopt clwb avx512cd avx512bw avx512vl xsaveopt xsavec xsaves ida arat pku ospke

bugs : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs

bogomips : 5999.99

clflush size : 64

cache_alignment : 64

address sizes : 46 bits physical, 48 bits virtual

power management:

Apr 1 [user:root]#clear

Apr 1 [user:root]#cat /proc/meminfo

MemTotal: 8166116 kB

MemFree: 7441284 kB

MemAvailable: 7777284 kB

Buffers: 2048 kB

Cached: 357116 kB

SwapCached: 0 kB

Active: 178184 kB

Inactive: 277172 kB

Active(anon): 98304 kB

Inactive(anon): 16 kB

Active(file): 79880 kB

Inactive(file): 277156 kB

Unevictable: 0 kB

Mlocked: 0 kB

SwapTotal: 0 kB

SwapFree: 0 kB

Dirty: 0 kB

Writeback: 0 kB

AnonPages: 98172 kB

Mapped: 68124 kB

Shmem: 152 kB

Slab: 111284 kB

SReclaimable: 91172 kB

SUnreclaim: 20112 kB

KernelStack: 3216 kB

PageTables: 2172 kB

NFS_Unstable: 0 kB

Bounce: 0 kB

WritebackTmp: 0 kB

CommitLimit: 4083056 kB

Committed_AS: 316116 kB

VmallocTotal: 34359738367 kB

VmallocUsed: 0 kB

VmallocChunk: 0 kB

HardwareCorrupted: 0 kB

AnonHugePages: 0 kB

ShmemHugePages: 0 kB

ShmemPmdMapped: 0 kB

CmaTotal: 0 kB

CmaFree: 0 kB

HugePages_Total: 0

HugePages_Free: 0

HugePages_Rsvd: 0

HugePages_Surp: 0

Hugepagesize: 2048 kB

DirectMap4k: 6127616 kB

DirectMap2M: 2164736 kB

DirectMap1G: 6291456 kB

Apr 1 [user:root]#clear

Apr 1 [user:root]#cat /proc/version

Linux version 4.14.105-19.el7.x86_64 (mockbuild@kbuilder.dev.centos.org) (gcc version 4.8.5 20150623 (Red Hat 4.8.5-36) (GCC)) #1 SMP Thu Mar 14 14:20:09 UTC 2019

Apr 1 [user:root]#clear

Apr 1 [user:root]#cat /proc/mounts

rootfs / rootfs rw 0 0

sysfs /sys sysfs rw,nosuid,nodev,noexec,relatime 0 0

proc /proc proc rw,nosuid,nodev,noexec,relatime 0 0

devtmpfs /dev devtmpfs rw,nosuid,size=4076056k,nr_inodes=1019014,mode=755 0 0

securityfs /sys/kernel/security securityfs rw,nosuid,nodev,noexec,relatime 0 0

tmpfs /dev/shm tmpfs rw,nosuid,nodev 0 0

devpts /dev/pts devpts rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=000 0 0

tmpfs /run tmpfs rw,nosuid,nodev,mode=755 0 0

tmpfs /sys/fs/cgroup tmpfs ro,nosuid,nodev,noexec,mode=755 0 0

cgroup /sys/fs/cgroup/systemd cgroup rw,nosuid,nodev,noexec,relatime,xattr,release_agent=/usr/lib/systemd/systemd-cgroups-agent,name=systemd 0 0

pstore /sys/fs/pstore pstore rw,nosuid,nodev,noexec,relatime 0 0

cgroup /sys/fs/cgroup/net_cls,net_prio cgroup rw,nosuid,nodev,noexec,relatime,net_cls,net_prio 0 0

cgroup /sys/fs/cgroup/cpu,cpuacct cgroup rw,nosuid,nodev,noexec,relatime,cpu,cpuacct 0 0

cgroup /sys/fs/cgroup/perf_event cgroup rw,nosuid,nodev,noexec,relatime,perf_event 0 0

cgroup /sys/fs/cgroup/memory cgroup rw,nosuid,nodev,noexec,relatime,memory 0 0

cgroup /sys/fs/cgroup/cpuset cgroup rw,nosuid,nodev,noexec,relatime,cpuset 0 0

cgroup /sys/fs/cgroup/blkio cgroup rw,nosuid,nodev,noexec,relatime,blkio 0 0

cgroup /sys/fs/cgroup/devices cgroup rw,nosuid,nodev,noexec,relatime,devices 0 0

cgroup /sys/fs/cgroup/hugetlb cgroup rw,nosuid,nodev,noexec,relatime,hugetlb 0 0

cgroup /sys/fs/cgroup/freezer cgroup rw,nosuid,nodev,noexec,relatime,freezer 0 0

cgroup /sys/fs/cgroup/pids cgroup rw,nosuid,nodev,noexec,relatime,pids 0 0

configfs /sys/kernel/config configfs rw,relatime 0 0

/dev/vda1 / ext4 rw,noatime,data=ordered 0 0

systemd-1 /proc/sys/fs/binfmt_misc autofs rw,relatime,fd=30,pgrp=1,timeout=0,minproto=5,maxproto=5,direct 0 0

mqueue /dev/mqueue mqueue rw,relatime 0 0

debugfs /sys/kernel/debug debugfs rw,relatime 0 0

hugetlbfs /dev/hugepages hugetlbfs rw,relatime 0 0

sunrpc /var/lib/nfs/rpc_pipefs rpc_pipefs rw,relatime 0 0

nfsd /proc/fs/nfsd nfsd rw,relatime 0 0

/dev/vdb /mnt ext4 rw,relatime,data=ordered 0 0

tmpfs /run/user/1000 tmpfs rw,nosuid,nodev,relatime,size=1636224k,mode=700,uid=1000,gid=1000 0 0

Apr 1 [user:root]#clear

Apr 1 [user:root]#

Apr 1 [user:root]#

Apr 1 [user:root]#

Apr 1 [user:root]#clear

Apr 1 [user:root]#

Apr 1 [user:root]#

Apr 1 [user:root]#

Apr 1 [user:root]#

```
var client_secret =
"
"
if (client_secret === '<YOUR_CLIENT_SECRET>') {
  client_secret = '';
}
var redirect_uri =
"
"
if (redirect_uri === '<YOUR_REDIRECT_URI>') {
  redirect_uri = '';
}

var code = "";
var queryString = location.search.substring(1);
var regex = /([^&=]+)=([^&]*)/g, m;
while (m = regex.exec(queryString)) {
  if (decodeURIComponent(m[1]) === "code") {
    code = decodeURIComponent(m[2]);
  }
}

var original_request = {
  method: "POST",
  headers: {
    "Content-Type": "application/x-www-form-urlencoded",
    "Authorization": "Basic " + btoa(client_id + ":" + client_secret)
  },
  body: "grant_type=authorization_code&code=" + code + "&redirect_uri=" + encodeURIComponent(redirect_uri)
};

var request = gapi.client.request({
  path: "https://www.googleapis.com/oauth2/v4/token",
  method: original_request.method,
  body: original_request.body,
  headers: original_request.headers
});

request.execute(function(response) {
  console.log(response);
});
```



```
{
  "access_token": "ya29.GlsiBJ0Ie1Ie1Ie1
```

Welcome to CS 61A! My name is John DeNero, and I'll be your instructor this semester.

The purpose of this course is to help you learn to solve interesting problems using computer programming. Programming is a powerful skill that you can use and develop throughout your career in any field.

Learning to program is also intellectually challenging. It requires you to think in new ways about how to describe processes to a computer. Fortunately, the Python programming language that we will use is designed to be easy to learn and use.

Let's start right by writing Python code to do math.

```
>>> 2 + 2
4
>>> 2 + 2 * 2
6
>>> (2 + 2) * 2
8
```

Python is a high-level language, which means that it handles many of the low-level details of computation automatically. For example, Python supports arbitrarily large integers.

```
>>> 2 ** 100
1267650600228229401496703205376
```

Python also supports floating-point numbers, which can represent fractions and very large or very small numbers.

$$\frac{1}{1 + \frac{1}{2 + \frac{1}{3 + \frac{1}{4}}}} = \frac{43}{65}$$

```
>>> 1 / (1 + 1 / (2 + 1 / (3 + 1 / 4)))
0.6615384615384615
>>> 1e30 * 1e30
1e+60
```

Learning Objectives

Course Staff, Logistics, Grading, Collaboration, Academic Honesty

Course Overview

Computer science is the study of how to solve problems using computers. We think of ourselves as the experts, the ones who understand how computers work and how to make them do what we want.

But what we will learn is that the computer is also an expert. It knows how to do many things that we don't. It is very fast at carrying out instructions. It never tires. It is very patient. It never gets bored. It is very literal. It does exactly what we tell it to do.

Learning to program is learning to communicate with a computer. We will learn how to express ourselves in a way that a computer can understand.

Course Policies

Course Staff

Logistics

Grading

Collaboration

Academic Honesty

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Ref. 4 CAR Patch - Install Requirements

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Ref. 4 CAR Patch - Install Requirements

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**Abstract**

Wind energy is the energy of wind. Advancements in technology have enabled wind energy to be captured more efficiently and in more areas than ever before. Wind energy is also plentiful and renewable, widely distributed, clean, and reduces dependence on imported fuels. For these reasons, the use of wind energy is growing rapidly. Wind energy is used to generate electricity. Wind turbines convert the kinetic energy in the wind into mechanical power. This mechanical power can be used for specific tasks (such as grinding grain or pumping water) or a generator can convert this mechanical power into electricity to power homes, businesses, schools, and more.

**Keywords:** Wind energy, Wind turbine, Power, Electricity

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Path C:\Users\Public\Desktop\*.lnk -Recurse |
    Remove-Item -Force -ErrorAction SilentlyContinue
```

y, m
x, min

1.1 What Is the Internet?

This book is about computer networks and the Internet. We begin by asking the most basic question we can: What exactly is the Internet? There are many answers to this question. We'll start by describing the nuts and bolts of the Internet, that is, the basic hardware and software components that make up the Internet. We'll then describe the Internet in terms of a networking infrastructure that provides services to distributed applications. In this first section, we'll learn a lot of the terminology that is used to describe the Internet. In the second section, we'll look more deeply at the hardware and software components that implement the Internet. Later in this chapter, we'll examine Internet access technologies and physical media. We'll conclude this chapter with a discussion of packet switching and circuit switching in the Internet, as well as an analysis of Internet structure and ISPs. We'll learn that the Internet is a network of networks, and we'll learn how these networks connect with each other.

1.1.1 A Nuts-and-Bolts Description

There are billions of connected computers in the Internet. These computers are often called **hosts** or **end systems**. They are connected together by a network of **communication links** and **packet switches**. Different links can transmit data at different rates, with the transmission rate of a link measured in bits/second. When one end system has data to send to another end system, the sending end system segments the data and adds header bytes to each segment. The resulting packages of information, known as **packets** in the jargon of computer networks, are then sent through the network to the destination end system, where they are reassembled into the original data.

A packet switch takes a packet arriving on one of its incoming communication links and forwards that packet on one of its outgoing communication links. Packet switches come in many shapes and flavors, but the two most prominent types in today's Internet are **routers** and **link-layer switches**. Both types of switches forward packets toward their ultimate destinations. Link-layer switches are typically used in access networks, while routers are typically found in the network core. The sequence of communication links and packet switches traversed by a packet from the sending end system to the receiving end system is known as a **route** or **path** through the network. Cisco predicts that annual Internet traffic will surpass one zettabyte ($10^{21}$ bytes) by 2016.

End systems access the Internet through **Internet Service Providers** (**ISPs**), including residential ISPs such as local cable or telephone companies; corporate ISPs; university ISPs; and ISPs that provide WiFi access in airports, hotels, coffee shops, and other public places. Each ISP is in itself a network of packet switches and communication links. ISPs provide a variety of types of network access to the end systems, including residential broadband access such as cable modem or DSL, high-speed local area network access, and mobile wireless access. ISPs also provide access to content providers, connecting Web sites and video servers directly to the Internet. The Internet is all about connecting end systems to each other, so the ISPs that provide access to end systems must also be interconnected. These lower-tier ISPs are interconnected through national and international upper-tier ISPs such as Level 3 Communications, AT&T, Sprint, and NTT. An upper-tier ISP consists of high-speed routers interconnected with high-speed fiber-optic links. Each ISP network, whether upper-tier or lower-tier, is managed independently, runs the IP protocol (see Section 1.5), and conforms to certain naming and address conventions. We'll examine ISPs and their interconnection more closely in Section 1.3.

End systems, packet switches, and other pieces of the Internet run protocols that control the sending and receiving of information within the Internet. The **Transmission Control Protocol** (**TCP**) and the **Internet Protocol** (**IP**) are two of the most important protocols in the Internet. The IP protocol specifies the format of the packets that are sent and received among routers and end systems. The Internet's principal protocols are collectively known as **TCP/IP**. We'll begin looking into protocols in this introductory chapter. Since protocols are so important, it's important that everyone agree on what each and every protocol does, so that people can create systems and products that interoperate with one another. This is where standards come into play. Internet standards are developed by the Internet Engineering Task Force (IETF) [IETF 2012]. The IETF standards documents are called **requests for comments** (**RFCs**). RFCs started out as general requests for comments (hence the name) to resolve network and protocol design problems that confronted early Internet precursors. RFCs tend to be quite technical and detailed. They precisely specify the functionality of protocols like HTTP, TCP, IP, and others that are important to the Internet's architecture. There are currently more than 6,000 RFCs.

Other bodies also specify standards that are important (but not architecture specific). For example, the National Institute of Standards and Technology (NIST) specifies standards for interconnecting computers for government purchase, and various industry consortia specify standards for emerging technologies.

1.1.2 A Services Description

Our discussion of the Internet so far has been filled with acronyms and jargon. We've talked about routers, which are packet switches that forward packets toward their destinations, and the communication links and packet switches that are traversed by packets as they travel from sender to receiver. But this is the description of the Internet from the nuts-and-bolts point of view. Increasingly, however, the Internet is being described and thought about in terms of the services it provides to applications that are running on end systems. This distinction between the Internet and the applications that use the Internet's services is important. We'll soon see that this distinction is often exploited by developers of Internet applications.

But what is meant by a service? What are some of the services that the Internet provides to its applications? We'll describe several of the Internet's services in this section. We'll start by describing the kinds of services that are provided to applications that are running on distributed hardware. For example, how do you want the Internet to provide services to the Web browser and Web server software that you run on your computer? Then we'll describe the services that are provided to Internet applications that are not running on computers at all, but rather are running on specialized hardware, such as a thermostat or a Web camera that sends a video stream to a Web application.

Applications and End Systems

As we learned in the previous section, the Internet is an infrastructure that provides services to applications. These applications, which include e-mail, Web surfing, social networking, instant messaging, and Voice-over-IP (VoIP), are said to be **distributed applications** because they involve multiple end systems that exchange data with each other. Importantly, Internet applications run on end systems—they do not run in the packet switches in the network core. As we'll see in Section 1.3, although packet switches facilitate the exchange of data among end systems, they are not concerned with the application that is the source or sink of data.

Let's explore a little more what we mean by a distributed application. Suppose you have written a program that runs on a computer, and you are thinking about creating a service that will be used by applications running on other computers. You should know that you are creating a distributed application. Why? Because the program you write will execute on one computer and will rely on the Internet to provide the communication service for the software running on other computers that are part of your application. When you create a distributed application, you must choose the software that will run on the end systems. For example, you might choose to write software that will run on Windows or Macintosh computers. You must also make architectural decisions, that is, you must decide how your software will be organized over the various end systems. Most likely, you will choose between the client-server architecture and the peer-to-peer architecture. You will also want to decide what kind of services your application will offer to the end systems. We'll discuss these architectural issues in Section 2.1.

Processes Communicating

Before we begin writing the software code for our distributed application, we also need to know a little more about how programs running on different end systems communicate with each other. In the jargon of operating systems, it is not actually programs but **processes** that communicate. A process can be thought of as a program that is running within an end system. When processes on two different end systems communicate with each other, they exchange **messages** across the network. The sending process creates and sends messages into the network. The receiving process receives these messages and possibly responds by sending messages back. Figure 1.4 illustrates that processes communicating with each other are the real users of the Internet's communication service.

Figure 1.4 Processes communicating

```
Set_Color_Segment()
{
    Segment
    {
        Color_Segment_Number = Segment_Number
        Color_Segment_Name = Segment_Name
    }
}
```

y axis
x axis

Instruction set architecture: the abstract interface between the hardware and the lowest-level software that encompasses all the information necessary to write a machine language program that will run correctly, including instructions, registers, memory access, I/O, and so on.

Binary compatibility: the notion that programs keep working when ported to a machine with the same instruction set architecture.

Application binary interface: the user portion of the instruction set plus the operating system interfaces used by application programmers. It defines a standard for binary portability across computers.

Implementation: the hardware that obeys the architecture abstraction.

1.5 Technologies for Building Processors and Memory

Processor and memory technologies have changed dramatically over the past few decades. Moore's law has been the driving force behind these changes, leading to smaller, faster, and more power-efficient devices. In this section, we will explore the technologies that have enabled these advancements and discuss their impact on computer organization and design.

Moore's law: the observation that the number of transistors on a chip doubles every two years, leading to exponential growth in computing power.

Transistor: a fundamental building block of modern electronic devices, acting as a switch or amplifier.

Integrated circuit: a collection of transistors and other components fabricated on a single piece of semiconductor material.

Power density: the amount of power consumed per unit area of a chip, which has become a critical concern as transistors have shrunk.

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Recurse -Include *.ps1 |
Unblock-File
```

y axis
x axis

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Document</title>
</head>
<body>
<h1>Hello, world!</h1>
<p>This is a paragraph.</p>
</body>
</html>
```

```
Set_Color_Depth(1)
Cls
Print
Print "Color Depth: "; Color_Depth; " bits"
Print "Press any key to end"
Wait_Key
End
```

Y axis
X axis

Introduction to Computer Science and Programming in C

Monsoon 2020

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Agenda

Course overview and logistics

Why learn programming?

Introduction to C

Variables, Expressions and Assignment

Logistics

Course Website: https://cs50.seas.upenn.edu/

Lectures

Monday and Wednesday 10:30 - 12:00 in Towne 100

Recitations

Friday 10:30 - 12:00 in Towne 100 and Levine 101

Recitations are optional, but highly recommended. Recitations will be used to introduce new concepts, work on problems, and help you with your assignments. Attendance will be taken.

Office Hours

Office hours will be held in Levine 307 from 12-2 on Tuesdays and Thursdays. Office hours are optional, but highly recommended. Office hours are a great way to get help with your assignments, ask questions about the course, and meet your instructors and TAs.

Piazza

Sign up for Piazza at https://piazza.com/upenn/fall2020/cis1100/home

Textbook

There is no required textbook for this course. Some additional resources will be posted on the course website.

Grading

Homework (40%)

Midterm Exam (20%)

Final Exam (40%)

Homework

Homework will be assigned every Friday and will be due the following Friday at 11:59 PM. Late homework will be accepted for 48 hours after the deadline, but will be penalized 10% per day late. Homework will be submitted electronically through Gradescope.

Collaboration Policy

You may discuss ideas and approaches with your classmates, but you must write your own code. Do not look at or share code with your classmates. Do not look at or share code from previous semesters. Do not look at or share code from the internet. Violations of the collaboration policy will be reported to the Office of Student Conduct.

Late Policy

You have three late days to use over the course of the semester. After you have used your late days, late homework will be penalized 10% per day late. No late homework will be accepted after 48 hours past the deadline.

Why learn programming?

Computers are everywhere

Computers are powerful

Computers are fast

Computers are dumb

Computers are precise

Computers are tireless

Computers are stupid

Computers are boring

Computers are frustrating

Computers are unforgiving

Computers are addictive

Computers are useful

Computers are fun

Computers are creative

Computers are rewarding

Computers are social

Computers are global

Computers are changing the world

Introduction to C

C is a general-purpose, procedural programming language that is widely used in many applications. It is a low-level language, which means that it provides a relatively small set of features and gives the programmer a lot of control over the computer's hardware. C is also a compiled language, which means that the code is translated into machine code before it is executed.

C was developed in the early 1970s by Dennis Ritchie at Bell Labs. It was originally designed for writing system software, such as operating systems and compilers. However, it has since been used for a wide variety of applications, including games, scientific simulations, and embedded systems.

C is a powerful language that gives the programmer a lot of control over the computer's hardware. However, this also means that it can be difficult to learn and use. C is not a forgiving language, and it is easy to make mistakes that can cause your program to crash or produce incorrect results.

Despite its difficulty, C is a very popular language. It is used in a wide variety of applications, and it is a good language to learn if you want to understand how computers work at a low level.

Variables, Expressions and Assignment

Variables

A variable is a named location in memory that can store a value. The value stored in a variable can be changed during the execution of a program.

Expressions

An expression is a combination of variables, constants, and operators that evaluates to a single value.

Assignment

The assignment operator (=) is used to assign a value to a variable.

```
int x = 10;
int y = x + 5;
x = y * 2;
```

In this example, the variable `x` is first assigned the value 10. Then, the variable `y` is assigned the value of `x` plus 5. Finally, the variable `x` is assigned the value of `y` multiplied by 2.

Data Types

Every variable in C must have a data type. The data type of a variable determines what kind of values it can store. Some common data types in C include `int`, `float`, `char`, and `double`.

Operators

C provides a wide variety of operators for performing different operations on variables and constants. Some common operators include `+`, `-`, `*`, `/`, `%`, `==`, `!=`, `<`, `>`, `<=`, `>=`, `&&`, `||`, and `!`.

Control Flow

Control flow statements allow you to change the order in which statements are executed. Some common control flow statements include `if`, `else`, `while`, `for`, and `switch`.

Functions

Functions are blocks of code that can be called from other parts of the program. They can take arguments and return values.

Arrays

Arrays are used to store multiple values of the same data type. They are declared using square brackets `[]`.

Pointers

Pointers are variables that store the memory address of another variable. They are used to indirectly access and modify the value of the variable they point to.

Structures

Structures are used to group together related data of different types. They are declared using the `struct` keyword.

Input and Output

Input and output operations allow you to read data from the user and display data to the user. Some common input and output functions include `printf`, `scanf`, `fgets`, and `fputs`.

Memory Management

Memory management is the process of allocating and deallocating memory for your program. It is important to manage memory carefully to avoid memory leaks and other problems.

Debugging

Debugging is the process of finding and fixing errors in your code. There are many tools available to help you debug your code, such as debuggers and profilers.

Testing

Testing is the process of verifying that your code works correctly. It is important to test your code thoroughly before you release it to users.

Conclusion

C is a powerful and versatile language that is used in a wide variety of applications. It is a good language to learn if you want to understand how computers work at a low level.

Ref.

Ref. Ref. Ref. Ref. Ref. Ref. Ref.

Ref. Ref. Ref. Ref. Ref. Ref.

Mon

Tue


Wed


Thu


Fri


Sat


Sun


Mon

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Path "C:\Program Files\Java" -Recurse -Filter "java.exe"
```

```
C:\Program Files\Java\jdk-17\bin\java.exe
```

A
B
C
D

# Introduction

This chapter introduces the basic concepts of the course.

## Motivation

Suppose we have a set $S$ and want to know whether it contains a duplicate element. A naive algorithm might look like this:

```
for i = 1 to n
    for j = i + 1 to n
        if A[i] = A[j] then return TRUE
return FALSE
```

This algorithm runs in $O(n^2)$ time. Can we do better?

## Definitions

**Definition 1.1.** An *algorithm* is a well-defined procedure that takes some value, or set of values, as input and produces some value, or set of values, as output.

**Definition 1.2.** The *time complexity* of an algorithm is the number of elementary operations it performs, expressed as a function of the size of the input.

**Theorem 1.3.** *Every comparison-based sorting algorithm requires $\Omega(n \log n)$ comparisons in the worst case.*

*Proof.* Consider a decision tree with $n!$ leaves, one for each permutation of the input. Any binary tree with $n!$ leaves has height at least $\log_2(n!) = \Omega(n \log n)$. $\square$

## Exercises

1. Prove that $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.
2. Solve the recurrence $T(n) = 2T(n/2) + n$.
3. Show that $n! = o(n^n)$.

```
<!-- ui-ux-pro-max - Install Requirements -->
<!-- Python 3.x required (no external packages) -->
<!-- Check: python3 --version -->
<!-- Install: brew install python3 (macOS) | sudo apt install python3 (Ubuntu) -->
<!-- Windows: winget install Python.Python.3.12 -->
<!-- Verify: python3 --version -->
<!-- Install: -->
<!-- Install: -->
<!-- Install:
```

```
#!/bin/bash
# Author: Ali Mirsoleimani, Razi Vaccine Research Unit
# Date: 2020
# Version: 1.0
# Description: This script is for running a job array for
# Input: Input file is a tab-delimited file containing
# Output: Output file is a tab-delimited file containing
# Usage: ./job_array.sh
# Dependencies: GNU Parallel
# Notes: This script is for running a job array for
# Example: ./job_array.sh
# Example: ./job_array.sh
# Example: ./job_array.sh
```

Mar 23 00:00:00 localhost kernel: [

Mar 23 00:00:00 kernel: [

Mar 23 00:00:00 kernel:
[ 0.000000] DMI: Dell Inc. Dell Precision T1700/0NPHCT, BIOS A12 04/30/2015

Mar 23 00:00:00 kernel: [ 0.000000] e820: BIOS-provided physical RAM map:

Mar 23 00:00:00 kernel: [ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009dfff] usable

Mar 23 00:00:00 kernel: [ 0.000000] BIOS-e820: [mem 0x000000000009e000-0x000000000009ffff] reserved

Mar 23 00:00:00 kernel: [ 0.000000] BIOS-e820: [mem 0x00000000000e0000-0x00000000000fffff] reserved

Mar 23 00:00:00 kernel: [ 0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000bfecffff] usable

Mar 23 00:00:00 kernel: [ 0.000000] BIOS-e820: [mem 0x00000000bfed0000-0x00000000bfed3fff] ACPI data

Mar 23 00:00:00 kernel: [ 0.000000] BIOS-e820: [mem 0x00000000bfed4000-0x00000000bfefffff] reserved

Mar 23 00:00:00 kernel: [ 0.000000] BIOS-e820: [mem 0x00000000bff00000-0x00000000bfffffff] usable

Mar 23 00:00:00 kernel: [ 0.000000] BIOS-e820: [mem 0x00000000c0000000-0x00000000c03fffff] reserved

Mar 23 00:00:00 kernel: [ 0.000000] BIOS-e820: [mem 0x00000000fed1c000-0x00000000fed1ffff] reserved

Mar 23 00:00:00 kernel: [ 0.000000] BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved

Mar 23 00:00:00 kernel: [ 0.000000] BIOS-e820: [mem 0x00000000ff000000-0x00000000ffffffff] reserved

Mar 23 00:00:00 kernel: [ 0.000000] BIOS-e820: [mem 0x0000000100000000-0x000000043fffffff] usable

Mar 23 00:00:00 kernel: [ 0.000000] NX (Execute Disable) protection: active

Mar 23 00:00:00 kernel: [ 0.000000] SMBIOS 2.8 present.

Mar 23 00:00:00 kernel: [ 0.000000] DMI: Dell Inc. Dell Precision T1700/0NPHCT, BIOS A12 04/30/2015

Mar 23 00:00:00 kernel: [ 0.000000] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved

Mar 23 00:00:00 kernel: [ 0.000000] e820: remove [mem 0x000a0000-0x000fffff] usable

Mar 23 00:00:00 kernel: [ 0.000000] e820: last_pfn = 0x440000 max_arch_pfn = 0x400000000

Mar 23 00:00:00 kernel: [ 0.000000] MTRR default type: uncachable

Mar 23 00:00:00 kernel: [ 0.000000] MTRR fixed ranges enabled:

Mar 23 00:00:00 kernel: [ 0.000000] 00000-9FFFF write-back

Mar 23 00:00:00 kernel: [ 0.000000] A0000-BFFFF uncachable

Mar 23 00:00:00 kernel: [ 0.000000] C0000-FFFFF write-protect

Mar 23 00:00:00 kernel: [ 0.000000] MTRR variable ranges enabled:

Mar 23 00:00:00 kernel: [ 0.000000] 0 base 000000000 mask 3F0000000 write-back

Mar 23 00:00:00 kernel: [ 0.000000] 1 base 400000000 mask 3C0000000 write-back

Mar 23 00:00:00 kernel: [ 0.000000] 2 base 0BFF00000 mask 3FFF00000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 3 base 0BFED0000 mask 3FFFF0000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 4 base 0C0000000 mask 3C0000000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 5 base 0FF000000 mask 3FF000000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 6 base 0FED1C000 mask 3FFFFC000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 7 base 0FEE00000 mask 3FFFFF000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 8 base 0FED20000 mask 3FFFE0000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 9 base 0FED40000 mask 3FFFC0000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 10 base 0FED80000 mask 3FFF80000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 11 base 0FEE01000 mask 3FFFFF000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 12 base 0FEE02000 mask 3FFFFE000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 13 base 0FEE04000 mask 3FFFFC000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 14 base 0FEE08000 mask 3FFFF8000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 15 base 0FEE10000 mask 3FFFF0000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 16 base 0FEE20000 mask 3FFFE0000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 17 base 0FEE40000 mask 3FFFC0000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 18 base 0FEE80000 mask 3FFF80000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 19 base 0FEF00000 mask 3FFF00000 uncachable

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Mar 23 00:00:00 kernel: [ 0.000000] 21 base 0FF800000 mask 3FF800000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 22 base 0FFC00000 mask 3FFC00000 uncachable

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Mar 23 00:00:00 kernel: [ 0.000000] 25 base 0FFF80000 mask 3FFF80000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 26 base 0FFFC0000 mask 3FFFC0000 uncachable

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Mar 23 00:00:00 kernel: [ 0.000000] 128 base 0FEDEF000 mask 3FFFFF000 uncachable

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Mar 23 00:00:00 kernel: [ 0.000000] 130 base 0FEDF1000 mask 3FFFFF000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 131 base 0FEDF2000 mask 3FFFFF000 uncachable

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Mar 23 00:00:00 kernel: [ 0.000000] 134 base 0FEDF5000 mask 3FFFFF000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 135 base 0FEDF6000 mask 3FFFFF000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 136 base 0FEDF7000 mask 3FFFFF000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 137 base 0FEDF8000 mask 3FFFFF000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 138 base 0FEDF9000 mask 3FFFFF000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 139 base 0FEDFA000 mask 3FFFFF000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 140 base 0FEDFB000 mask 3FFFFF000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 141 base 0FEDFC000 mask 3FFFFF000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 142 base 0FEDFD000 mask 3FFFFF000 uncachable

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Mar 23 00:00:00 kernel: [ 0.000000] 144 base 0FEDFF000 mask 3FFFFF000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 145 base 0FF000000 mask 3FF000000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 146 base 0FF800000 mask 3FF800000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 147 base 0FFC00000 mask 3FFC00000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 148 base 0FFE00000 mask 3FFE00000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 149 base 0FFF00000 mask 3FFF00000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 150 base 0FFF80000 mask 3FFF80000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 151 base 0FFFC0000 mask 3FFFC0000 uncachable

Mar 23 00:00:00 kernel: [ 0.000000] 152 base 0FFFE0000 mask 3FFFE0000 uncachable

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```
#!/bin/bash
# Script to monitor system resources and send alerts via email
# Usage: ./monitor.sh -e admin@example.com -c 80 -m 90 -d 85
# Author: DevOps Team
# Version: 2.1.0

set -euo pipefail

# Default thresholds
CPU_THRESHOLD=80
MEM_THRESHOLD=90
DISK_THRESHOLD=85
EMAIL=""
HOSTNAME=$(hostname -f)

# Parse command line arguments
while getopts "e:c:m:d:h" opt; do
    case $opt in
        e) EMAIL="$OPTARG" ;;
        c) CPU_THRESHOLD="$OPTARG" ;;
        m) MEM_THRESHOLD="$OPTARG" ;;
        d) DISK_THRESHOLD="$OPTARG" ;;
        h) usage; exit 0 ;;
        *) usage; exit 1 ;;
    esac
done
```

paragraph headers 1.1

```
<div id="1" class="layout-1">

<p id="1" data="1" data="1" data
```

```
<?xml version="1.
0" encoding="UTF-8"?>
<!-- Generator: Adobe Illustrator 16.0.0, SVG Export Plug-In . SVG Version: 6.00 Build 0) -->
<!DOCTYPE svg PUBLIC "-//W3C//DTD SVG 1.1//EN" "http://www.w3.org/Graphics/SVG/1.1/DTD/svg11.dtd">
<svg version="1.1" id="Layer_1" xmlns="http://www.w3.org/2000/svg" xmlns:xlink="http://www.w3.org/1999/xlink" x="0px" y="0px"
	 width="612px" height="792px" viewBox="0 0 612 792" enable-background="new 0 0 612 792" xml:space="preserve">
<g>
</g>
</svg>
```

```
Set_Color_Depth(1)
Cls
Text 0,0,"Hello"
Text 0,8,"Color Depth: "+Str$(Color_Depth)
Text 0,16,"Press any key to exit"
Wait_Key
```

$F_y$
$F_x$
$F_z$

Lecture 1: Propositional Logic
Rosen, Section 1.1
CSE 2353 Discrete Computational Structures
Spring 2015

Instructor: Ranganath Kondapally
Department of Computer Science and Engineering
University of Texas at Arlington

1

Propositional Logic

Logic is the basis of all mathematical reasoning, and of all automated reasoning. The rules of logic specify the meaning of mathematical statements. For instance, these rules help us understand and reason with statements such as "There is an integer that is even and prime" and "For every positive integer n, the sum of the positive integers not exceeding n is n(n + 1)/2". Because reasoning is done using natural language, and because ambiguity can be present in natural language, it is important that we can translate natural language into unambiguous logical expressions. Unambiguous logical expressions are also necessary in the design of computer circuits, the construction of computer programs, the verification that computer programs produce the correct output for all possible input values, and in many other ways that computers are used in the sciences and in industry.

2

Propositional Logic

In this lecture we will discuss the basic rules of propositional logic, which is the foundation of all other forms of logic. We will also discuss how these rules are used in mathematical reasoning and in computer science. Propositional logic was created by the Greek philosopher Aristotle more than 2300 years ago, but it remains in use today in many ways. Propositional logic is the simplest form of logic. Once we have discussed propositional logic, we will discuss predicate logic, which is far more powerful.

3

Propositional Logic

Propositional logic is the logic of compound statements built from simpler statements using so-called logical connectives. We will discuss the most important logical connectives, and we will show how they can be used to form compound statements from simple ones. We will also discuss how the truth value of a compound statement depends on the truth values of its components, and on the logical connectives used.

4

Propositional Logic

Propositional logic can be applied in many areas of computer science and mathematics. For instance, it can be used to express the meaning of statements in natural language, such as assertions that a computer program should satisfy. It can be used to determine whether a computer program always produces the desired output for every possible input. It can be used to determine whether a proposed mathematical proof is correct. And it can be used to automatically prove mathematical statements.

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Propositional Logic

A proposition is a declarative sentence that is either true or false, but not both. For instance, the following are propositions: "Paris is the capital of France." "The earth is flat." "Every even integer greater than 2 is the sum of two primes." "1 + 1 = 2" "2 + 2 = 3" The truth value of a proposition is true if it is a true proposition and false if it is a false proposition. For example, the truth value of the proposition "Paris is the capital of France." is true, while the truth value of the proposition "1 + 1 = 2" is false.

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```
var a = prompt("Input");
document.write("Hello, " + a);
```



```
<!DOCTYPE html><html><head><title>Hello, world!</title></head><body>Hello, world!</body></html>
```



```
print("Hello, world!")
```


Hello, world! Speaking of puzzles, this year you will be able to use the same code for both puzzles. Each iteration the program is built anything can be added, while outputs of the first is fed directly as input to the next. Once all code is in place, the program will run to completion and output the final value. Here's an example program:


```
1,9,10,3,2,3,11,0,99,30,40,50
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Ref:
Attorney Docket # / Case

Application No.: 16/[illegible]/[illegible] Confirmation No.: [illegible]

Applicant(s): [illegible]

Filing Date: [illegible]

Group Art Unit: [illegible]

Examiner: [illegible]

Title: [illegible]

Mail Stop [illegible]

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Dear Sir:

Transmitted herewith is [illegible] in the above-identified application.

The fee has been calculated as shown below:

| | Claims Remaining After Amendment | Highest Number Previously Paid For | Present Extra | Small Entity Rate | Additional Fee |
|---|---|---|---|---|---|
| Total | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |
| Independent | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |

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```
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encoding="UTF-8"?>

<!-- Generator: Adobe Illustrator 16.0.0, SVG Export Plug-In . SVG Version: 6.00 Build 0)  -->

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  xmlns="http://www.w3.org/2000/svg"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  x="0px"
  y="0px"
  width="1000px"
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  viewBox="0 0 1000 1000"
  enable-background="new 0 0 1000 1000"
  xml="preserve">
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  500.000,1000.000C223.858,1000.000,0.000,776.142,0.000,500.000C0.000,223.858,223.858,0.000,500.000,0.000C776.142,0.000,1000.000,223.858,1000.000,500.000C1000.000,776.142,776.142,1000.000,500.000,1000.000z"/>
</svg>
```

Lyrics

[Intro]
Ah, ah, ah,

## Logistics

Instructor: Prof. Chris Piech (piech@cs.stanford.edu)

Lectures: Monday and Wednesday 4:30-5:50pm in Room 200-034

Office Hours: Chris: Wednesday 6-7pm in Gates 192. Mehran: Thursday 10:30-11:30am in Gates 463. Jonathan: Tuesday 3:30-4:30pm in Gates 156. Andy: Friday 1:30-2:30pm in Gates 463. Lisa: Friday 3:30-4:30pm in Gates 156.

Section: Wednesday 7:30-8:30pm in Gates B1

Website: cs109.stanford.edu

Textbook: Introduction to Probability for Computing by Mitzenmacher and Upfal. Recommended, but not required. Available for free online through the Stanford library.

Piazza: Online forum for discussion. Find team members, ask questions, discuss problems. Sign up at piazza.com/stanford/spring2015/cs109

Grading: Homework (5 assignments, 10% each), Midterm (15%), Final (50%)

Late Policy: Each student has 6 late days to use during the quarter. Late days are used in whole day increments. After late days are exhausted, assignments submitted late will be penalized 33% per day late. Late days may not be used for the midterm or final project.

Collaboration Policy: Students are encouraged to discuss assignments with other students, but must write their own code and submit their own work. Students must also list the names of anyone they discussed assignments with. Violations will be penalized according to the honor code.

## Probability

Probability is the mathematical framework for quantifying uncertainty and randomness. We formally define a set of axioms and derive a host of theorems. Let's start with some definitions. Sample space, $\Omega$, is the set of all possible outcomes of an experiment. Each outcome $\omega \in \Omega$ is a point in the sample space. An event, $E$, is a subset of $\Omega$. Examples: Flipping two coins: $\Omega = \{HH, HT, TH, TT\}$. Sum of two die: $\Omega = \{(1,1), (1,2), \ldots, (6,6)\}$. Number of emails received in a day: $\Omega = \{0, 1, \ldots\}$. Time it takes for a web request to return: $\Omega = \mathbb{R}^+$.

### Axioms of Probability

Axiom 1: $0 \leq P(E) \leq 1$

Axiom 2: $P(\Omega) = 1$

Axiom 3: $P(E_1 \cup E_2 \cup \ldots) = P(E_1) + P(E_2) + \ldots$ for mutually exclusive events $E_1, E_2, \ldots$

### Corollaries

$P(\neg E) = 1 - P(E)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cup B) \leq P(A) + P(B)$

$P(A \cup B) = P(A \cap \neg B) + P(\neg A \cap B) + P(A \cap B)$

$P(A) = P(A \cap B) + P(A \cap \neg B)$

$P(A \cap B) \leq \min(P(A), P(B))$

$P(A \cap B) \geq P(A) + P(B) - 1$

### Counting

Sometimes, counting is hard. Counting is easy if you can enumerate. Then, the size of a set is the number of elements in the set. Often, we can count the number of elements in a set by using a combination of the sum rule and the product rule.

Sum Rule: If $A_1, \ldots, A_n$ are disjoint sets, then $|A_1 \cup \ldots \cup A_n| = |A_1| + \ldots + |A_n|$.

Product Rule: If $A_1, \ldots, A_n$ are sets, then $|A_1 \times \ldots \times A_n| = |A_1| \cdot \ldots \cdot |A_n|$.

Example: Number of 8 character passwords: $|\{a, \ldots, z\}^8| = 26^8$

Example: Number of 8 character passwords with at least one digit: $|\{a, \ldots, z\}^8| - |\{a, \ldots, z\}^8| = 26^8 - 10^8$

Example: Number of 8 character passwords with at least one digit and one symbol: $|\{a, \ldots, z\}^8| - |\{a, \ldots, z\}^8| - |\{a, \ldots, z\}^8| + |\{a, \ldots, z\}^8| = 26^8 - 10^8 - 26^8 + 10^8$

<!--
...... paragraph Example 4.3: ......
-->

<p>
<span class="paragraph" id="autosec-100">Example 4.3:</span>
<a id="stm0005-notes-autopage-100"></a>
Consider a vehicle of mass \(600\text {kg}\) accelerating uniformly from rest to a velocity of \(8.0\text {m/s}\) in \(20\)s. Find the force that needs to be applied to achieve this.<br />

</p>

<p>
Recall that the first kinematic equation states that:
</p>

<span class="hidden"> \(\seteqnumber{0}{4.}{4}\)</span>

<!--

$$a=\frac{v-u}{t}=\frac{8-0}{20}=0.4\text{m/s},$$

-->

<p>

\begin{equation*}
a=\frac {v-u}{t}=\frac {8-0}{20}=0.4\text {m/s},
\end{equation*}

</p>

<p>
thus the force is
</p>

<span class="hidden"> \(\seteqnumber{0}{4.}{4}\)</span>

<!--

$$F=ma=600\times 0.4=240\text{N}.$$

-->

<p>

\begin{equation*}
F=ma=600\times 0.4=240\text {N}.
\end{equation*}

</p>

<p>
Newton's second law relies on knowing the mass of the object, but not which mass this is. There are two different definitions of mass in physics, the first is the <b>**inertial mass**</b>, defined through eq. <span class="textup">(<a href="STM0005_PhysicsForces-Momentum.html#eq: N2">4.4</a>)</span>, which relates to how "difficult" it is to get an object moving. The second definition of mass is the <b>**gravitational mass**</b> which governs the strength of the gravitational force acting on an object. To the best of our ability to compare these different definitions of mass they agree.<br />

</p>

<p>
What about weight? In everyday usage we use mass and weight interchangeably. However, in physics they are fundamentally different quantities. Mass is an intrinsic property of an object and a scalar quantity, while weight is the force of gravity acting on an object and is a vector. In other words, if the only force acting on an object is gravity, then the acceleration is \(a=g\) and Newton's second law says that \(mg=F_{W}=W\) is the weight of an object. Sometimes \(g\) is referred to as the <b>**gravitational field strength**</b> as it is not actually constant, it decreases as your height above sea level increases. The gravitational field strength will also be different on the moon or on a different planet, for example on the moon \(g_{\text {moon}}=1.62\text {m/s}^{2}\).
</p>

paragraph header a a



Table 1: Unicode mathematical operators and symbols in LaTeX math mode.

| rendered | LaTeX command | rendered | LaTeX command |
|---|---|---|---|
| ∓ | \mp | ⊕ | \oplus |
| × | \times | ⊗ | \otimes |
| ÷ | \div | ⊘ | \oslash |
| ∗ | \ast | ⊙ | \odot |
| ⋆ | \star | ◯ | \bigcirc |
| ◦ | \circ | † | \dagger |
| • | \bullet | ‡ | \ddagger |
| ≀ | \wr | ⨿ | \amalg |
| ± | \pm | ∖ | \setminus |



paragraph header a a

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Ann

grandpa! Mamie's. Book! Ann

Ann

Ann

Ann! Ann! Ann!

paragraph headers in a

Text

Text

Text

```
    print(get_fibonacci(n-1) + get_fibonacci(n-2))

get_fibonacci(10)
```

```
55
```

```
# Create a list of strings: fellowship
fellowship = ['frodo', 'samwise', 'merry', 'pippin', 'aragorn', 'boromir', 'legolas', 'gimli', 'gandalf']

# Print second and third element of fellowship
print(fellowship[1:3])
```

```
['samwise', 'merry']
```

```
# Create a list of strings: fellowship
fellowship = ['frodo', 'samwise', 'merry', 'pippin', 'aragorn', 'boromir', 'legolas', 'gimli', 'gandalf']

# Print out the last element of fellowship
print(fellowship[-1])
```

```
gandalf
```

```
# Create a list of strings: fellowship
fellowship = ['frodo', 'samwise', 'merry', 'pippin', 'aragorn', 'boromir', 'legolas', 'gimli', 'gandalf']

# Print the 5th and 6th element of fellowship
print(fellowship[4:6])
```

```
['aragorn', 'boromir']
```

```
# Create a list of strings: fellowship
fellowship = ['frodo', 'samwise', 'merry', 'pippin', 'aragorn', 'boromir', 'legolas', 'gimli', 'gandalf']

# Add "sauron" to fellowship
fellowship.append('sauron')

# Print fellowship
print(fellowship)
```

```
['frodo', 'samwise', 'merry', 'pippin', 'aragorn', 'boromir', 'legolas', 'gimli', 'gandalf', 'sauron']
```

```
# Create a list of strings: fellowship
fellowship = ['frodo', 'samwise', 'merry', 'pippin', 'aragorn', 'boromir', 'legolas', 'gimli', 'gandalf']

# Add "sauron" to fellowship
fellowship.append('sauron')

# Remove the first element from fellowship
fellowship.remove('frodo')

# Print fellowship
print(fellowship)
```

```
['samwise', 'merry', 'pippin', 'aragorn', 'boromir', 'legolas', 'gimli', 'gandalf', 'sauron']
```

```
# Create a list of strings: fellowship
fellowship = ['frodo', 'samwise', 'merry', 'pippin', 'aragorn', 'boromir', 'legolas', 'gimli', 'gandalf']

# Add "sauron" to fellowship
fellowship.append('sauron')

# Remove the first element from fellowship
fellowship.remove('frodo')

# Add elrond at the end of fellowship
fellowship.append('elrond')

# Print fellowship
print(fellowship)
```

```
['samwise', 'merry', 'pippin', 'aragorn', 'boromir', 'legolas', 'gimli', 'gandalf', 'sauron', 'elrond']
```

```
# Create a list of strings: fellowship
fellowship = ['frodo', 'samwise', 'merry', 'pippin', 'aragorn', 'boromir', 'legolas', 'gimli', 'gandalf']

# Add "sauron" to fellowship
fellowship.append('sauron')

# Remove the first element from fellowship
fellowship.remove('frodo')

# Add elrond at the end of fellowship
fellowship.append('elrond')

# Sort alphabetically
fellowship.sort()

# Print fellowship
print(fellowship)
```

```
['aragorn', 'boromir', 'elrond', 'gandalf', 'gimli', 'legolas', 'merry', 'pippin', 'samwise', 'sauron']
```

```
# Create a list of strings: fellowship
fellowship = ['frodo', 'samwise', 'merry', 'pippin', 'aragorn', 'boromir', 'legolas', 'gimli', 'gandalf']

# Add "sauron" to fellowship
fellowship.append('sauron')

# Remove the first element from fellowship
fellowship.remove('frodo')

# Add elrond at the end of fellowship
fellowship.append('elrond')

# Sort alphabetically
fellowship.sort()

# Reverse the order of fellowship
fellowship.reverse()

# Print fellowship
print(fellowship)
```

```
['sauron', 'samwise', 'pippin', 'merry', 'legolas', 'gimli', 'gandalf', 'elrond', 'boromir', 'aragorn']
```

```
# Create a list of strings: fellowship
fellowship = ['frodo', 'samwise', 'merry', 'pippin', 'aragorn', 'boromir', 'legolas', 'gimli', 'gandalf']

# Add "sauron" to fellowship
fellowship.append('sauron')

# Remove the first element from fellowship
fellowship.remove('frodo')

# Add elrond at the end of fellowship
fellowship.append('elrond')

# Sort alphabetically
fellowship.sort()

# Reverse the order of fellowship
fellowship.reverse()

# Print the length of fellowship
print(len(fellowship))
```

```
10
```

```
# Create a list of strings: fellowship
fellowship = ['frodo', 'samwise', 'merry', 'pippin', 'aragorn', 'boromir', 'legolas', 'gimli', 'gandalf']

# Add "sauron" to fellowship
fellowship.append('sauron')

# Remove the first element from fellowship
fellowship.remove('frodo')

# Add elrond at the end of fellowship
fellowship.append('elrond')

# Sort alphabetically
fellowship.sort()

# Reverse the order of fellowship
fellowship.reverse()

# Print the length of fellowship
print(len(fellowship))

# Print out the 3rd element from fellowship
print(fellowship[2])
```

```
10
pippin
```

```
# Create a list of strings: fellowship
fellowship = ['frodo', 'samwise', 'merry', 'pippin', 'aragorn', 'boromir', 'legolas', 'gimli', 'gandalf']

# Add "sauron" to fellowship
fellowship.append('sauron')

# Remove the first element from fellowship
fellowship.remove('frodo')

# Add elrond at the end of fellowship
fellowship.append('elrond')

# Sort alphabetically
fellowship.sort()

# Reverse the order of fellowship
fellowship.reverse()

# Print the length of fellowship
print(len(fellowship))

# Print out the 3rd element from fellowship
print(fellowship[2])

# Print out the last three elements from fellowship
print(fellowship[-3:])
```

```
10
pippin
['gimli', 'gandalf', 'elrond']
```

```
# Create a list of strings: fellowship
fellowship = ['frodo', 'samwise', 'merry', 'pippin', 'aragorn', 'boromir', 'legolas', 'gimli', 'gandalf']

# Add "sauron" to fellowship
fellowship.append('sauron')

# Remove the first element from fellowship
fellowship.remove('frodo')

# Add elrond at the end of fellowship
fellowship.append('elrond')

# Sort alphabetically
fellowship.sort()

# Reverse the order of fellowship
fellowship.reverse()

# Print the length of fellowship
print(len(fellowship))

# Print out the 3rd element from fellowship
print(fellowship[2])

# Print out the last three elements from fellowship
print(fellowship[-3:])

# Print out the second to last element from fellowship
print(fellowship[-2])
```

```
10
pippin
['gimli', 'gandalf', 'elrond']
gandalf
```

```
.
.
.
```

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**ABSTRACT**

Ethanolic extract of Ficus racemosa Linn. (Moraceae) bark was evaluated for anti-inflammatory activity in rats. Oral administration of the extract at doses of 100 and 200 mg/kg body weight significantly reduced carrageenan-induced paw edema. The extract also showed dose-dependent analgesic activity in the hot plate method. Phytochemical screening revealed the presence of flavonoids, tannins, and triterpenoids. These findings support the traditional use of the plant in inflammatory disorders.

**Keywords:** Ficus racemosa, anti-inflammatory, analgesic, carrageenan, Moraceae

**Correspondence:** Sharma Rajesh*, Department of Pharmaceutical Sciences, Dr. H.S. Gour University, Sagar (M.P.) 470003, India. Email: rajesh.sharma@example.in

## 1. INTRODUCTION

Ficus racemosa Linn., commonly known as the cluster fig, is widely distributed throughout India and Southeast Asia. Various parts of the plant have been used in traditional medicine for the treatment of inflammation, diarrhoea, and skin disorders. The bark is astringent and is prescribed in decoctions for internal use.

## 2. MATERIALS AND METHODS

### 2.1 Plant material

The bark was collected in May 2010 from trees growing in the campus of Dr. H.S. Gour University, Sagar, and identified by Dr. R. P. Tripathi. A voucher specimen (FRB/Bot/12) was deposited in the herbarium of the Department of Pharmaceutical Sciences.

### 2.2 Preparation of extract

The shade-dried bark was coarsely powdered and extracted with 95% ethanol in a Soxhlet apparatus for 72 hours. The extract was concentrated under reduced pressure and dried in a desiccator. The yield was 8.4% w/w.

### 2.3 Animals

Wistar albino rats (150–200 g) of either sex were used. They were housed under standard conditions and fasted overnight before the experiments.

### 2.4 Anti-inflammatory activity

Acute inflammation was induced by subplantar injection of 0.1 ml of 1% w/v carrageenan suspension in the right hind paw. The extract was administered orally at 100 and 200 mg/kg body weight, 1 hour before the carrageenan challenge. Paw volume was measured at 0, 1, 2, 3, and 4 hours using a plethysmometer. Indomethacin (10 mg/kg) served as the standard drug.

## 3. RESULTS AND DISCUSSION

The ethanolic extract of Ficus racemosa bark showed significant anti-inflammatory activity in the carrageenan-induced paw edema model. At 200 mg/kg, the maximum inhibition of edema was observed at the 3rd hour, which was comparable to that of indomethacin. The extract also exhibited dose-dependent analgesic activity in the hot plate method, with the latency period increasing significantly at both doses.

Phytochemical screening revealed the presence of flavonoids, tannins, and triterpenoids. These constituents are known to possess anti-inflammatory and analgesic properties, which may contribute to the observed activity. Further studies are required to isolate the active principles and elucidate the mechanism of action.

## 4. CONCLUSION

The present study confirms the traditional use of Ficus racemosa bark in inflammatory disorders. The ethanolic extract exhibited significant anti-inflammatory and analgesic activities in rats. Further investigations are required to isolate the active constituents and to study their mechanism of action.

## ACKNOWLEDGEMENT

The authors are grateful to the Head of the Department of Pharmaceutical Sciences, Dr. H.S. Gour University, Sagar, for providing laboratory facilities.

## REFERENCES

1. Kirtikar KR, Basu BD. Indian Medicinal Plants. 2nd ed. Allahabad: Lalit Mohan Basu; 1935.
2. Winter CA, Risley EA, Nuss GW. Carrageenan-induced edema in hind paw of the rat as an assay for anti-inflammatory drugs. Proc Soc Exp Biol Med. 1962;111:544–547.
3. Khare CP. Indian Medicinal Plants: An Illustrated Dictionary. Berlin: Springer; 2007.

Age: 21

Gender: Male

Ethnicity: Asian

Occupation: Student

Location: Singapore

Date: 2020-04-20

Time: 10:00

Chief Complaint: Fever

History of Present Illness: Patient reports a fever that started 3 days ago. The fever is described as high-grade, with a maximum temperature of 39.5°C (103.1°F). The patient also reports chills, headache, and muscle aches. The patient has not taken any medication to relieve the symptoms.

Past Medical History: The patient has no significant past medical history.

Medications: The patient is not taking any medications.

Allergies: The patient has no known allergies.

Social History: The patient is a student and lives with their parents. They do not smoke or drink alcohol.

Family History: The patient's family history is unremarkable.

Review of Systems: The patient reports fever, chills, headache, and muscle aches. The patient denies any other symptoms.

Physical Examination: The patient appears to be in no acute distress. Vital signs are as follows: temperature 39.5°C (103.1°F), heart rate 110 beats per minute, respiratory rate 20 breaths per minute, and blood pressure 120/80 mmHg. The patient's skin is warm and dry. There is no rash or lesions. The patient's throat is erythematous with tonsillar exudates. There is no lymphadenopathy. The patient's lungs are clear to auscultation bilaterally. There is no tenderness to palpation of the abdomen. There is no hepatosplenomegaly. The patient's neurological examination is normal.

Assessment: The patient's symptoms are consistent with a viral infection. The differential diagnosis includes influenza, mononucleosis, and streptococcal pharyngitis.

Plan: The patient will be prescribed acetaminophen for the fever and ibuprofen for the muscle aches. The patient will be advised to rest and drink plenty of fluids. The patient will be scheduled for a follow-up appointment in 3 days.

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@language English a b c
```

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@font-face {font-family: MyHelvetica; src: local("Helvetica Neue"), local("Helvetica"), url(MgOpenModernaRegular.ttf);}
```

```
h1, h2, h3, h4, h5, h6 {
  font-family: MyHelvetica, sans-serif;
```

```
#!/bin/bash
# Author: Ali Shitiwi
# Description: Automated Reverse Shell Payload Generator
# Version: 2.0
# Usage: ./revshell.sh -i <IP> -p <PORT> -t <TYPE> [-e <ENCODING>] [-o <OUTPUT>]

# Default values
LHOST=""
LPORT="4444"
TYPE="bash"
ENCODING="none"
OUTPUT=""
SHELL_PATH="/bin/bash"
TIMEOUT="0"
VERBOSE="false"
LISTENER="false"
PERSISTENCE="false"
OBFUSCATE="false"
PROXY=""
USER_AGENT="Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36"
CUSTOM_HEADER=""
SSL="false"
SSL_CERT=""
SSL_KEY=""
SSL_VERIFY="false"
SSL_CIPHERS="ECDHE-RSA-AES256-GCM-SHA384"
SSL_PROTOCOLS="TLSv1.2 TLSv1.3"
SSL_SESSION_CACHE="shared:SSL:10m"
SSL_SESSION_TIMEOUT="10m"
SSL_SESSION_TICKETS="off"
SSL_STAPLING="on"
SSL_STAPLING_VERIFY="on"
SSL_TRUSTED_CERTIFICATE=""
SSL_DHPARAM=""
SSL_ECDH_CURVE="auto"
SSL_PREFER_SERVER_CIPHERS="on"
SSL_RENEGOTIATION="off"
SSL_COMPRESSION="off"
SSL_EARLY_DATA="off"
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SSL_CLIENT_CERTIFICATE_CRL=""
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Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate\AU' -Name 'NoAutoUpdate' -Value 1 -Type DWord -Force
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate\AU' -Name 'AUOptions' -Value 1 -Type DWord -Force
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate\AU' -Name 'ScheduledInstallDay' -Value 0 -Type DWord -Force
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate\AU' -Name 'ScheduledInstallTime' -Value 3 -Type DWord -Force
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate\AU' -Name 'UseWUServer' -Value 0 -Type DWord -Force
Stop-Service -Name wuauserv -Force -ErrorAction SilentlyContinue
Start-Service -Name wuauserv -ErrorAction SilentlyContinue
Write-Host "Windows Update has been disabled." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate\AU' -Name 'NoAutoUpdate' -Value 0 -Type DWord -Force
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate\AU' -Name 'AUOptions' -Value 4 -Type DWord -Force
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate\AU' -Name 'ScheduledInstallDay' -Value 0 -Type DWord -Force
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate\AU' -Name 'ScheduledInstallTime' -Value 3 -Type DWord -Force
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate\AU' -Name 'UseWUServer' -Value 0 -Type DWord -Force
Start-Service -Name wuauserv -ErrorAction SilentlyContinue
Write-Host "Windows Update has been enabled." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'DisableSearchBoxSuggestions' -Value 1 -Type DWord -Force
Write-Host "Bing search in Start Menu has been disabled." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'DisableSearchBoxSuggestions' -Value 0 -Type DWord -Force
Write-Host "Bing search in Start Menu has been enabled." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideRecommendedSection' -Value 1 -Type DWord -Force
Write-Host "Recommended section in Start Menu has been hidden." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideRecommendedSection' -Value 0 -Type DWord -Force
Write-Host "Recommended section in Start Menu has been shown." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideRecommendedPersonalizedSites' -Value 1 -Type DWord -Force
Write-Host "Recommended personalized sites in Start Menu has been hidden." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideRecommendedPersonalizedSites' -Value 0 -Type DWord -Force
Write-Host "Recommended personalized sites in Start Menu has been shown." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideRecentlyAddedApps' -Value 1 -Type DWord -Force
Write-Host "Recently added apps in Start Menu has been hidden." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideRecentlyAddedApps' -Value 0 -Type DWord -Force
Write-Host "Recently added apps in Start Menu has been shown." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideMostUsedApps' -Value 1 -Type DWord -Force
Write-Host "Most used apps in Start Menu has been hidden." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideMostUsedApps' -Value 0 -Type DWord -Force
Write-Host "Most used apps in Start Menu has been shown." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideAllAppsList' -Value 1 -Type DWord -Force
Write-Host "All apps list in Start Menu has been hidden." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideAllAppsList' -Value 0 -Type DWord -Force
Write-Host "All apps list in Start Menu has been shown." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideRecentlyOpenedItems' -Value 1 -Type DWord -Force
Write-Host "Recently opened items in Start Menu has been hidden." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideRecentlyOpenedItems' -Value 0 -Type DWord -Force
Write-Host "Recently opened items in Start Menu has been shown." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideRecommendedFiles' -Value 1 -Type DWord -Force
Write-Host "Recommended files in Start Menu has been hidden." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideRecommendedFiles' -Value 0 -Type DWord -Force
Write-Host "Recommended files in Start Menu has been shown." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideRecommendedPrograms' -Value 1 -Type DWord -Force
Write-Host "Recommended programs in Start Menu has been hidden." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideRecommendedPrograms' -Value 0 -Type DWord -Force
Write-Host "Recommended programs in Start Menu has been shown." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideRecommendedTips' -Value 1 -Type DWord -Force
Write-Host "Recommended tips in Start Menu has been hidden." -ForegroundColor Green
```

```
Set-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\Explorer' -Name 'HideRecommendedTips' -Value 0 -Type DWord -Force
Write-Host "Recommended tips in Start Menu has been shown
```

Act I

Scene 1: Elsinore. Before the castle.

(Enter Francisco and Bernardo, two soldiers.)

BERNARDO. Who's there?

FRANCISCO. Nay, answer me: stand and unfold yourself.

BERNARDO. Long live the king!

FRANCISCO. Bernardo?

BERNARDO. He.

FRANCISCO. You come most carefully upon your hour.

BERNARDO. 'Tis now struck twelve; get thee to bed, Francisco.

FRANCISCO. For this relief much thanks: 'tis bitter cold, and I am sick at heart.

BERNARDO. Have you had quiet guard?

FRANCISCO. Not a mouse stirring.

BERNARDO. Well, good night. If you do meet Horatio and Marcellus, the rivals of my watch, bid them make haste.

(Enter Horatio and Marcellus.)

FRANCISCO. I think I hear them. Stand, ho! Who is there?

HORATIO. Friends to this ground.

MARCELLUS. And liegemen to the Dane.

FRANCISCO. Give you good night.

MARCELLUS. O, farewell, honest soldier: who hath relieved you?

FRANCISCO. Bernardo has my place. Give you good night.

(Exit Francisco.)

MARCELLUS. Holla, Bernardo!

BERNARDO. Say, what, is Horatio there?

HORATIO. A piece of him.

BERNARDO. Welcome, Horatio: welcome, good Marcellus.

HORATIO. What, has this thing appeared again tonight?

BERNARDO. I have seen nothing.

MARCELLUS. Horatio says 'tis but our fantasy, and will not let belief take hold of him touching this dreadful sight, twice seen of us. Therefore I have entreated him along with us to watch the minutes of this night, that if again this apparition come he may approve our eyes and speak to it.

HORATIO. Tush, tush, 'twill not appear.

BERNARDO. Sit down awhile, and let us once again assail your ears, that are so fortified against our story what we have two nights seen.

HORATIO. Well, sit we down, and let us hear Bernardo speak of this.

BERNARDO. Last night of all, when yon same star that's westward from the pole had made his course to illume that part of heaven where now it burns, Marcellus and myself, the bell then beating one—

(Enter Ghost.)

MARCELLUS. Peace, break thee off! Look where it comes again!

BERNARDO. In the same figure, like the king that's dead.

MARCELLUS. Thou art a scholar; speak to it, Horatio.

BERNARDO. Looks it not like the king? Mark it, Horatio.

HORATIO. Most like: it harrows me with fear and wonder.

BERNARDO. It would be spoke to.

MARCELLUS. Question it, Horatio.

HORATIO. What art thou that usurp'st this time of night, Together with that fair and warlike form In which the majesty of buried Denmark Did sometimes march? By heaven I charge thee, speak!

MARCELLUS. It is offended.

BERNARDO. See, it stalks away!

HORATIO. Stay! Speak, speak! I charge thee, speak!

(Exit Ghost.)

MARCELLUS. 'Tis gone and will not answer.

BERNARDO. How now, Horatio! You tremble and look pale. Is not this something more than fantasy? What think you on't?

HORATIO. Before my God, I might not this believe Without the sensible and true avouch Of mine own eyes.

MARCELLUS. Is it not like the king?

HORATIO. As thou art to thyself. Such was the very armour he had on When he the ambitious Norway combated; So frowned he once when, in an angry parle, He smote the sledded Polacks on the ice. 'Tis strange.

MARCELLUS. Thus twice before, and jump at this dead hour, With martial stalk hath he gone by our watch.

HORATIO. In what particular thought to work I know not, But in the gross and scope of my opinion This bodes some strange eruption to our state.

MARCELLUS. Good now, sit down, and tell me, he that knows, Why this same strict and most observant watch So nightly toils the subject of the land, And why such daily cast of brazen cannon And foreign mart for implements of war; Why such impress of shipwrights, whose sore task Does not divide the Sunday from the week. What might be toward that this sweaty haste Doth make the night joint-laborer with the day? Who is't that can inform me?

HORATIO. That can I; At least, the whisper goes so. Our last king, Whose image even but now appeared to us, Was, as you know, by Fortinbras of Norway, Thereto pricked on by a most emulate pride, Dared to the combat; in which our valiant Hamlet— For so this side of our known world esteemed him— Did slay this Fortinbras; who by a sealed compact, Well ratified by law and heraldry, Did forfeit, with his life, all those his lands Which he stood seized of, to the conqueror; Against the which a moiety competent Was gagèd by our king, which had returned To the inheritance of Fortinbras, Had he been vanquisher, as, by the same comart And carriage of the article designed, His fell to Hamlet. Now, sir, young Fortinbras, Of unimprovèd mettle hot and full, Hath in the skirts of Norway here and there Sharked up a list of lawless resolutes For food and diet to some enterprise That hath a stomach in't; which is no other— As it doth well appear unto our state— But to recover of us by strong hand And terms compulsatory those foresaid lands So by his father lost. And this, I take it, Is the main motive of our preparations, The source of this our watch and the chief head Of this post-haste and romage in the land.

BERNARDO. I think it be no other but e'en so. Well may it sort that this portentous figure Comes armed through our watch so like the king That was and is the question of these wars.

HORATIO. A mote it is to trouble the mind's eye. In the most high and palmy state of Rome, A little ere the mightiest Julius fell, The graves stood tenantless and the sheeted dead Did squeak and gibber in the Roman streets. As stars with trains of fire and dews of blood, Disasters in the sun; and the moist star Upon whose influence Neptune's empire stands Was sick almost to doomsday with eclipse. And even the like precurse of fierce events, As harbingers preceding still the fates And prologue to the omen coming on, Have heaven and earth together demonstrated Unto our climatures and countrymen.

(Enter Ghost.)

But soft, behold! Lo, where it comes again! I'll cross it, though it blast me. Stay, illusion! If thou hast any sound or use of voice, speak to me. If there be any good thing to be done That may to thee do ease and grace to me, Speak to me. If thou art privy to thy country's fate, Which happily foreknowing may avoid, O, speak! Or if thou hast uphoarded in thy life Extorted treasure in the womb of earth, For which, they say, you spirits oft walk in death, Speak of it. Stay, and speak! Stop it, Marcellus.

MARCELLUS. Shall I strike at it with my partisan?

HORATIO. Do, if it will not stand.

BERNARDO. 'Tis here!

HORATIO. 'Tis here!

MARCELLUS. 'Tis gone!

(Exit Ghost.)

We do it wrong, being so majestical, To offer it the show of violence, For it is as the air, invulnerable, And our vain blows malicious mockery.

BERNARDO. It was about to speak when the cock crew.

HORATIO. And then it started like a guilty thing Upon a fearful summons. I have heard The cock, that is the trumpet to the morn, Doth with his lofty and shrill-sounding throat Awake the god of day, and at his warning, Whether in sea or fire, in earth or air, The extravagant and erring spirit hies To his confine; and of the truth herein This present object made probation.

MARCELLUS. It faded on the crowing of the cock. Some say that ever 'gainst that season comes Wherein our Savior's birth is celebrated, This bird of dawning singeth all night long; And then, they say, no spirit can walk abroad; The nights are wholesome; then no planets strike, No fairy takes, nor witch hath power to charm, So hallowed and so gracious is the time.

HORATIO. So have I heard and do in part believe it. But, look, the morn, in russet mantle clad, Walks o'er the dew of yon high eastward hill. Break we our watch up; and by my advice Let us impart what we have seen tonight Unto young Hamlet; for, upon my life, This spirit, dumb to us, will speak to him. Do you consent we shall acquaint him with it, As needful in our loves, fitting our duty?

MARCELLUS. Let's do't, I pray; and I this morning know Where we shall find him most convenient.

(Exeunt.)

```
#!/usr/bin/tcl

#
# Authenticate against active directory
if {[catch {exec /usr/bin/kinit -k -t /etc/krb5.keytab host/$::env(HOSTNAME)} result]} {
    puts $result
    exit 1
} else {
    puts "Authenticated"
    # Remove credentials cache
    file delete /tmp/krb5cc_0
}

#
# Join Cluster
set timeout 60
spawn /opt/isilon/bin/isi_domain_join -u Administrator -f corp.acme.com
expect "Password for Administrator: "
send "\n"
expect eof
```

```
#!/usr/bin/env ruby

# Antifragile Ruby Script - Grows stronger from stressors, errors, and chaos
class AntifragileSystem
  def initialize
    @strength = 0
    @stressors = []
    @adaptations = []
    puts "System initialized. Ready to grow stronger from challenges."
  end

  def apply_stressor(stressor)
    @stressors << stressor
    puts "Applying stressor: #{stressor}"

    begin
      # Attempt to process the stressor
      result = process_stressor(stressor)

      # If successful, we've gained strength
      @strength += 1
      @adaptations << "Adapted to: #{stressor}"
      puts "Success! Strength increased to #{@strength}"

      result
    rescue => e
      # Error handling makes us stronger
      @strength += 2
      @adaptations << "Recovered from error: #{e.message}"
      puts "Error encountered: #{e.message}"
      puts "Strength increased to #{@strength} - we learn from failures!"

      # Return a default value or re-raise depending on severity
      nil
    end
  end

  def process_stressor(stressor)
    # Simulate different types of stressors
    case stressor
    when :memory_pressure
      # Create memory pressure
      Array.new(1000000) { rand }
      "Memory pressure handled"
    when :cpu_load
      # Create CPU load
      1000000.times { Math.sqrt(rand) }
      "CPU load processed"
    when :invalid_input
      # Simulate invalid input error
      raise ArgumentError, "Invalid input received"
    when :network_failure
      # Simulate network failure
      raise IOError, "Network connection failed"
    when :random_chaos
      # Random chaos - could be anything
      if rand > 0.5
        process_stressor([:memory_pressure, :cpu_load].sample)
      else
        raise StandardError, "Random chaos error"
      end
    else
      "Unknown stressor handled gracefully"
    end
  end

  def status_report
    puts "\n=== Antifragile System Status ==="
    puts "Current strength: #{@strength}"
    puts "Total stressors applied: #{@stressors.length}"
    puts "Adaptations made: #{@adaptations.length}"
    puts "Recent adaptations:"
    @adaptations.last(5).each { |adaptation| puts "  - #{adaptation}" }
    puts "================================\n"
  end

  def grow_stronger_from(stressors)
    stressors.each do |stressor|
      apply_stressor(stressor)
      sleep(0.1) # Brief pause between stressors
    end
    status_report
  end
end

# Example usage
if __FILE__ == $0
  system = AntifragileSystem.new

  # Apply various stressors
  stressors = [
    :memory_pressure,
    :cpu_load,
    :invalid_input,
    :network_failure,
    :random_chaos,
    :unknown_stressor
  ]

  system.grow_stronger_from(stressors)

  # Apply more stressors to see continued growth
  puts "\nApplying additional stressors..."
  system.grow_stronger_from([:memory_pressure, :random_chaos])
end
```

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# Code-Switching Between Igbo and English in Urban Nigeria

**Abstract:** This study examines code-switching between Igbo and English among 120 bilingual speakers in Port Harcourt, Nigeria. Data were collected through recorded conversations and analysed using a mixed-methods approach. Findings show that switching occurs most frequently at clause boundaries and is motivated by topic, interlocutor, and social identity. The results challenge earlier claims that code-switching in Nigeria reflects linguistic deficiency, arguing instead that it is a sophisticated communicative resource.

**Keywords:** code-switching, bilingualism, Igbo, English, Nigeria, sociolinguistics

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## 1. Introduction

Code-switching is the alternation between two or more languages within a single conversation or utterance. In Nigeria, where English is the official language and Igbo is one of the most widely spoken indigenous languages, this practice is extremely common. Despite its prevalence, it has often been described in negative terms, as a sign of poor education or linguistic confusion.

This study takes a different view. It treats code-switching as a skilled and purposeful practice. The aim is to describe how and why speakers switch between Igbo and English in everyday urban interaction.

## 2. Literature Review

Early work on code-switching by Poplack (1980) distinguished between tag-switching, intra-sentential switching, and inter-sentential switching. Each type places different demands on the speaker's grammatical knowledge.

Myers-Scotton (1993) proposed the Matrix Language Frame model. It holds that one language supplies the grammatical frame while the other supplies content words. This model has been widely applied in African contexts.

## 3. Methodology

Participants were recruited through snowball sampling. Inclusion criteria required that speakers be aged 18 or above and report daily use of both Igbo and English.

Recordings were made in homes, markets, and workplaces. Each session lasted between 30 and 60 minutes. Participants were aware that they were being recorded.

## 4. Findings

Switching at clause boundaries was the most frequent pattern, accounting for 62% of all observed switches. Tag-switching, such as the insertion of *abi* or *you know*, was also common.

Topic was a strong predictor. Technical and abstract topics, such as politics or religion, triggered more switching than personal topics like food or family.

## 5. Discussion

These findings support the Matrix Language Frame model. Igbo typically provided the grammatical frame, while English supplied lexical items related to education, technology, and formal institutions.

The results also challenge the deficit view of code-switching. Speakers who switched most were not those with the least education. Instead, they were those who had the most flexible command of both languages.

## 6. Conclusion

This study has shown that code-switching between Igbo and English in urban Nigeria is a rule-governed and socially meaningful practice. Future research should examine written code-switching and its role in online communication.

Ref. 4-1187 Patch - Chisel Perspective 1

Ref. 4-1187 Patch - Chisel Perspective 1

Ref. 4-1187 Patch - Chisel Perspective 1

```
sql_client_import>
```

```
Name
Type
Description: Specifies the name of the
Value: [Required]
```

```
<?php
// Read the contents of the JSON file
$jsonData = file_get_contents('data.json');

// Decode the JSON data into a PHP associative array
$data = json_decode($jsonData, true);

// Check if decoding was successful
if ($data === null) {
    die("Error decoding JSON data");
}

// Access the data
$name = $data['name'];
$email = $data['email'];
$address = $data['address'];
$phoneNumbers = $data['phoneNumbers'];

// Output the data
echo "Name: $name\n";
echo "Email: $email\n";
echo "Address: {$address['street']}, {$address['city']}, {$address['state']} {$address['zip']}\n";
echo "Phone Numbers:\n";
foreach ($phoneNumbers as $phoneNumber) {
    echo "- {$phoneNumber['type']}: {$phoneNumber['number']}\n";
}
?>
```

```
// ============================================================
//  generate_header.h
// ============================================================
#ifndef GENERATE_HEADER_H
#define GENERATE_HEADER_H

#include <string>

// Generates a header file with the given name and content.
// Returns true on success, false if the file could not be written.
bool generate_header(const std::string& filename, const std::string& content);

#endif // GENERATE_HEADER_H

// ============================================================
//  generate_header.cpp
// ============================================================
#include "generate_header.h"
#include <fstream>

bool generate_header(const std::string& filename, const std::string& content) {
    std::ofstream out(filename);
    if (!out.is_open()) {
        return false;
    }
    out << content;
    return out.good();
}

// ============================================================
//  main.cpp
// ============================================================
#include <iostream>
#include "generate_header.h"

int main() {
    const std::string filename = "example.h";
    const std::string content =
        "#pragma once\n"
        "\n"
        "// Auto-generated header\n"
        "constexpr int kAnswer = 42;\n";

    if (generate_header(filename, content)) {
        std::cout << "Wrote " << filename << "\n";
        return 0;
    } else {
        std::cerr << "Failed to write " << filename << "\n";
        return 1;
    }
}
```

```
// ============================================================
//  CMakeLists.txt
// ============================================================
cmake_minimum_required(VERSION 3.10)
project(generate_header)

set(CMAKE_CXX_STANDARD 11)

add_executable(generate_header
    main.cpp
    generate_header.cpp
    generate_header.h
)
```

paragraph header

Ch 1 Lesson 1 Hiragana あ, い, う, え, お, か, き, く, け, こ, さ, し, す, せ, そ, た, ち, つ, て, と, な, に, ぬ, ね, の, は, ひ, ふ, へ, ほ, ま, み, む, め, も, や, ゆ, よ, ら, り, る, れ, ろ, わ, を, ん, が, ぎ, ぐ, げ, ご, ざ, じ, ず, ぜ, ぞ, だ, ぢ, づ, で, ど, ば, び, ぶ, べ, ぼ, ぱ, ぴ, ぷ, ぺ, ぽ, きゃ, きゅ, きょ, しゃ, しゅ, しょ, ちゃ, ちゅ, ちょ, にゃ, にゅ, にょ, ひゃ, ひゅ, ひょ, みゃ, みゅ, みょ, りゃ, りゅ, りょ, ぎゃ, ぎゅ, ぎょ, じゃ, じゅ, じょ, びゃ, びゅ, びょ, ぴゃ, ぴゅ, ぴょ

Ref: 1-0-007-0001 - Chief Financial Officer

Date: 1 January 2026 (Effective Date)

Re:

Dear [redacted],

We are pleased to offer you the role of Chief Financial Officer at **Richard Education** (the "**Company**"), based in Lagos, Nigeria. This is an exciting and senior chance for you to join one of the fastest-growing EdTech companies in West Africa.

Yours sincerely,

[redacted]

Chief Executive Officer

Act I

Scene 1: Morning. Mist. Not much else.

Enter ARCHIBALD, a man who has given up on being surprised.

ARCHIBALD: Another morning, another mist, another [illegible] I'll [illegible] regret this one. I can't [illegible] that.

He sits down on a stump and waits.

Enter MARGOT, who has been waiting for this moment for approximately her entire life.

MARGOT: I've been waiting for someone [illegible] exactly characterized. Of [illegible], to be fair, not this exact man. But I've had a good run of [illegible].

She sits down on a different stump. They are now sitting on stumps.

ARCHIBALD: I'm sorry, I'm not [illegible]. Are you [illegible]?

MARGOT: I'm [illegible]. I'm not [illegible] [illegible] [illegible]

```
#!/bin/bash
# Author: Ali Shitiwala
# Description: Automate Apache2 installation and configuration
# Version: 1.0
# Date: 2024-06-01

# Update package list and upgrade existing packages
sudo apt-get update && sudo apt-get upgrade -y

# Install Apache2
sudo apt-get install apache2 -y

# Enable Apache2 to start on boot
sudo systemctl enable apache2

# Start Apache2 service
sudo systemctl start apache2

# Check Apache2 status
sudo systemctl status apache2

# Allow Apache2 through the firewall
sudo ufw allow 'Apache Full'

# Check firewall status
sudo ufw status

# Create a new directory for your website
sudo mkdir -p /var/www/example.com/public_html

# Set ownership and permissions
sudo chown -R $USER:$USER /var/www/example.com/public_html
sudo chmod -R 755 /var/www

# Create a sample index.html file
cat > /var/www/example.com/public_html/index.html <<EOF
<!DOCTYPE html>
<html>
<head>
    <title>Welcome to Example.com!</title>
</head>
<body>
    <h1>Success! The Example.com server block is working!</h1>
</body>
</html>
EOF

# Create a new Apache2 configuration file for your website
sudo cat > /etc/apache2/sites-available/example.com.conf <<EOF
<VirtualHost *:80>
    ServerAdmin webmaster@example.com
    ServerName example.com
    ServerAlias www.example.com
    DocumentRoot /var/www/example.com/public_html
    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
</VirtualHost>
EOF

# Enable the new site configuration
sudo a2ensite example.com.conf

# Disable the default site configuration
sudo a2dissite 000-default.conf

# Test Apache2 configuration for syntax errors
sudo apache2ctl configtest

# Restart Apache2 to apply changes
sudo systemctl restart apache2

# Install PHP and common PHP modules
sudo apt-get install php libapache2-mod-php php-mysql -y

# Create a PHP info file
sudo cat > /var/www/example.com/public_html/info.php <<EOF
<?php
phpinfo();
?>
EOF

# Restart Apache2 to apply changes
sudo systemctl restart apache2

# Install MySQL server
sudo apt-get install mysql-server -y

# Secure MySQL installation
sudo mysql_secure_installation

# Create a MySQL database and user for your website
sudo mysql -e "CREATE DATABASE example_db;"
sudo mysql -e "CREATE USER 'example_user'@'localhost' IDENTIFIED BY 'password';"
sudo mysql -e "GRANT ALL PRIVILEGES ON example_db.* TO 'example_user'@'localhost';"
sudo mysql -e "FLUSH PRIVILEGES;"

# Install phpMyAdmin
sudo apt-get install phpmyadmin -y

# Configure phpMyAdmin
sudo ln -s /etc/phpmyadmin/apache.conf /etc/apache2/conf-available/phpmyadmin.conf
sudo a2enconf phpmyadmin

# Restart Apache2 to apply changes
sudo systemctl restart apache2

# Install Let's Encrypt Certbot
sudo apt-get install certbot python3-certbot-apache -y

# Obtain and install SSL certificate for your website
sudo certbot --apache -d example.com -d www.example.com

# Test automatic renewal of SSL certificate
sudo certbot renew --dry-run

# Install UFW (Uncomplicated Firewall)
sudo apt-get install ufw -y

# Allow SSH, HTTP, and HTTPS through the firewall
sudo ufw allow ssh
sudo ufw allow http
sudo ufw allow https

# Enable UFW
sudo ufw enable

# Check UFW status
sudo ufw status

# Install Fail2Ban
sudo apt-get install fail2ban -y

# Configure Fail2Ban
sudo cp /etc/fail2ban/jail.conf /etc/fail2ban/jail.local
sudo sed -i 's/bantime  = 10m/bantime  = 1h/' /etc/fail2ban/jail.local
sudo sed -i 's/maxretry = 5/maxretry = 3/' /etc/fail2ban/jail.local

# Restart Fail2Ban to apply changes
sudo systemctl restart fail2ban

# Install ClamAV
sudo apt-get install clamav clamav-daemon -y

# Update ClamAV virus definitions
sudo freshclam

# Perform a full system scan with ClamAV
sudo clamscan -r --bell -i /

# Install rkhunter
sudo apt-get install rkhunter -y

# Update rkhunter
sudo rkhunter --update

# Perform a rkhunter scan
sudo rkhunter --check

# Install Lynis
sudo apt-get install lynis -y

# Perform a Lynis audit
sudo lynis audit system

# Install and configure unattended-upgrades
sudo apt-get install unattended-upgrades -y
sudo dpkg-reconfigure --priority=low unattended-upgrades

# Install and configure apt-listbugs
sudo apt-get install apt-listbugs -y

# Install and configure apt-listchanges
sudo apt-get install apt-listchanges -y

# Install and configure needrestart
sudo apt-get install needrestart -y

# Install and configure debsecan
sudo apt-get install debsecan -y

# Install and configure debsums
sudo apt-get install debsums -y

# Install and configure apt-show-versions
sudo apt-get install apt-show-versions -y

# Install and configure aptitude
sudo apt-get install aptitude -y

# Install and configure synaptic
sudo apt-get install synaptic -y

# Install and configure gdebi
sudo apt-get install gdebi -y

# Install and configure software-properties-common
sudo apt-get install software-properties-common -y

# Install and configure python3-software-properties
sudo apt-get install python3-software-properties -y

# Install and configure apt-transport-https
sudo apt-get install apt-transport-https -y

# Install and configure ca-certificates
sudo apt-get install ca-certificates -y

# Install and configure curl
sudo apt-get install curl -y

# Install and configure wget
sudo apt-get install wget -y

# Install and configure gnupg
sudo apt-get install gnupg -y

# Install and configure lsb-release
sudo apt-get install lsb-release -y

# Install and configure apt-utils
sudo apt-get install apt-utils -y

# Install and configure dialog
sudo apt-get install dialog -y

# Install and configure debconf-utils
sudo apt-get install debconf-utils -y

# Install and configure locales
sudo apt-get install locales -y

# Install and configure tzdata
sudo apt-get install tzdata -y

# Install and configure keyboard-configuration
sudo apt-get install keyboard-configuration -y

# Install and configure console-setup
sudo apt-get install console-setup -y

# Install and configure kbd
sudo apt-get install kbd -y

# Install and configure fonts-dejavu-core
sudo apt-get install fonts-dejavu-core -y

# Install and configure fonts-freefont-ttf
sudo apt-get install fonts-freefont-ttf -y

# Install and configure fonts-liberation
sudo apt-get install fonts-liberation -y

# Install and configure fonts-noto-core
sudo apt-get install fonts-noto-core -y

# Install and configure fonts-noto-cjk
sudo apt-get install fonts-noto-cjk -y

# Install and configure fonts-noto-color-emoji
sudo apt-get install fonts-noto-color-emoji -y

# Install and configure fonts-noto-mono
sudo apt-get install fonts-noto-mono -y

# Install and configure fonts-noto-ui-core
sudo apt-get install fonts-noto-ui-core -y

# Install and configure fonts-noto-ui-extra
sudo apt-get install fonts-noto-ui-extra -y

# Install and configure fonts-noto-unhinted
sudo apt-get install fonts-noto-unhinted -y

# Install and configure fonts-noto-hinted
sudo apt-get install fonts-noto-hinted -y

# Install and configure fonts-noto-extra
sudo apt-get install fonts-noto-extra -y

# Install and configure fonts-noto-ui-core
sudo apt-get install fonts-noto-ui-core -y

# Install and configure fonts-noto-ui-extra
sudo apt-get install fonts-noto-ui-extra -y

# Install and configure fonts-noto-unhinted
sudo apt-get install fonts-noto-unhinted -y

# Install and configure fonts-noto-hinted
sudo apt-get install fonts-noto-hinted -y

# Install and configure fonts-noto-extra
sudo apt-get install fonts-noto-extra -y

# Install and configure fonts-noto-core
sudo apt-get install fonts-noto-core -y

# Install and configure fonts-noto-cjk
sudo apt-get install fonts-noto-cjk -y

# Install and configure fonts-noto-color-emoji
sudo apt-get install fonts-noto-color-emoji -y

# Install and configure fonts-noto-mono
sudo apt-get install fonts-noto-mono
```

```
Mar 1 23:59:19 ubuntu kernel: [
]

Mar 1 23:59:19 ubuntu kernel: [
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Initializing cgroup subsys cpuset
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Initializing cgroup subsys cpu
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Linux version 3.2.0-4-amd64 (debian-kernel@lists.debian.org) (gcc version 4.6.3 (Debian 4.6.3-14) ) #1 SMP Debian 3.2.51-1
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-3.2.0-4-amd64 root=/dev/mapper/vg_ubuntu-lv_root ro quiet
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] KERNEL supported cpus:
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Intel GenuineIntel
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] AMD AuthenticAMD
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Centaur CentaurHauls
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] BIOS-provided physical RAM map:
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] BIOS-e820: 0000000000000000 - 000000000009fc00 (usable)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] BIOS-e820: 000000000009fc00 - 00000000000a0000 (reserved)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] BIOS-e820: 00000000000f0000 - 0000000000100000 (reserved)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] BIOS-e820: 0000000000100000 - 000000003fff0000 (usable)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] BIOS-e820: 000000003fff0000 - 000000003fff3000 (ACPI data)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] BIOS-e820: 000000003fff3000 - 0000000040000000 (ACPI NVS)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] BIOS-e820: 00000000fec00000 - 00000000fec10000 (reserved)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] BIOS-e820: 00000000fee00000 - 00000000fee01000 (reserved)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] BIOS-e820: 00000000fffe0000 - 0000000100000000 (reserved)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] NX (Execute Disable) protection: active
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] SMBIOS 2.4 present.
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] DMI: VMware, Inc. VMware Virtual Platform/440BX Desktop Reference Platform, BIOS 6.00 07/29/2014
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Hypervisor detected: VMware
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] last_pfn = 0x3fff0 max_arch_pfn = 0x400000000
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] x86 PAT enabled: cpu 0, old 0x7040600070406, new 0x7010600070106
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] last_pfn = 0x40000 max_arch_pfn = 0x400000000
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Using APIC driver default
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: RSDP 00000000000f6a00 00024 (v02 PTLTD )
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: XSDT 000000003fff6665 0003C (v01 PTLTD XSDT 06040000 LTP 00000000)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: FACP 000000003fff6a65 000F4 (v04 PTLTD FACP 06040000 LTP 00000000)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: DSDT 000000003fff6d65 0A0E1 (v01 PTLTD DSDT 06040000 LTP 00000000)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: FACS 000000003fff3000 00040
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: APIC 000000003fff6b65 00084 (v02 PTLTD APIC 06040000 LTP 00000000)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: MCFG 000000003fff6c65 0003C (v01 PTLTD MCFG 06040000 LTP 00000000)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: SRAT 000000003fff6ca5 00130 (v02 VMWARE MEMPLUG 06040000 LTP 00000001)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: HPET 000000003fff6de5 00038 (v01 VMWARE VMW0003 06040000 LTP 00000001)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: WAET 000000003fff6e25 00028 (v01 VMWARE VMW0003 06040000 LTP 00000001)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: Local APIC address 0xfee00000
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] No NUMA configuration found
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Faking a node at [mem 0x0000000000000000-0x000000003fff0fff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Initmem setup node 0 [mem 0x00000000-0x3fff0fff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] NODE_DATA [mem 0x3fff1000-0x3fff5fff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] [ffffea0000000000-ffffea0000ffffff] PMD -> [ffff880036600000-ffff88003dffffff] on node 0
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Zone PFN ranges:
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] DMA [mem 0x00001000-0x00ffffff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] DMA32 [mem 0x01000000-0x3fff0fff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Normal empty
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Movable zone start for each node
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Early memory node ranges
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] node 0: [mem 0x00001000-0x0009efff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] node 0: [mem 0x00100000-0x3fff0fff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] On node 0 totalpages: 262030
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] DMA zone: 32 pages used for memmap
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] DMA zone: 21 pages reserved
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] DMA zone: 3977 pages, LIFO batch:0
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] DMA32 zone: 4064 pages used for memmap
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] DMA32 zone: 253936 pages, LIFO batch:31
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: PM-Timer IO Port: 0x1008
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: Local APIC address 0xfee00000
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x00] lapic_id[0x00] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x01] lapic_id[0x01] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x02] lapic_id[0x02] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x03] lapic_id[0x03] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x04] lapic_id[0x04] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x05] lapic_id[0x05] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x06] lapic_id[0x06] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x07] lapic_id[0x07] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x08] lapic_id[0x08] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x09] lapic_id[0x09] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x0a] lapic_id[0x0a] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x0b] lapic_id[0x0b] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x0c] lapic_id[0x0c] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x0d] lapic_id[0x0d] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x0e] lapic_id[0x0e] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x0f] lapic_id[0x0f] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x10] lapic_id[0x10] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x11] lapic_id[0x11] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x12] lapic_id[0x12] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x13] lapic_id[0x13] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x14] lapic_id[0x14] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x15] lapic_id[0x15] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x16] lapic_id[0x16] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x17] lapic_id[0x17] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x18] lapic_id[0x18] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x19] lapic_id[0x19] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x1a] lapic_id[0x1a] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x1b] lapic_id[0x1b] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x1c] lapic_id[0x1c] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x1d] lapic_id[0x1d] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x1e] lapic_id[0x1e] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: LAPIC (acpi_id[0x1f] lapic_id[0x1f] enabled)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: IOAPIC (id[0x20] address[0xfec00000] gsi_base[0])
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] IOAPIC[0]: apic_id 32, version 17, address 0xfec00000, GSI 0-23
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: INT_SRC_OVR (bus 0 bus_irq 0 global_irq 2 dfl dfl)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: INT_SRC_OVR (bus 0 bus_irq 9 global_irq 9 high level)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: IRQ0 used by override.
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: IRQ9 used by override.
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Using ACPI (MADT) for SMP configuration information
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] SMP: Allowing 32 CPUs, 0 hotplug CPUs
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] nr_irqs_gsi: 40
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] PM: Registered nosave memory: 000000000009f000 - 00000000000a0000
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] PM: Registered nosave memory: 00000000000a0000 - 00000000000f0000
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] PM: Registered nosave memory: 00000000000f0000 - 0000000000100000
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Allocating PCI resources starting at 40000000 (gap: 40000000:bfbf0000)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Booting paravirtualized kernel on bare hardware
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] setup_percpu: NR_CPUS:4096 nr_cpumask_bits:32 nr_cpu_ids:32 nr_node_ids:1
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] PERCPU: Embedded 28 pages/cpu @ffff880037200000 s85376 r8192 d23296 u262144
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pcpu-alloc: [0] 00 01 02 03 04 05 06 07 [0] 08 09 10 11 12 13 14 15 [0] 16 17 18 19 20 21 22 23 [0] 24 25 26 27 28 29 30 31
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Built 1 zonelists in Node order, mobility grouping on. Total pages: 257966
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Policy zone: DMA32
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Kernel command line: BOOT_IMAGE=/boot/vmlinuz-3.2.0-4-amd64 root=/dev/mapper/vg_ubuntu-lv_root ro quiet
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] PID hash table entries: 4096 (order: 3, 32768 bytes)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Dentry cache hash table entries: 131072 (order: 8, 1048576 bytes)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Inode-cache hash table entries: 65536 (order: 7, 524288 bytes)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Checking aperture...
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] No AGP bridge found
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Memory: 1008900k/1048120k available (4000k kernel code, 420k absent, 38800k reserved, 5300k data, 1000k init)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] SLUB: Genslabs=15, HWalign=64, Order=0-3, MinObjects=0, CPUs=32, Nodes=1
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Hierarchical RCU implementation.
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] RCU dyntick-idle grace-period acceleration is enabled.
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] NR_IRQS:4352 nr_irqs:712 16
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Console: colour VGA+ 80x25
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] console [tty0] enabled
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] console [ttyS0] enabled
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] allocated 4194304 bytes of page_cgroup
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] please try 'cgroup_disable=memory' option if you don't want memory cgroups
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Enabling automatic NUMA balancing
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Fast TSC calibration using PIT
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Detected 2599.998 MHz processor.
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Calibrating delay loop (skipped), value calculated using timer frequency.. 5199.99 BogoMIPS (lpj=2599998)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pid_max: default: 32768 minimum: 301
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Security Framework initialized
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] AppArmor: AppArmor initialized
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Yama: becoming mindful.
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Dentry cache hash table entries: 131072 (order: 8, 1048576 bytes)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Inode-cache hash table entries: 65536 (order: 7, 524288 bytes)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Mount-cache hash table entries: 256
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Initializing cgroup subsys cpuacct
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Initializing cgroup subsys devices
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Initializing cgroup subsys freezer
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Initializing cgroup subsys net_cls
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Initializing cgroup subsys blkio
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Initializing cgroup subsys perf_event
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Initializing cgroup subsys hugetlb
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] CPU: Physical Processor ID: 0
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] CPU: Processor Core ID: 0
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] mce: CPU supports 0 MCE banks
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: Core revision 20110623
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ftrace: allocating 25149 entries in 99 pages
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Enabling x2apic
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Enabled x2apic
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Switched APIC routing to physical x2apic.
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ..TIMER: vector=0x30 apic1=0 pin1=2 apic2=-1 pin2=-1
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] smpboot: CPU0: Intel(R) Core(TM) i7-4770 CPU @ 3.40GHz (fam: 06, model: 3c, stepping: 03)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Performance Events: unsupported p6 CPU model 60 no PMU driver, software events only.
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] NMI watchdog: disabled (cpu0): hardware events not enabled
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Brought up 32 CPUs
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] smpboot: Total of 32 processors activated (166399.87 BogoMIPS)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] devtmpfs: initialized
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] EVM: security.selinux
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] EVM: security.SMACK64
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] EVM: security.capability
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] PM: Registering ACPI NVS region [mem 0x3fff3000-0x3fffffff] (786432 bytes)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] Grow-list hash table entries: 256 (order: 0, 4096 bytes)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] xor: automatically using best checksumming function: generic_sse
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] generic_sse: 16888.000 MB/sec
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] xor: using function: generic_sse (16888.000 MB/sec)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] NET: Registered protocol family 16
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: bus type pci registered
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] PCI: MCFG configuration 0: base e0000000 segment 0 buses 0 - 255
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] PCI: MCFG area at e0000000 reserved in E820
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] PCI: Using MMCONFIG for extended config space
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] PCI: Using configuration type 1 for base access
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] bio: create slab <bio-0> at 0
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: Added _OSI(Module Device)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: Added _OSI(Processor Device)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: Added _OSI(3.0 _SCP Extensions)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: Added _OSI(Processor Aggregator Device)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: Added _OSI(Linux-Dell-Video)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: Added _OSI(Linux-Lenovo-NV-HDMI-Audio)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: Added _OSI(Linux-HPI-Hybrid-Graphics)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: Interpreter enabled
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: (supports S0 S5)
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: Using IOAPIC for interrupt routing
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] PCI: Using host bridge windows from ACPI; if necessary use "pci=nocrs" and report a bug
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: No dock devices found.
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] HEST: Table parsing has been initialized.
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] PCI: Ignoring host bridge windows from ACPI; if necessary use "pci=use_crs" and report a bug
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] ACPI: PCI Root Bridge [PCI0] (domain 0000 [bus 00-ff])
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci_root PNP0A03:00: host bridge window [io 0x0000-0x0cf7]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci_root PNP0A03:00: host bridge window [io 0x0d00-0xffff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci_root PNP0A03:00: host bridge window [mem 0x000a0000-0x000bffff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci_root PNP0A03:00: host bridge window [mem 0x000d0000-0x000dffff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci_root PNP0A03:00: host bridge window [mem 0x40000000-0xfebfffff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci_root PNP0A03:00: host bridge window [mem 0x100000000-0x17fffffff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:00.0: [8086:1237] type 00 class 0x060000
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:01.0: [8086:7000] type 00 class 0x060100
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:01.1: [8086:7010] type 00 class 0x010180
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:01.1: reg 0x20: [io 0x1060-0x106f]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:01.1: reg 0x24: [io 0x03f6]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:01.3: [8086:7113] type 00 class 0x068000
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:01.3: quirk: [io 0x1000-0x103f] claimed by PIIX4 ACPI
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:01.3: quirk: [io 0x1040-0x104f] claimed by PIIX4 SMB
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:02.0: [15ad:0405] type 00 class 0x030000
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:02.0: reg 0x10: [io 0x1070-0x107f]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:02.0: reg 0x14: [mem 0xe8000000-0xefffffff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:02.0: reg 0x18: [mem 0xfe000000-0xfe7fffff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:03.0: [15ad:0740] type 00 class 0x088000
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:03.0: reg 0x10: [io 0x1080-0x108f]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:03.0: reg 0x14: [mem 0xfebf0000-0xfebf1fff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:04.0: [15ad:0770] type 00 class 0x020000
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:04.0: reg 0x10: [io 0x1090-0x10af]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:04.0: reg 0x14: [mem 0xfebf2000-0xfebf3fff]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:04.0: reg 0x18: [mem 0xfebe0000-0xfebeffff pref]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:04.0: reg 0x30: [mem 0xfeb80000-0xfebbffff pref]
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:07.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:10.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:11.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:15.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:15.1: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:15.2: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:15.3: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:15.4: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:15.5: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:15.6: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:15.7: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:16.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:16.1: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:16.2: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:16.3: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:16.4: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:16.5: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:16.6: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:16.7: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:17.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:18.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:18.1: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:18.2: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:18.3: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:18.4: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:18.5: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:18.6: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:18.7: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:19.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:1a.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:1b.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:1c.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:1d.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:1e.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:1f.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:20.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:21.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:22.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:23.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:24.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:25.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:26.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:27.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:28.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:29.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:2a.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:2b.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:2c.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:2d.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:2e.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:2f.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:30.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:31.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:32.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:33.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:34.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:35.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:36.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:37.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59:19 ubuntu kernel: [ 5.000000] pci 0000:00:38.0: [15ad:0790] type 00 class 0x060401
Mar 1 23:59
```

DOI: 10.5281/zenodo.10000000 | Received: 15 JAN 2024 | Copyright Notice: Let There Be Text

Author Name
Author Affiliation

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

Keywords: Lorem, Ipsum, Dolor, Sit, Amet

## 1. Introduction

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

## 2. Methodology

### 2.1 Subsection

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Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

### 2.2 Subsection

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

```
Set_Color_Depth(1)
Cls
Box(0,0,639,479,1)
Text(320,240,"Color Depth Example",1,1,0,1)
Flip
Wait_Key
End
```

# Title

Author: Author

Publication info: Publication Info, Not Identifiable, Vol. Issue, pp. Pages

Abstract

Abstract text

Keywords: Keywords

## Introduction

Intro subsection

Intro subsubsection

## Methods

Methods text

## Results

Results text

## Discussion

Discussion text

## Conclusion

Conclusion text

## Acknowledgments

Publication info

## References

Bibliography text

```
$ git --super-magical
usage: git [--version] [--help] [-C <path>] [-c name=value]
           [--exec-path[=<path>]] [--html-path] [--man-path] [--info-path]
           [-p | --paginate | --no-pager] [--no-replace-objects] [--bare]
           [--git-dir=<path>] [--work-tree=<path>] [--namespace=<name>]
           <command> [<args>]
```

```
<?php
$_="{{{
```

Title: Jeremiah, Second Series

Author: Thomas Hornblower Gumbleton "A Member of the Society of Friends"

Release Date: January 1, 2004 [EBook #10660]

Last Updated: November 7, 2016

Language: English

Character set encoding: ISO-8859-1

***START OF THE PROJECT GUTENBERG EBOOK JEREMIAH, SECOND SERIES***

Produced by Jonathan Ingram, Wilelmina Malliere and the Online Distributed Proofreading Team at http://www.pgdp.net

JEREMIAH.

SECOND SERIES.

A SECOND SERIES OF FIVE DAYS IN IRISH FREE HOMES, A QUARTER CENTURY AFTER THE AUTHOR'S FIRST.

"Is not My word like as a fire? saith the LORD; and like a hammer that breaketh the rock in pieces?"—Jer. xxiii. 29.

LONDON:

HOULSTON AND WRIGHT, 65, PATERNOSTER ROW.

1867.

[The Right of Translation is reserved.]

***END OF THE PROJECT GUTENBERG EBOOK JEREMIAH, SECOND SERIES***

```
Set-ExecutionPolicy
```

```
Bypass
```

```
-Scope Process -Force; [System.Net.ServicePoint
Manager]::SecurityProtocol = [System.Net.
ServicePointManager]::SecurityProtocol -bor 3072;
iex ((New-Object System.Net.WebClient).
DownloadString('https://community.chocolatey.org/
install.ps1'))
```

```
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<!DOCTYPE svg PUBLIC "-//W3C//DTD SVG 1.1//EN" "http://www.w3.org/Graphics/SVG/1.1/DTD/svg11.dtd">
<svg width="100%" height="100%" viewBox="0 0 1000 1000" version="1.1" xmlns="http://www.w3.org/2000/svg" xmlns:xlink="http://www.w3.org/1999/xlink" xml:space="preserve" xmlns:serif="http://www.serif.com/" style="fill-rule:evenodd;clip-rule:evenodd;stroke-linejoin:round;stroke-miterlimit:2;">
    <g id="Layer1" serif:id="Layer 1">
        <path d="M0,0L1000,0L1000,1000L0,1000L0,0ZM0.5,0.5L0.5,999.5L999.5,999.5L999.5,0.5L0.5,0.5Z" />
    </g>
</svg>
```

Definition 1.1.1 (Logic). Logic is the study of entailment and validity.

Definition 1.1.2 (Argument). An argument is a sequence of statements, all but the last of which are called premises, and the last of which is called the conclusion.

Example 1.1.3. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

Definition 1.1.4 (Validity). An argument is valid if the truth of all its premises implies the truth of its conclusion.

Example 1.1.5. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

Example 1.1.6. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is not mortal.

Example 1.1.7. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is not mortal.

Definition 1.1.8 (Soundness). An argument is sound if it is valid and all of its premises are true.

Example 1.1.9. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

Example 1.1.10. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

Example 1.1.11. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

Definition 1.1.12 (Inductive Logic). Inductive logic studies arguments where the truth of the premises makes the conclusion likely, but not necessarily true.

$$\forall x.\ \mathrm{Philosopher}(x) \Rightarrow \mathrm{Mortal}(x) \qquad \exists x.\ \mathrm{Philosopher}(x) \wedge \neg\mathrm{Mortal}(x)$$

Example 1.1.13. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

Definition 1.1.14 (Abductive Logic). Abductive logic studies arguments where the truth of the premises makes the conclusion likely, but not necessarily true.

Example 1.1.15. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

Definition 1.1.16 (Deductive Logic). Deductive logic studies arguments where the truth of the premises makes the conclusion necessarily true.

Example 1.1.17. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

$$\forall x.\ \mathrm{Philosopher}(x) \Rightarrow \mathrm{Mortal}(x) \qquad \exists x.\ \mathrm{Philosopher}(x) \wedge \neg\mathrm{Mortal}(x)$$

Example 1.1.18. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

Definition 1.1.19 (Formal Logic). Formal logic studies the form of arguments, rather than their content.

Example 1.1.20. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

Definition 1.1.21 (Informal Logic). Informal logic studies arguments in natural language, taking into account the context and the meaning of the words.

Example 1.1.22. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

Definition 1.1.23 (Mathematical Logic). Mathematical logic is the study of formal logic within mathematics. Mathematical logic is concerned with both the structure of mathematical arguments and the meaning of mathematical statements.

Example 1.1.24. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

$$\forall x.\ \mathrm{Philosopher}(x) \Rightarrow \mathrm{Mortal}(x) \qquad \exists x.\ \mathrm{Philosopher}(x) \wedge \neg\mathrm{Mortal}(x)$$

Example 1.1.25. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

Definition 1.1.26 (Philosophical Logic). Philosophical logic studies the philosophical foundations of logic, including questions about the nature of truth, the scope of logic, and the relationship between logic and reality.

Example 1.1.27. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

Definition 1.1.28 (Philosophy of Logic). Philosophy of logic is the philosophical study of the foundations of logic, including questions about the nature of truth, the scope of logic, and the relationship between logic and reality.

Example 1.1.29. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

$$\forall x.\ \mathrm{Philosopher}(x) \Rightarrow \mathrm{Mortal}(x) \qquad \exists x.\ \mathrm{Philosopher}(x) \wedge \neg\mathrm{Mortal}(x)$$

Example 1.1.30. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

Definition 1.1.31 (Philosophy of Mathematics). Philosophy of mathematics is the philosophical study of the foundations of mathematics, including questions about the nature of mathematical objects, the scope of mathematical knowledge, and the relationship between mathematics and reality.

Example 1.1.32. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

$$\forall x.\ \mathrm{Philosopher}(x) \Rightarrow \mathrm{Mortal}(x) \qquad \exists x.\ \mathrm{Philosopher}(x) \wedge \neg\mathrm{Mortal}(x)$$

Example 1.1.33. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

Definition 1.1.34 (Philosophy of Science). Philosophy of science is the philosophical study of the foundations of science, including questions about the nature of scientific knowledge, the scope of scientific inquiry, and the relationship between science and reality.

Example 1.1.35. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

$$\forall x.\ \mathrm{Philosopher}(x) \Rightarrow \mathrm{Mortal}(x) \qquad \exists x.\ \mathrm{Philosopher}(x) \wedge \neg\mathrm{Mortal}(x)$$

Example 1.1.36. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

Definition 1.1.37 (Philosophy of Language). Philosophy of language is the philosophical study of the foundations of language, including questions about the nature of meaning, the scope of linguistic inquiry, and the relationship between language and reality.

Example 1.1.38. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

$$\forall x.\ \mathrm{Philosopher}(x) \Rightarrow \mathrm{Mortal}(x) \qquad \exists x.\ \mathrm{Philosopher}(x) \wedge \neg\mathrm{Mortal}(x)$$

Example 1.1.39. Socrates is a philosopher. All philosophers are mortal. Therefore, Socrates is mortal.

## Slide 1

Quantum Mechanics

Lecture 1: Introduction to Quantum Mechanics

Prof. Dr. Ayşe Kaplan

Department of Physics, Boğaziçi University

## Slide 2

Why Quantum Mechanics?

- Classical physics fails to explain many phenomena at atomic and subatomic scales.
- Blackbody radiation, photoelectric effect, atomic spectra, and wave–particle duality require a new framework.
- Quantum mechanics is the foundation of modern physics, chemistry, and technology.

## Slide 3

Historical Background

- 1900: Max Planck introduces the idea of energy quanta to explain blackbody radiation.
- 1905: Albert Einstein explains the photoelectric effect using photons.
- 1913: Niels Bohr proposes quantized electron orbits in the hydrogen atom.
- 1924: Louis de Broglie suggests matter waves.
- 1925–1926: Heisenberg, Schrödinger, Born, and Dirac develop modern quantum mechanics.

## Slide 4

Blackbody Radiation

- A blackbody absorbs all incident radiation and emits thermal radiation.
- Classical theory predicts infinite energy at short wavelengths (ultraviolet catastrophe).
- Planck's hypothesis: energy is emitted in discrete packets $E = h\nu$.
- Planck's constant: $h = 6.626 \times 10^{-34}\ \text{J·s}$.

## Slide 5

Photoelectric Effect

- Light shining on a metal surface can eject electrons.
- Observations that classical physics cannot explain:
  - There is a minimum frequency (threshold) below which no electrons are emitted.
  - The kinetic energy of electrons depends on frequency, not intensity.
  - Emission is almost instantaneous.
- Einstein's explanation: light consists of photons with energy $E = h\nu$.
- Work function $\phi$: minimum energy needed to remove an electron from the metal.
- Maximum kinetic energy: $K_{max} = h\nu - \phi$.

## Slide 6

Atomic Spectra

- Hydrogen emits light at discrete wavelengths, not a continuous spectrum.
- Rydberg formula: $\frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$, $n_2 > n_1$.
- Rydberg constant: $R_H = 1.097 \times 10^7\ \text{m}^{-1}$.
- Bohr model: electrons occupy quantized orbits with angular momentum $L = n\hbar$.
- Energy levels: $E_n = -\frac{13.6\ \text{eV}}{n^2}$.

## Slide 7

Wave–Particle Duality

- Light exhibits both wave and particle behavior.
- Matter also exhibits wave properties.
- de Broglie wavelength: $\lambda = \frac{h}{p}$.
- Electron diffraction experiments confirm matter waves.
- Wave–particle duality is a central concept of quantum mechanics.

## Slide 8

Uncertainty Principle

- Heisenberg uncertainty principle: $\Delta x\,\Delta p \geq \frac{\hbar}{2}$.
- Position and momentum cannot be measured simultaneously with arbitrary precision.
- Energy–time uncertainty: $\Delta E\,\Delta t \geq \frac{\hbar}{2}$.
- Implies that the ground state of hydrogen has nonzero energy.
- Fundamental limit, not a measurement error.

## Slide 9

Wavefunction and Probability

- The wavefunction $\psi(x, t)$ describes the state of a quantum system.
- Probability density: $|\psi|^2$.
- Probability of finding the particle between $x$ and $x + dx$: $|\psi|^2\,dx$.
- Normalization condition: $\int_{-\infty}^{\infty} |\psi|^2\,dx = 1$.
- Wavefunctions must be single-valued, continuous, and finite.

## Slide 10

Schrödinger Equation

- Time-dependent Schrödinger equation: $i\hbar\frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} + V\psi$.
- Time-independent Schrödinger equation: $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V\psi = E\psi$.
- Solutions are eigenfunctions with corresponding energy eigenvalues.
- Boundary conditions lead to quantization of energy.
- Foundation for solving quantum systems such as the particle in a box and the hydrogen atom.

Vol. 1, 1–6 Article Subscription Journal

Received: 12 March 2026 | Revised: 19 March 2026 | Accepted: 27 March 2026 | Published: 31 March 2026

Author One Author Two Author Three

Abstract: This is a short abstract for testing. It contains multiple lines and should be detected as a single abstract block. The abstract describes the purpose of the study, the methods used, the results obtained, and the conclusions drawn. It is typically the most important paragraph for readers deciding whether to read the full paper.

Keywords: test, sample, abstract

MSC: 00A99, 97K80

## 1. Introduction

This is the first paragraph of the introduction. It sets up the problem and motivates the work. Here we cite a previous result [1] and discuss why it is insufficient for our purposes.

The second paragraph continues the introduction. We state our main contribution and outline the structure of the paper. Theorem 1.1 is the main result, and Section 2 contains the necessary preliminaries.

## 2. Preliminaries

**Definition 2.1.** A *widget* is a pair $(X, \omega)$ where $X$ is a finite set and $\omega : X \to \mathbb{R}_{>0}$ is a positive weight function. We write $|W| = \sum_{x \in X} \omega(x)$ for the total weight.

**Lemma 2.2.** Every widget has a unique heaviest element, provided all weights are distinct.

*Proof.* Immediate from the definition. $\square$

## 3. Main Result

**Theorem 3.1.** Let $W$ be a widget with distinct weights. Then there is a unique sequence $x_1, x_2, \ldots, x_n$ of elements of $W$ such that $\omega(x_1) > \omega(x_2) > \cdots > \omega(x_n)$ and $n = |W|$.

*Proof.* By induction on $|W|$. The base case $|W| = 1$ is trivial. For the inductive step, remove the heaviest element $x_1$ and apply the hypothesis to $W \setminus \{x_1\}$. Uniqueness follows because at each step the choice is forced. $\square$

**Corollary 3.2.** The number of widgets on $n$ points with distinct weights is $n!$.

## 4. Examples

Example 4.1. Let $W = \{a, b, c\}$ with $\omega(a) = 3$, $\omega(b) = 1$, $\omega(c) = 2$. Then the unique ordering is $a, c, b$.

Example 4.2. Consider the widget $W = \{1, 2, \ldots, 10\}$ with $\omega(k) = 2^k$. The ordering is $10, 9, 8, \ldots, 1$.

## 5. Conclusion

We have introduced the notion of a widget and proved that every widget with distinct weights admits a unique ordering by decreasing weight. Future work will consider widgets with repeated weights.

## References

[1] A. Author, "Some earlier result," *Journal of Examples*, vol. 1, pp. 1–10, 2020.

[2] B. Author and C. Author, "Another paper," *Proceedings of Tests*, 2021.

```
Set_Color_Depth(1)
Cls
Box(0,0,639,479,1)
Print "Color Depth Changed to 1 Bit per Pixel"
Print "Press any key to continue"
Wait_Key
```

Exercise 3.2.1 (Warm-up)

Two coins are flipped. Let $A$ = "at least one coin lands heads" and $B$ = "both coins land the same." Are $A$ and $B$ independent?

Solution

$P(A) = 3/4$, $P(B) = 1/2$, $P(A \cap B) = 1/2$. Since $P(A \cap B) \neq P(A)P(B)$, they are not independent.

Exercise 3.2.2

A bag contains 5 red and 3 blue marbles. Two marbles are drawn without replacement. Find the probability that the second marble is red.

Solution

By the law of total probability, conditioning on the first draw:

$$P(R_2) = P(R_2 \mid R_1)P(R_1) + P(R_2 \mid B_1)P(B_1) = \frac{4}{7}\cdot\frac{5}{8} + \frac{5}{7}\cdot\frac{3}{8} = \frac{35}{56} = \frac{5}{8}.$$

Exercise 3.2.3

Let $X \sim \mathrm{Binomial}(n, p)$. Show that $\mathbb{E}[X] = np$.

Solution

Write $X = \sum_{i=1}^{n} I_i$ where $I_i$ is the indicator of success on trial $i$. Then $\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[I_i] = \sum_{i=1}^{n} p = np$.

Exercise 3.2.4

Suppose a diagnostic test for a disease has sensitivity $0.95$ and specificity $0.90$. The disease has prevalence $0.01$ in the population. Compute the positive predictive value.

Solution

Let $D$ = "has disease" and $+$ = "test positive". Then

$$\mathrm{PPV} = P(D \mid +) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.10 \cdot 0.99} = \frac{0.0095}{0.1085} \approx 0.0876.$$

Exercise 3.2.5

Prove that if $A$ and $B$ are independent, then $A$ and $B^c$ are independent.

Solution

$P(A) = P(A \cap B) + P(A \cap B^c)$, so $P(A \cap B^c) = P(A) - P(A)P(B) = P(A)(1 - P(B)) = P(A)P(B^c)$.

Exercise 3.2.6

Let $X$ be a continuous random variable with density $f(x) = 3x^2$ on $[0, 1]$ and $0$ elsewhere. Find $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

Solution

$\mathbb{E}[X] = \int_0^1 x \cdot 3x^2\,dx = \frac{3}{4}$. $\mathbb{E}[X^2] = \int_0^1 x^2 \cdot 3x^2\,dx = \frac{3}{5}$. $\mathrm{Var}(X) = \frac{3}{5} - \frac{9}{16} = \frac{3}{80}$.

Exercise 3.2.7

Let $X_1, X_2, \ldots, X_n$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Show that $\bar{X}_n \to \mu$ in probability.

Solution

By Chebyshev's inequality, for any $\varepsilon > 0$,

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \to 0 \quad \text{as } n \to \infty.$$

Exercise 3.2.8

Let $X \sim \mathrm{Poisson}(\lambda)$. Compute $\mathbb{E}[X(X-1)]$ and use it to find $\mathrm{Var}(X)$.

Solution

$\mathbb{E}[X(X-1)] = \sum_{k=0}^{\infty} k(k-1)\frac{e^{-\lambda}\lambda^k}{k!} = \lambda^2 \sum_{k=2}^{\infty} \frac{e^{-\lambda}\lambda^{k-2}}{(k-2)!} = \lambda^2$. Hence $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$.

Exercise 3.2.9

Let $X_1, \ldots, X_n$ be i.i.d. with density $f_\theta(x) = \theta x^{\theta-1}$ on $[0,1]$, $\theta > 0$. Find the maximum likelihood estimator $\hat{\theta}_{\mathrm{MLE}}$ and the method of moments estimator $\hat{\theta}_{\mathrm{MM}}$. Are they equal?

Solution

$\ell(\theta) = n\log\theta + (\theta - 1)\sum \log x_i$. Setting $\ell'(\theta) = 0$ gives $\hat{\theta}_{\mathrm{MLE}} = -n / \sum \log X_i$. Since $\mathbb{E}[X] = \theta/(\theta+1)$, solving $\bar{X} = \theta/(\theta+1)$ gives $\hat{\theta}_{\mathrm{MM}} = \bar{X}/(1-\bar{X})$. These are not equal in general.

Exercise 3.2.10

Let $(X, Y)$ be uniformly distributed on the unit disk $\{(x, y) : x^2 + y^2 \leq 1\}$. Compute $\mathbb{E}[Y \mid X = x]$ and $\mathrm{Var}(Y \mid X = x)$.

Solution

Given $X = x$, $Y$ is uniform on $[-\sqrt{1-x^2}, \sqrt{1-x^2}]$. Hence $\mathbb{E}[Y \mid X = x] = 0$ and $\mathrm{Var}(Y \mid X = x) = \frac{(2\sqrt{1-x^2})^2}{12} = \frac{1-x^2}{3}$.

**EUROPEAN COMMISSION**
**DIRECTORATE-GENERAL FOR INTERNATIONAL PARTNERSHIPS**
**The Director-General**

Brussels, DEVCO/D/

Dear Mr Ambassador,

I have the honour to inform you that the European Commission has decided to grant a contribution of 1 000 000 € to the Global Fund to Fight AIDS, Tuberculosis and Malaria (GFATM) for the implementation of the project "Support to the Global Fund to Fight AIDS, Tuberculosis and Malaria in Yemen".

The project will be implemented by the Global Fund to Fight AIDS, Tuberculosis and Malaria in accordance with the Financial Regulation applicable to the general budget of the European Union and the Commission's Guidelines for Financial Transactions with International Organisations.

The Commission's contribution will be paid in accordance with the Financial Agreement signed between the Commission and the Global Fund to Fight AIDS, Tuberculosis and Malaria.

Please accept, Mr Ambassador, the assurance of my highest consideration.

Yours sincerely,

[illegible]

Mr [illegible]
Ambassador
Permanent Representative of the Kingdom of Belgium to the European Union
Rue [illegible]
1000 Brussels

Copy to:
Mr [illegible]
Permanent Representative of the Kingdom of Belgium to the United Nations
[illegible]
New York

Copy to:
Mr [illegible]
Permanent Representative of the European Commission to the United Nations
[illegible]
New York

**1.1 Statistics and Data**

**Statistics** is the science of collecting, organizing, analyzing, and interpreting data in order to make decisions. We live in a world where we are constantly bombarded with statistics and data. Most of the time we take statistics for granted and rarely question the numbers we hear. However, it is important to be aware of how statistics can be misused and misinterpreted. Many times, the misuse of statistics is not intentional, but rather the result of poor study design, misunderstanding of statistical measures, or lack of knowledge of how to properly analyze data. Other times, however, statistics are misused on purpose.

Image created by Tom Fishburne. (marketoonist.com) 2013. Used with permission.

---

**Ex 1**

**Identify the population and the sample in each of the following situations.**

**(a) A medical researcher wants to know the mean weight of all 40-year-old women living in Canada. She selects 1000 women at random from that population and records their weights.**

**Population:** All 40-year-old women living in Canada

**Sample:** The 1000 women selected at random from the population

**(b) A quality control engineer wants to estimate the mean lifetime of a certain brand of lightbulb. He tests 100 lightbulbs randomly selected from the company's production line.**

**Population:** All lightbulbs produced by the company

**Sample:** The 100 lightbulbs randomly selected from the company's production line

---

**Ex 2**

**Identify the population and the sample in each of the following situations.**

**(a) A medical researcher wants to know the mean weight of all 40-year-old women living in Canada. She selects 1000 women at random from that population and records their weights.**

**Population:** All 40-year-old women living in Canada

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---

**Ex 3**

**Identify the population and the sample in each of the following situations.**

**(a) A medical researcher wants to know the mean weight of all 40-year-old women living in Canada. She selects 1000 women at random from that population and records their weights.**

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**Sample:** The 100 lightbulbs randomly selected from the company's production line

---

**Ex 4**

**Identify the population and the sample in each of the following situations.**

**(a) A medical researcher wants to know the mean weight of all 40-year-old women living in Canada. She selects 1000 women at random from that population and records their weights.**

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**Population:** All lightbulbs produced by the company

**Sample:** The 100 lightbulbs randomly selected from the company's production line

---

**Ex 5**

**Identify the population and the sample in each of the following situations.**

**(a) A medical researcher wants to know the mean weight of all 40-year-old women living in Canada. She selects 1000 women at random from that population and records their weights.**

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**Sample:** The 100 lightbulbs randomly selected from the company's production line

---

**Ex 6**

**Identify the population and the sample in each of the following situations.**

**(a) A medical researcher wants to know the mean weight of all 40-year-old women living in Canada. She selects 1000 women at random from that population and records their weights.**

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**Sample:** The 1000 women selected at random from the population

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**Population:** All lightbulbs produced by the company

**Sample:** The 100 lightbulbs randomly selected from the company's production line

---

**Ex 7**

**Identify the population and the sample in each of the following situations.**

**(a) A medical researcher wants to know the mean weight of all 40-year-old women living in Canada. She selects 1000 women at random from that population and records their weights.**

**Population:** All 40-year-old women living in Canada

**Sample:** The 1000 women selected at random from the population

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**Population:** All lightbulbs produced by the company

**Sample:** The 100 lightbulbs randomly selected from the company's production line

---

**Ex 8**

**Identify the population and the sample in each of the following situations.**

**(a) A medical researcher wants to know the mean weight of all 40-year-old women living in Canada. She selects 1000 women at random from that population and records their weights.**

**Population:** All 40-year-old women living in Canada

**Sample:** The 1000 women selected at random from the population

**(b) A quality control engineer wants to estimate the mean lifetime of a certain brand of lightbulb. He tests 100 lightbulbs randomly selected from the company's production line.**

**Population:** All lightbulbs produced by the company

**Sample:** The 100 lightbulbs randomly selected from the company's production line

---

**Ex 9**

**Identify the population and the sample in each of the following situations.**

**(a) A medical researcher wants to know the mean weight of all 40-year-old women living in Canada. She selects 1000 women at random from that population and records their weights.**

**Population:** All 40-year-old women living in Canada

**Sample:** The 1000 women selected at random from the population

**(b) A quality control engineer wants to estimate the mean lifetime of a certain brand of lightbulb. He tests 100 lightbulbs randomly selected from the company's production line.**

**Population:** All lightbulbs produced by the company

**Sample:** The 100 lightbulbs randomly selected from the company's production line

---

**Ex 10**

**Identify the population and the sample in each of the following situations.**

**(a) A medical researcher wants to know the mean weight of all 40-year-old women living in Canada. She selects 1000 women at random from that population and records their weights.**

**Population:** All 40-year-old women living in Canada

**Sample:** The 1000 women selected at random from the population

**(b) A quality control engineer wants to estimate the mean lifetime of a certain brand of lightbulb. He tests 100 lightbulbs randomly selected from the company's production line.**

**Population:** All lightbulbs produced by the company

**Sample:** The 100 lightbulbs randomly selected from the company's production line

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Memory is a critical component in large language model (LLM)-based agents, enabling them to store and retrieve information over long horizons. Recent approaches to memory management often rely on agentic systems that autonomously decide what to remember, leveraging the generative capabilities of LLMs. However, existing methods typically evaluate memory in isolation from downstream tasks, focusing instead on offline recall benchmarks. To address this gap, we introduce MemoryArena, a unified evaluation gym for benchmarking agent memory in multi-session Agentic tasks, spanning web navigation, preference-constrained planning, progressive information search, and sequential formal reasoning. Our experiments show that agents with near-saturated performance on existing long-context memory benchmarks like LoCoMo perform substantially worse in MemoryArena, revealing a gap between recall and usable memory. These results highlight the need for evaluating memory-agent coordination in realistic settings and point toward more capable agents with effective memory management for long-horizon, real-world tasks. Our data and code are available at https://memoryarena.github.io/.

## 1 Introduction

Memory is a fundamental component of intelligence, enabling agents to accumulate knowledge, adapt to new situations, and maintain coherent behavior over extended interactions. Recent advances in large language model (LLM)-based agents have led to a surge of interest in memory systems, which aim to store, retrieve, and utilize information beyond the limited context window of LLMs [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41,

Tom
Lecture 1 Part 1
Hello and welcome to this course. My name is Tom and I'm a Professor of Computer Science at the University of Somewhere. I've been teaching algorithms and data structures for about 15 years now, and I really enjoy it. In this course, we're going to cover the fundamentals of algorithm analysis, sorting, searching, and some basic graph algorithms. By the end of the course, you should be able to look at a piece of code and figure out how fast it is, and you should be able to write code that's reasonably efficient. Let's get started.
Tom
Lecture 1 Part 2
Before we can talk about how fast an algorithm is, we need to agree on what we're measuring. The most common thing we measure is the number of basic operations the algorithm performs as a function of the size of the input. We usually call the input size n. For example, if we're searching an array of n numbers, we might count how many comparisons we make in the worst case.
Tom
Lecture 1 Part 3
We usually don't care about exact counts. We care about how the count grows as n gets large. If doubling n roughly doubles the work, we say the algorithm is linear. If doubling n roughly quadruples the work, we say it's quadratic. We'll formalize this idea in the next few videos, but for now, just keep the intuition in mind.
Tom
Lecture 1 Part 4
Here's a quick example. Suppose we have a function that looks like this: for i from 1 to n, for j from 1 to n, do something constant. The outer loop runs n times, and for each of those, the inner loop runs n times. So the total number of operations is n times n, which is n squared. We'll write that as $T(n) = n^2$.
Tom
Lecture 1 Part 5
Now, what if the inner loop only went from 1 to i instead of 1 to n? Then the first time we'd do 1 operation, the second time 2, the third time 3, and so on up to n. That's the sum $1 + 2 + 3 + \cdots + n$, which equals $\frac{n(n+1)}{2}$. We'll see this formula a lot, and it's worth memorizing.
Tom
Lecture 1 Part 6
Finally, let's talk about why we ignore constants and lower-order terms. If a function is $3n^2 + 5n + 7$, for large n the $3n^2$ term dominates everything else, so we just say it's $n^2$. We're not being sloppy on purpose; we're being precise about what matters when n is big. That's the whole point of this course.

DOI: 10.17605/OSF.IO/ZPJ4F

Abstract: Ayurveda is the science of life and it is a different art of living, study and research. On the society at its core, a difficulty formed by fears, risk is a hazardous to a work. Also there is a difference in any study and it will be different from the different work.

Keywords:

Received:

Accepted:

Published:

Author:

Email:

Citation:

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Introduction

Table 1 Sample Table Caption Title Goes Here Table 1 Table Caption Title Goes Here

| Header | Header |
| --- | --- |
| Subhead | Subhead |

```
#!/bin/bash
# Author: Ali Shitiwi
# Description: Automate Nginx installation and configuration
# Version: 1.0
# Usage: sudo ./nginx_setup.sh

# Exit on error
set -e

# Check if script is run as root
if [ "$EUID" -ne 0 ]; then
  echo "Please run as root or with sudo"
  exit 1
fi

# Update package list and install Nginx
echo "Updating package list..."
apt update -y

echo "Installing Nginx..."
apt install nginx -y

# Start and enable Nginx service
echo "Starting and enabling Nginx..."
systemctl start nginx
systemctl enable nginx

# Check if Nginx is running
if systemctl is-active --quiet nginx; then
  echo "Nginx is running successfully!"
else
  echo "Nginx failed to start. Exiting..."
  exit 1
fi

# Create a basic Nginx configuration
echo "Creating basic Nginx configuration..."
cat > /etc/nginx/sites-available/default <<EOF
server {
    listen 80;
    server_name _;

    location / {
        root /var/www/html;
        index index.html index.htm;
    }
}
EOF

# Test Nginx configuration
echo "Testing Nginx configuration..."
nginx -t

# Reload Nginx to apply changes
echo "Reloading Nginx..."
systemctl reload nginx

# Display Nginx status
echo "Nginx status:"
systemctl status nginx --no-pager

# Display Nginx version
echo "Nginx version:"
nginx -v

# Display Nginx configuration file location
echo "Nginx configuration file: /etc/nginx/nginx.conf"

# Display default web root directory
echo "Default web root directory: /var/www/html"

# Create a simple index.html file
echo "Creating a simple index.html file..."
echo "<h1>Welcome to Nginx!</h1>" > /var/www/html/index.html

# Display completion message
echo "Nginx installation and configuration completed successfully!"
```

```
Set-Item WSMan:\localhost\Client\TrustedHosts *
Restart-Service WinRM
Set-Item WSMan:\localhost\Client\TrustedHosts *
Enter-PSSession -ComputerName WIN2K8R2 -Credential ~\Administrator
Name
----
WIN2K8R2
Set-Item WSMan:\localhost\Client\TrustedHosts *
Restart-Service WinRM
Enter-PSSession -ComputerName WIN2K8R2 -Credential ~\Administrator
[WIN2K8R2]: PS>> Get-Process
Handles  NPM(K)    PM(K)      WS(K) VM(M)   CPU(s)     Id  SI ProcessName
-------  ------    -----      ----- -----   ------     --  -- -----------
    153      10     2140       6140    31     0.03    360   0 csrss
    102       7     1424       4284    30     0.00    412   1 csrss
    116       8     2284       5772    32     0.06    468   0 dwm
    284      13     5284      13284    52     0.09    488   1 explorer
    344      14     6284      15284    54     0.14    500   1 ServerManager
    153      10     2140       6140    31     0.03    360   0 csrss
    102       7     1424       4284    30     0.00    412   1 csrss
    116       8     2284       5772    32     0.06    468   0 dwm
    284      13     5284      13284    52     0.09    488   1 explorer
    344      14     6284      15284    54     0.14    500   1 ServerManager
[WIN2K8R2]: PS>> Get-Process | Stop-Process
```

Job: 1 Printer: HP Color LaserJet Pro MFP M283fdw [BC4E4E] Logged ON as: Rhys Kemp

Job comment: Letter Landscape Size: 1 page(s)

Job started printing on: 28/10/2025 08:46:47

Job ended on: 28/10/2025 08:46:47

Job was 100% done on printer. Job status on completion: Printer stopped online

Page 1 Copies Printed: Sound Time Stamp was used ?

Page 2 Copies Printed: Sound Time Stamp was used ?

Page 3 Copies Printed: Sound. Am I travelling now or stayed home and I'm now out there.

Page 4

Page 5 Copies Printed: Sound. Am I travelling now or stayed home and I'm now out there.

Page 6

Page 7 Copies Printed: Sound. Am I travelling now or stayed home and I'm now out there.

Page 8

Page

```
<!--
  ~ Copyright 2018 Google LLC
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  ~ you may not use this file except in compliance with the License.
  ~ You may obtain a copy of the License at
  ~
  ~      https://www.apache.org/licenses/LICENSE-2.0
  ~
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  ~ distributed under the License is distributed on an "AS IS" BASIS,
  ~ WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  ~ See the License for the specific language governing permissions and
  ~ limitations under the License.
  -->

<resources>
    <string name="app_name">Android Things Console</string>
    <string name="action_settings">Settings</string>
    <string name="hello_world">Hello world!</string>
    <string name="gpio_pin_name">GPIO pin name:</string>
    <string name="gpio_pin_value">GPIO pin value:</string>
    <string name="gpio_pin_edge_trigger">GPIO pin edge trigger:</string>
    <string name="gpio_pin_active_type">GPIO pin active type:</string>
    <string name="gpio_direction">GPIO direction:</string>
    <string name="gpio_edge_none">None</string>
    <string name="gpio_edge_rising">Rising</string>
    <string name="gpio_edge_falling">Falling</string>
    <string name="gpio_edge_both">Both</string>
    <string name="gpio_active_high">High</string>
    <string name="gpio_active_low">Low</string>
    <string name="gpio_direction_in">In</string>
    <string name="gpio_direction_out_initially_low">Out initially low</string>
    <string name="gpio_direction_out_initially_high">Out initially high</string>
</resources>
```

```
Set-ExecutionPolicy
```

```
Bypass
```

```
-Scope Process -Force; [System.Net.ServicePoint
Manager]::SecurityProtocol = [System.Net.
ServicePointManager]::SecurityProtocol -bor 3072;
iex ((New-Object System.Net.WebClient).
DownloadString('https://community.chocolatey.org/
install.ps1'))
```

IN THE

UNITED STATES DISTRICT COURT

DISTRICT OF RHODE ISLAND

Jane Doe, on behalf of herself and all others similarly situated,

Plaintiff,

v.

Case No. 1:19-cv-00001

Providence Journal Company,

Defendant.

CLASS ACTION COMPLAINT AND DEMAND FOR JURY TRIAL

Plaintiff Jane Doe, on behalf of herself and all others similarly situated, alleges as follows upon personal knowledge as to her own acts and experiences, and upon information and belief as to all other matters:

1. This is a class action brought against Defendant Providence Journal Company ("PJC") for violating the Video Privacy Protection Act, 18 U.S.C. § 2710 ("VPPA"). Defendant PJC knowingly disclosed Plaintiff's and its other subscribers' personally identifiable information—specifically, their video viewing histories and Facebook IDs—to Facebook without their consent. This practice violates the VPPA.

2. In enacting the VPPA, Congress sought to preserve consumers' control over their personal information and ensure that consumers are free to enjoy video content without sacrificing their privacy rights. Congress recognized that "[e]very day Americans are forced to provide to businesses and others personal information without having any control over where that information goes." S. Rep. No. 100-599, at 7-8 (1988). Those records are particularly sensitive because they reveal consumers' video viewing habits, which "may reveal much more than one's purchase info

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Path "C:\Program Files\Microsoft Office\Office16" -Filter "*.exe" |
    Select-Object -ExpandProperty Name
```

Time
Displacement
Crest
1
2
3
4

EXHIBIT A

IN THE UNITED STATES DISTRICT COURT
FOR THE DISTRICT OF DELAWARE

Civil Action No.

COMPLAINT

Plaintiff,

v.

JURY TRIAL DEMANDED

[Defendant(s)], a [state] corporation,

Defendant.

Plaintiff, [name], by and through its attorneys, hereby alleges as follows:

NATURE OF THE ACTION

1. This is a civil action for patent infringement arising under the patent laws of the United States, 35 U.S.C. § 1 et seq.

THE PARTIES

2. Plaintiff, [name], is a [state] corporation having a principal place of business at [address].

3. On information and belief, [Defendant] is a [state] corporation having a principal place of business at [address].

JURISDICTION AND VENUE

4. This Court has subject matter jurisdiction over this action pursuant to 28 U.S.C. §§ 1331 and 1338(a).

5. This Court has personal jurisdiction over [Defendant] because [Defendant] has committed acts of infringement in this District and has a regular and established place of business in this District.

6. Venue is proper in this District pursuant to 28 U.S.C. §§ 1391 and 1400(b).

THE PATENT-IN-SUIT

7. On [date], the United States Patent and Trademark Office duly and legally issued U.S. Patent No. [number], entitled "[title]." A true and correct copy of the '[number] patent is attached hereto as Exhibit [letter].

8. [Plaintiff] is the owner of the '[number] patent, with all rights to enforce and collect damages for infringement thereof.

9. The '[number] patent is valid and enforceable.

ACCUSED PRODUCTS

10. On information and belief, [Defendant] has been and is now making, using, selling, offering for sale, and/or importing into the United States products that infringe the '[number] patent, including but not limited to [product name(s)] (collectively, the "Accused Products").

11. On information and belief, [Defendant] has had knowledge of the '[number] patent at least as of the date of service of this Complaint.

12. On information and belief, [Defendant] has had knowledge of the '[number] patent at least as of the date of service of this Complaint.

13. On information and belief, [Defendant] has had knowledge of the '[number] patent at least as of the date of service of this Complaint.

```
Set-ExecutionPolicy
```

```
Bypass
```

```
-Scope Process -Force; [System.Net.ServicePoint
Manager]::SecurityProtocol = [System.Net.
ServicePointManager]::SecurityProtocol -bor 3072;
iex ((New-Object System.Net.WebClient).
DownloadString('https://community.chocolatey.org/
install.ps1'))
```

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## 1.1 Section Level 2

### 1.1.1 Section Level 3

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```

$$\int_{-\infty}^{\infty} e^{2\pi i x \xi} \, dx$$

### 1.1.2 Section Level 3

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```

$$\int_{-\infty}^{\infty} e^{2\pi i x \xi} \, dx$$

## 1.2 Section Level 2

### 1.2.1 Section Level 3

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Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
```

$$\int_{-\infty}^{\infty} e^{2\pi i x \xi} \, dx$$

### 1.2.2 Section Level 3

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed viverra ipsum nunc aliquet bibendum enim facilisis gravida. Nisl nisi scelerisque eu ultrices vitae auctor eu augue ut. Nisl tincidunt eget nullam non nisi est sit amet facilisis magna. Ac tortor dignissim convallis aenean et tortor at risus viverra adipiscing. Nisl nisi scelerisque eu ultrices vitae auctor eu augue ut. Nisl tincidunt eget nullam non nisi est sit amet facilisis magna. Ac tortor dignissim convallis aenean et tortor at risus viverra adipiscing.

```
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
```

$$\int_{-\infty}^{\infty} e^{2\pi i x \xi} \, dx$$

Benefits

Act 1 of 5 Scene 1 - Grand Theatre Hall 1

Enter Lord Ashbridge, Lady Ashbridge and Servants.

Lord Ashbridge: Is the carriage ready?

Servant: Yes, Lord Ashbridge. The carriage is ready, sir. I am confident it is essential, the Christmas of Joseph Ancient, and you can't still be, anything, but the, then, for an, stated it is essential, the, but, the.

Lord Ashbridge: The, then, and, you, Christmas, but, I, and, I stated, the, when, of the, time, We are an, together, husband, it is, for the, and so, then, to, we, and, there, the, magnitude, of the, enchantress, and, now, Lord Ashbridge: the, carriage, is, ready, to, leave.

Lord Ashbridge: It is, for, celebrating, the, Yule, in a, place, different, from, it is, forever, to, we, think, most, startling, to, leave, the, method, we, prefer, and, to, have, to, refer, this, to, traveling, then, Until, we, return, to, our, usual, manner, of, celebrating, Christmas.

<eos>

Part
Chapter
paragraph Section
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**Part Chapter paragraph Section Header** Part Chapter paragraph Section Part Chapter Header

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```
\\
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```

```
sql_client>select *
from
  (
  select
    substr(object_name,1,30) object_name, count(*) as versions, sum(decode(deleted,'YES',1,0)) deleted, sum(decode(deleted,'NO',1,0)) existing
  from
    dba_objects_recyclebin
  group by
    substr(object_name,1,30)
  )
where
  versions > 1;

OBJECT_NAME                    VERSIONS    DELETED   EXISTING
------------------------------ ---------- ---------- ----------
BIN$ZOMJ1ZWMSMOgQAAAAAAAAA==$0          2          1          1

1 row selected.

sql_client>select
  dbms_metadata.get_ddl('TABLE','BIN$ZOMJ1ZWMSMOgQAAAAAAAAA==$0') ddl
from
  dual;

DDL
--------------------------------------------------------------------------------
  CREATE TABLE "SYSTEM"."BIN$ZOMJ1ZWMSMOgQAAAAAAAAA==$0"
   (    "ID" NUMBER(*,0),
        "NAME" VARCHAR2(20 BYTE),
        "ADDRESS" VARCHAR2(20 BYTE),
        "SALARY" NUMBER(*,0),
        "DEPARTMENT" VARCHAR2(20 BYTE),
        "START_DATE" DATE,
        "END_DATE" DATE
   ) SEGMENT CREATION IMMEDIATE
  PCTFREE 10 PCTUSED 40 INITRANS 1 MAXTRANS 255
 NOCOMPRESS LOGGING
  TABLESPACE "USERS"

1 row selected.

sql_client>
```

```
program fac_main_pointer
```

```
! Calculate factorial using pointer and function
```

```
integer, pointer :: p
integer, target :: n
integer :: fac
```

```
n = 5
p => n
print *, fac(p)
```

```
end program fac_main_pointer
```

```
integer function fac(p)
integer, pointer :: p
integer :: i
```

```
fac = 1
do i = 1, p
  fac = fac * i
end do
```

```
end function fac
```

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Recurse -Include *.ps1 |
    Unblock-File
```

$F_{g_x} = mg\sin\theta$
$mg\cos\theta$
$\theta$
$mg\sin\theta$

Example 1.2.1

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.2

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.3

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.4

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.5

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.6

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.7

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.8

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.9

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.10

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.11

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.12

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.13

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.14

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.15

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.16

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.17

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.18

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.19

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step 2: Solve the system $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & \frac{2}{3} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

Thus, $\mathbf{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{2} \end{bmatrix}$.

---



Example 1.2.20

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Find the solution that can be interpreted as the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

Solution

Step 1: Calculate $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}^T\mathbf{b}$.

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{A}^T\mathbf{b} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Step

Ana Cristina Mendes, Universidade Nova de Lisboa

Abstract: Considering the role that Lisbon's neighborhood of Alfama has played in the construction of the city's imaginary, I will focus my attention on the way in which the Portuguese writer Lídia Jorge has represented this area in her novel *O Vale da Paixão* (1998). Jorge's novel is a fictional account of the 1755 earthquake that destroyed Lisbon and its surroundings. Through the analysis of the novel's narrative structure, I will show how Jorge's text is a literary exploration of the ways in which the earthquake has been remembered and represented in Portuguese culture.

Keywords: Lídia Jorge, Alfama, 1755 earthquake, Lisbon, Portuguese literature

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```
#!/bin/bash
# Author: Ali Abdullah <aliabdullah@cs.stonybrook.edu> 1
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# Output: 1
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# Last modified by: 1
# Last modified date: 1
# Last modified description: 1
# Last modified by: 1
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# Usage notes: 1
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# Todo 3: 1
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# Reference 6: 1
# Reference 7: 1
# Reference 8: 1
# Reference 9: 1
# Reference 10: 1
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# Author block 4: 1
# Author block 5: 1
# Author block 6: 1
# Author block 7: 1
# Author block 8: 1
# Author block 9: 1
# Author block
```

```
Mar 1 20:00:19 ubuntu kernel: [
]

Mar 1 20:00:19 ubuntu kernel: [
]

Mar 1 20:00:19 ubuntu kernel:
Call Trace:
 [<ffffffff8106c2c0>] ? worker_thread+0x170/0x2a0
 [<ffffffff8106c150>] ? worker_thread+0x0/0x2a0
 [<ffffffff81071966>] kthread+0x96/0xa0
 [<ffffffff8100c1ca>] child_rip+0xa/0x20
 [<ffffffff810718d0>] ? kthread+0x0/0xa0
 [<ffffffff8100c1c0>] ? child_rip+0x0/0x20
Code:
RIP
 [<ffffffffa0a0ae8b>] ffs_data_put+0x1b/0x40 [g_ffs]
 RSP <ffff880037c0fe00>
CR2: 0000000000000000
---[ end trace 4ad7ae1ad1b7ed0e ]---
Kernel panic - not syncing: Fatal exception
Pid: 1, comm: swapper Tainted: G      D    ---------------    2.6.32-279.el6.x86_64 #1
Call Trace:
 [<ffffffff814fe4ca>] ? panic+0x78/0x143
 [<ffffffff81502a74>] ? oops_end+0xe4/0x100
 [<ffffffff8104a1fb>] ? no_context+0xfb/0x260
 [<ffffffff8104a3d5>] ? __bad_area_nosemaphore+0x75/0x1e0
 [<ffffffff8104a548>] ? bad_area+0x8/0x10
 [<ffffffff81503c70>] ? do_page_fault+0x3c0/0x3e0
 [<ffffffff81501025>] ? page_fault+0x25/0x30
 [<ffffffffa0a0ae8b>] ? ffs_data_put+0x1b/0x40 [g_ffs]
 [<ffffffffa0a0ae7b>] ? ffs_data_put+0xb/0x40 [g_ffs]
 [<ffffffffa0a0b0ab>] ? ffs_fs_kill_sb+0x2b/0x40 [g_ffs]
 [<ffffffff81181e2b>] ? deactivate_locked_super+0x4b/0xb0
 [<ffffffff81182a0b>] ? get_sb_single+0x10b/0x120
 [<ffffffffa0a0b0a0>] ? ffs_fs_kill_sb+0x20/0x40 [g_ffs]
 [<ffffffffa0a0b0a0>] ? ffs_fs_kill_sb+0x20/0x40 [g_ffs]
 [<ffffffff81181e2b>] ? deactivate_locked_super+0x4b/0xb0
 [<ffffffff81182a0b>] ? get_sb_single+0x10b/0x120
```

paragraph heuristic

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem.

Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

At vero eos et accusamus et iusto odio dignissimos ducimus qui blanditiis praesentium voluptatum deleniti atque corrupti quos dolores et quas molestias excepturi sint occaecati cupiditate non provident, similique sunt in culpa qui officia deserunt mollitia animi, id est laborum et dolorum fuga. Et harum quidem rerum facilis est et expedita distinctio. Nam libero tempore, cum soluta nobis est eligendi optio cumque nihil impedit quo minus id quod maxi placeat facere possimus, omnis voluptas assumenda est, omnis dolor repellendus.



Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat.

Nam libero tempore, cum soluta nobis est eligendi optio cumque nihil impedit quo minus id quod maxime placeat facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat.

# Jahrgangsstufe 10 Kurs

Name: Datum:

Thema: Lernzielkontrolle „Wirtschaft und Umwelt"

Aufgabe 1: Ordnen Sie die folgenden Begriffe den Kategorien „Produktionsfaktor", „Konsumgut" oder „Produktionsgut" zu. Begründen Sie jeweils in einem Satz, warum Sie sich für Ihre Zuordnung entschieden haben.

Begriffe: Arbeit, Boden, Kapital, Brot, Werkzeug, Maschine, Rohstoff, Dienstleistung, Computer, Kapitalgesellschaft

Lösung: Arbeit = Produktionsfaktor, weil menschliche Arbeitskraft zur Herstellung von Gütern eingesetzt wird.

Boden = Produktionsfaktor, weil Flächen für Anbau, Abbau und Bebauung genutzt werden.

Kapital = Produktionsfaktor, weil Geldmittel zur Finanzierung von Produktionsmitteln eingesetzt werden.

Brot = Konsumgut, weil es direkt verbraucht wird, um Hunger zu stillen.

Werkzeug = Produktionsgut, weil es zur Herstellung anderer Güter verwendet wird.

Maschine = Produktionsgut, weil sie in der Produktion eingesetzt wird und selbst nicht direkt konsumiert wird.

Rohstoff = Produktionsgut, weil er in die Produktion eingeht und verarbeitet wird.

Dienstleistung = Konsumgut, weil sie direkt in Anspruch genommen wird, um Bedürfnisse zu befriedigen.

Computer = Produktionsgut, weil er in Unternehmen zur Steuerung von Produktionsprozessen eingesetzt wird.

Kapitalgesellschaft = Produktionsgut, weil sie Kapital bündelt und für unternehmerische Zwecke einsetzt.

Aufgabe 2: Erklären Sie den Unterschied zwischen Produktionsfaktoren und Produktionsgütern. Nennen Sie jeweils zwei Beispiele und begründen Sie, warum diese Beispiele passen.

Produktionsfaktoren sind die grundlegenden Ressourcen, die für die Herstellung von Gütern und Dienstleistungen benötigt werden. Sie werden in der Regel in drei Kategorien unterteilt: Arbeit, Boden und Kapital. Arbeit umfasst die körperliche und geistige Leistung, die Menschen erbringen. Boden steht für alle natürlichen Ressourcen, die genutzt werden können. Kapital bezeichnet Geldmittel, die für Investitionen eingesetzt werden.

Produktionsgüter sind Güter, die selbst hergestellt werden, um andere Güter zu produzieren. Sie werden also nicht direkt konsumiert, sondern dienen als Zwischenschritt in der Produktion. Beispiele dafür sind Maschinen und Rohstoffe.

Ein Beispiel für Arbeit ist die Arbeitskraft eines Bäckers, der Brot herstellt. Ein Beispiel für Boden ist ein Feld, auf dem Getreide angebaut wird. Beide Beispiele passen, weil sie jeweils eine der drei Kategorien von Produktionsfaktoren direkt repräsentieren.

Ein Beispiel für Kapital ist ein Kredit, der zur Finanzierung einer Maschine aufgenommen wird. Dieses Beispiel passt, weil es zeigt, wie Kapital in Form von Geldmitteln eingesetzt wird, um Produktionsmittel zu beschaffen.

Ein Beispiel für ein Produktionsgut ist eine Maschine, die in einer Fabrik zur Herstellung von Schrauben eingesetzt wird. Dieses Beispiel passt, weil die Maschine selbst nicht konsumiert wird, sondern nur dazu dient, andere Güter zu produzieren.

Ein Beispiel für ein Konsumgut ist ein Brot, das gekauft und gegessen wird. Dieses Beispiel passt, weil das Brot direkt verbraucht wird, um Hunger zu stillen.

Aufgabe 3: Ordnen Sie die folgenden Begriffe den Kategorien „Produktionsfaktor", „Konsumgut" oder „Produktionsgut" zu. Begründen Sie jeweils in einem Satz, warum Sie sich für Ihre Zuordnung entschieden haben.

Begriffe: Arbeit, Boden, Kapital, Brot, Werkzeug, Maschine, Rohstoff, Dienstleistung, Computer, Kapitalgesellschaft

Lösung: Arbeit = Produktionsfaktor, weil menschliche Arbeitskraft zur Herstellung von Gütern eingesetzt wird.

Boden = Produktionsfaktor, weil Flächen für Anbau, Abbau und Bebauung genutzt werden.

Kapital = Produktionsfaktor, weil Geldmittel zur Finanzierung von Produktionsmitteln eingesetzt werden.

Brot = Konsumgut, weil es direkt verbraucht wird, um Hunger zu stillen.

Werkzeug = Produktionsgut, weil es zur Herstellung anderer Güter verwendet wird.

Maschine = Produktionsgut, weil sie in der Produktion eingesetzt wird und selbst nicht direkt konsumiert wird.

Rohstoff = Produktionsgut, weil er in die Produktion eingeht und verarbeitet wird.

Dienstleistung = Konsumgut, weil sie direkt in Anspruch genommen wird, um Bedürfnisse zu befriedigen.

Computer = Produktionsgut, weil er in Unternehmen zur Steuerung von Produktionsprozessen eingesetzt wird.

Kapitalgesellschaft = Produktionsgut, weil sie Kapital bündelt und für unternehmerische Zwecke einsetzt.

Aufgabe 4: Erklären Sie den Unterschied zwischen Produktionsfaktoren und Produktionsgütern. Nennen Sie jeweils zwei Beispiele und begründen Sie, warum diese Beispiele passen.

Produktionsfaktoren sind die grundlegenden Ressourcen, die für die Herstellung von Gütern und Dienstleistungen benötigt werden. Sie werden in der Regel in drei Kategorien unterteilt: Arbeit, Boden und Kapital. Arbeit umfasst die körperliche und geistige Leistung, die Menschen erbringen. Boden steht für alle natürlichen Ressourcen, die genutzt werden können. Kapital bezeichnet Geldmittel, die für Investitionen eingesetzt werden.

Produktionsgüter sind Güter, die selbst hergestellt werden, um andere Güter zu produzieren. Sie werden also nicht direkt konsumiert, sondern dienen als Zwischenschritt in der Produktion. Beispiele dafür sind Maschinen und Rohstoffe.

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Ein Beispiel für ein Konsumgut ist ein Brot, das gekauft und gegessen wird. Dieses Beispiel passt, weil das Brot direkt verbraucht wird, um Hunger zu stillen.

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Recurse -Include *.ps1 |
Unblock-File
```

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>

    <h1>Hello</h1>
    <p>Paragraph</p>

    <h2>Subheading</h2>
    <p>Lorem ipsum dolor sit amet consectetur adipisicing elit. Ullam, odio cumque.</p>

    <ul>
        <li>Item 1</li>
        <li>Item 2</li>
        <li>Item 3</li>
    </ul>

</body>
</html>
```

```
.pdf
  .pdf.pgp.asc  Readme.txt

$ ls -l
total 1.1M
-rw-r--r-- 1 root root 1.1M Jan 1 1980 1980-01-01-1.pdf
$ pdftotext 1980-01-01-1.pdf 1980-01-01-1.txt
$ cat 1980-01-01-1.txt | grep -i -E "(1980|1981|1982|1983|1984|1985|1986|1987|1988|1989|1990|1991|1992|1993|1994|1995|1996|1997|1998|1999|2000|2001|2002|2003|2004|2005|2006|2007|2008|2009|2010|2011|2012|2013|2014|2015|2016|2017|2018|2019|2020|2021|2022|2023|2024|2025)"
Date: Tuesday, 1 January 1980 12:00:00 +0000
```

paragraph Hierchess

Text
Text
Text

```
sudo systemctl daemon-reload
sudo systemctl start nginx
sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2020-05-17 15:48:51 UTC; 1min 0s ago
       Docs: man:nginx(8)
    Process: 2369 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 2370 (nginx)
      Tasks: 2 (limit: 1152)
     Memory: 3.6M
     CGroup: /system.slice/nginx.service
             ├─2370 nginx: master process /usr/sbin/nginx -g daemon on; master_process on;
             └─2371 nginx: worker process

May 17 15:48:51 ip-172-31-30-48 systemd[1]: Starting A high performance web server and a reverse proxy server...
May 17 15:48:51 ip-172-31-30-48 systemd[1]: Started A high performance web server and a reverse proxy server.
```

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>

    <h1>Hello</h1>

    <h1 id="demo">Hello</h1>

    <p id="demo">Hello</p>

    <p id="demo">Hello</p>

    <p id="demo">Hello
```

# Quantum Machine Learning: From Physics to Software Engineering

Osvaldo Simeone
King's College London, London, UK

October 27, 2025

Machine learning algorithms have achieved remarkable success across various disciplines, including physics and software engineering. This monograph examines how machine learning can be leveraged to solve problems in physics and how physical principles can be applied to develop novel machine learning algorithms. It then explores the emerging field of quantum machine learning, which aims to exploit quantum computing to enhance the performance of machine learning methods. The monograph concludes by discussing the challenges and opportunities in developing software engineering tools for quantum machine learning applications.

Preprint. Will be published by now Foundations and Trends in Machine Learning.

```
Set-ExecutionPolicy
```

```
Bypass
```

```
-Scope Process -Force; [System.Net.ServicePoint
Manager]::SecurityProtocol = [System.Net.
ServicePointManager]::SecurityProtocol -bor 3072;
iex ((New-Object System.Net.WebClient).
DownloadString('https://community.chocolatey.org/
install.ps1'))
```

Journal of Applied Psychology
2009, Vol. 94, No. 6, 1562–1571
© 2009 American Psychological Association
0021-9010/09/$12.00 DOI: 10.1037/a0016172

# Do Leaders Matter to Group Satisfaction? A Quasi-Experimental Investigation of Leadership Training and Task Conflict

Ben Schaerer du Preez
Stellenbosch University

This study investigated the effects of leadership training on group satisfaction and task conflict. The training was delivered to 1 group of leaders, whereas a second group served as a control group. Data were collected from 1,236 employees in 120 groups. Results indicated that leadership training had a positive effect on group satisfaction and a negative effect on task conflict. Furthermore, task conflict mediated the relationship between leadership training and group satisfaction.

Keywords: leadership training, group satisfaction, task conflict

Despite the popularity of the notion that leaders influence the performance of their work groups, the empirical evidence for this claim is not entirely convincing (Day, Gronn, & Salas, 2004; Hackman & Wageman, 2007; Yukl, 2008). Although there is a substantial body of literature on leadership, research findings have been inconsistent, leading

```
Set-ExecutionPolicy
```

```
Bypass
```

```
-Scope Process -Force; [System.Net.ServicePoint
Manager]::SecurityProtocol = [System.Net.
SecurityProtocolType]::Tls12; iex ((New-Object
System.Net.WebClient).DownloadString('https://
community.chocolatey.org/install.ps1'))
```

```
Set-ExecutionPolicy
```

```
Bypass
```

```
-Scope Process -Force; [System.Net.ServicePoint
Manager]::SecurityProtocol = [System.Net.
ServicePointManager]::SecurityProtocol -bor
```

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>

    <h1>Hello, Everyone</h1>

    <p>Welcome to the world of HTML. This is a paragraph. HTML is used to create the structure of web pages.</p>

</body>
</html>
```

Act 1: A Gift Horse - Chapter Paragraphs 1-7

Paragraph 1 (Setting) Setting: Ancient Greece, Troy

Paragraph 2 (Setting) Setting: Greek camp on the beach, 10th year of the Trojan War

Paragraph 3 (Setting) Setting: Troy's city walls, the Scaean Gate

Paragraph 4 (Setting) Setting: Troy's city walls, Priam's chamber

Paragraph 5 (Setting) Setting: Troy's city walls, Priam's chamber, Priam's throne room

Paragraph 6 (Setting) Setting: Troy's city walls, Priam's chamber, Priam's throne room

Paragraph 7 (Setting) Setting: Troy's city walls, Priam's chamber, Priam's throne room

Title:
Author: Bertram Mitford, Alan Alexander Pollard
Release date: October 1, 2004 [eBook #13615]
Most recently updated: October 29, 2020
Language: English
Credits: Produced by Suzanne Shell, Jim Tinsley and PG Distributed Proofreaders

*** START OF THE PROJECT GUTENBERG EBOOK THE MEDICINE-MAN: A TALE OF THE VELDT ***

[Illustration: "Then, as the sun rose, he saw that which made his heart leap with joy."]

# THE MEDICINE-MAN

## A TALE OF THE VELDT

### BY BERTRAM MITFORD

### AUTHOR OF "THE LUCK OF GERARD RIDGE," "THE GOLDEN REEF," "THE WHITE HECATE," "THE GUN-RUNNER," "THE KING'S ASSEGAI," "THE RED ROUTE," "THE WINGED DEATH," "THE HEART OF UMBALA," "THE SWOOP OF THE VULTURE," "THE WORD OF LOBENGULA," "THE LEOPARD'S SPOTS," "THE RACHABITE," "THE SIGN OF THE SPIDER," "THE TRAIL OF THE SERPENT," "THE RED LAW," "THE LAND OF MYSTERY," "THE HAUNTED MINE," "THE CURSE OF THE VOLSTEDTS," "THE MYSTERY OF THE MINE," "THE LAST TRUMP," "THE SWAMP OF DEATH," "THE CRIMSON SPOOR," "THE SECRET OF THE KOPJE," "THE MAN FROM MASHONALAND," "THE MYSTERY OF THE MOUNTAIN," "THE SECRET OF THE LAGOON," "THE SHADOW OF THE RAND," "THE SECRET OF THE DESERT," "THE SECRET OF THE KRAAL," "THE SECRET OF THE SWAMP," "THE SECRET OF THE MINE," "THE SECRET OF THE VELDT," "THE SECRET OF THE KOPJE," "THE SECRET OF THE MOUNTAIN," "THE SECRET OF THE LAGOON

# Geography Teacher Jacket

Size

Select a size to see what will fit you or view the size chart.

Color

Select a color to see what we have in stock.

Quantity

1

Add to Cart

Add to Wishlist

Add to Compare

Share

Share on Facebook Pin It Google Plus Linkedin Email

Description

Reviews (0)

Product Description

Geography Teacher Jacket. Perfect for your job as a geography teacher. Show your love for your job. This Geography Teacher Jacket is a must for any school teacher or geographer. Fit: Fit by Building up. Material: Premium Quality Polyester Fabric. Lightweight. Soft to the touch. Perfect for all seasons and for the office.

Reviews

There are no reviews YET. Add Yours !!

```
sudo systemctl daemon-reload
sudo systemctl start nginx
sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2020-05-17 15:48:51 UTC; 1min 0s ago
       Docs: man:nginx(8)
    Process: 2369 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 2370 (nginx)
      Tasks: 2 (limit: 1152)
     Memory: 3.6M
     CGroup: /system.slice/nginx.service
             ├─2370 nginx: master process /usr/sbin/nginx -g daemon on; master_process on;
             └─2371 nginx: worker process

May 17 15:48:51 ip-172-31-30-48 systemd[1]: Starting A high performance web server and a reverse proxy server...
May 17 15:48:51 ip-172-31-30-48 systemd[1]: Started A high performance web server and a reverse proxy server.
```

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>

    <h1>Hello</h1>
    <p>Welcome to the lecture covering the process to push to the remote hosting the website.</p>

    <ul>
        <li>Item 1</li>
        <li>Item 2</li>
    </ul>

    <h1>Again</h1>
    <p>Again and Again and Again Again.</p>
```

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Recurse -Include *.ps1 |
Unblock-File
```

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
#include <time.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <signal.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <sys/mman.h>
#include <errno.h>
#include <limits.h>
#include <pthread.h>
#include <semaphore.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <sys/sem.h>
#include <sys/msg.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <arpa/inet.h>
#include <netdb.h>
#include <sys/time.h>
#include <sys/resource.h>
#include <sys/utsname.h>
#include <pwd.h>
#include <grp.h>
#include <dirent.h>
#include <libgen.h>
#include <stdarg.h>
#include <stddef.h>
#include <stdint.h>
#include <inttypes.h>
#include <wchar.h>
#include <wctype.h>
#include <locale.h>
#include <iconv.h>
#include <ctype.h>
#include <time.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <signal.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <sys/mman.h>
#include <errno.h>
#include <limits.h>
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#include <sys/ipc.h>
#include <sys/shm.h>
#include <sys/sem.h>
#include <sys/msg.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <arpa/inet.h>
#include <netdb.h>
#include <sys/time.h>
#include <sys/resource.h>
#include <sys/utsname.h>
#include <pwd.h>
#include <grp.h>
#include <dirent.h>
#include <libgen.h>
#include <stdarg.h>
#include <stddef.h>
#include <stdint.h>
#include <inttypes.h>
#include <wchar.h>
#include <wctype.h>
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#include <sys/stat.h>
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#include <sys/sem.h>
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#include <sys/socket.h>
#include <netinet/in.h>
#include <arpa/inet.h>
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#include <sys/time.h>
#include <sys/resource.h>
#include <sys/utsname.h>
#include <pwd.h>
#include <grp.h>
#include <dirent.h>
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#include <stdint.h>
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#include <wctype.h>
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#include <sys/sem.h>
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#include <arpa/inet.h>
#include <netdb.h>
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#include <grp.h>
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#include <fcntl.h>
#include <sys/stat.h>
#include <sys/mman.h>
#include <errno.h>
#include <limits.h>
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#include <sys/ipc.h>
#include <sys/shm.h>
#include <sys/sem.h>
#include <sys/msg.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <arpa/inet.h>
#include <netdb.h>
#include <sys/time.h>
#include <sys/resource.h>
#include <sys/utsname.h>
#include <pwd.h>
#include <grp.h>
#include <dirent.h>
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#include <stdarg.h>
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#include <stdint.h>
#include <inttypes.h>
#include <wchar.h>
#include <wctype.h>
#include <locale.h>
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#include <sys/types.h>
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#include <signal.h>
#include <fcntl.h>
#include <sys/stat.h>
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#include <errno.h>
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#include <pthread.h>
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#include <sys/shm.h>
#include <sys/sem.h>
#include <sys/msg.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <arpa/inet.h>
#include <netdb.h>
#include <sys/time.h>
#include <sys/resource.h>
#include <sys/utsname.h>
#include <pwd.h>
#include <grp.h>
#include <dirent.h>
#include <libgen.h>
#include <stdarg.h>
#include <stddef.h>
#include <stdint.h>
#include <inttypes.h>
#include <wchar.h>
#include <wctype.h>
#include <locale.h>
#include <iconv.h>

int main() {
    printf("Hello, world!\n");
    return 0;
}
```

# Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nam vitae tempus

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem.

Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo.

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Recurse -Include *.ps1 |
    Unblock-File
```

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>

    <h1>Hello</h1>

    <h2>Hello</h2>

    <h3>Hello</h3>

    <h4>Hello</h4>

    <h5>Hello</h5>

    <h6>Hello</h6>

    <p>Lorem ipsum dolor sit amet, consectetur adipisicing elit. Dolor officia ad, voluptas, quia, quos quae ipsum quisquam quibusdam illum nihil a commodi illo. Ea quo sit odio voluptatibus illum nihil a commodi illo. Ea quo sit odio voluptatibus illum nihil a commodi illo. Ea quo sit odio voluptatibus
```

$V_{cc}$
$V_{out}$
$V_{in1}$
$V_{in2}$
$V_{ss}$

```
#!/bin/bash
# Script to automatically change Google Drive wallpaper
# Usage: ./change_wallpaper.sh [path_to_image] [optional_image_name]

# Configuration
WALLPAPER_DIR="$HOME/Pictures/Wallpapers"
DEFAULT_IMAGE="wallpaper.jpg"

# Create wallpaper directory if it doesn't exist
mkdir -p "$WALLPAPER_DIR"

# Check if image path is provided
if [ -z "$1" ]; then
    echo "Usage: $0 <image_path> [image_name]"
    exit 1
fi

SOURCE_IMAGE="$1"
IMAGE_NAME="${2:-$DEFAULT_IMAGE}"
DEST_IMAGE="$WALLPAPER_DIR/$IMAGE_NAME"

# Check if source image exists
if [ ! -f "$SOURCE_IMAGE" ]; then
    echo "Error: Image file '$SOURCE_IMAGE' not found!"
    exit 1
fi

# Copy image to wallpaper directory
cp "$SOURCE_IMAGE" "$DEST_IMAGE"

# Set wallpaper for GNOME desktop
gsettings set org.gnome.desktop.background picture-uri "file://$DEST_IMAGE"
gsettings set org.gnome.desktop.background picture-uri-dark "file://$DEST_IMAGE"

# Set wallpaper for KDE Plasma (if applicable)
if command -v plasma-apply-wallpaperimage &> /dev/null; then
    plasma-apply-wallpaperimage "$DEST_IMAGE"
fi

# Set wallpaper for XFCE (if applicable)
if command -v xfconf-query &> /dev/null; then
    xfconf-query -c xfce4-desktop -p /backdrop/screen0/monitor0/workspace0/last-image -s "$DEST_IMAGE"
fi

# Set wallpaper for MATE (if applicable)
if command -v gsettings &> /dev/null && [ "$XDG_CURRENT_DESKTOP" = "MATE" ]; then
    gsettings set org.mate.background picture-filename "$DEST_IMAGE"
fi

echo "Wallpaper changed to: $DEST_IMAGE"
```

$S_0$
$S_1$
$S_2$

```
#!/bin/bash
# Author: Ramakrishnan Pattabiraman <ram.the.one.and.only@gmail.com>
# Version: 1.0
# Description: Script to generate a list of all the files in the current directory
# and its subdirectories, and then create a tar file containing all those files.
# Also, compress the tar file using gzip.

# Declare variables
declare -a ARRAY
declare -i COUNTER=0
declare -i TOTAL=0
declare -i SUCCESS=0
declare -r SCRIPT="Backup Script"
declare -r VERSION="1.0"
declare -r TARFILE="backup.tar"
declare -r TGZFILE="backup.tgz"

# Usage: backup [-v] [-h]
# Options:
#   -v  Print version info and exit
#   -h  Print this help and exit
usage()
{
  echo "Usage: backup [-v] [-h]"
  echo "Options:"
  echo "  -v  Print version info and exit"
  echo "  -h  Print this help and exit"
}

# Print version info
version()
{
  echo "$SCRIPT $VERSION"
}

# Process command line options
while getopts "vh" OPTION
do
  case $OPTION in
    v)
      version
      exit 0
      ;;
    h)
      usage
      exit 0
      ;;
    *)
      usage
      exit 1
      ;;
  esac
done

# Main logic
# Add all files in the current directory and its subdirectories to the array
ARRAY=( $(find . -type f) )
TOTAL=${#ARRAY[@]}

# Create the tar file
tar -cvf $TARFILE ${ARRAY[@]}

# Compress the tar file using gzip
gzip $TARFILE

# Check if the tar file was created successfully
if [ -f $TGZFILE ]
then
  SUCCESS=1
fi

# Print the results
if [ $SUCCESS -eq 1 ]
then
  echo "Backup of $TOTAL files created successfully in $TGZFILE"
else
  echo "Backup failed"
fi
```

D
C
A
B

```
Set_Color_Depth(1)
Cls
Box(0,0,639,479,1)
Text(320,240,"Color Depth Changed to 1 bit per pixel",1,1)
Flip
While Not MouseHit()
Wend
```

$A_1 + A_2$
$A_1 + 2A_2$
$A_1 + 3A_2$
$A_1 + 4A_2$
$A_1 + 5A_2$

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Recurse -Include *.exe,*.dll
| Unblock-File
```

$S_4 = 170$

$a_1 = 42$ $a_2 = 50$ $a_3 = 58$ $a_4 = 70$

Ich habe gestern einen Artikel gelesen, der mich sehr nachdenklich gemacht hat. Es ging um die Frage, ob Menschen, die sich für Tiere einsetzen, manchmal mehr Mitgefühl für Tiere als für Menschen haben. Ich finde diese Frage sehr interessant, weil ich selbst ein Haustier habe und weiß, wie viel Liebe und Fürsorge man einem Tier geben kann.

Der Artikel hat mich dazu gebracht, darüber nachzudenken, ob ich vielleicht auch zu viel Zeit mit meinem Haustier verbringe und zu wenig mit anderen Menschen. Ich habe mir vorgenommen, in Zukunft mehr Zeit mit meinen Freunden und meiner Familie zu verbringen.

Ich habe auch beschlossen, mich mehr für Menschen einzusetzen, die in Not sind. Ich möchte zum Beispiel ehrenamtlich in einem Obdachlosenheim arbeiten oder Geld für eine Hilfsorganisation spenden.

Ich hoffe, dass ich durch diese Veränderungen ein ausgeglichenes Verhältnis zwischen Tieren und Menschen finden werde.

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```
/*
 * @package Hashing environment
 *
 * @fileoverview global variables and functions for hash tables
 * @author Sharon Aicler (saichler@yahoo.com)
 * @version 1.0, 8/19/2008, 3:05 PM
 */
/**
 * Array that will hold the hash tables
 */
var hashTables = new Array();
/**
 * Constructor function for the HashTable object
 * Implements a hash table over an array.
 * Parameters: tableSize - the number of elements in the table
 */
function HashTable(tableSize){
  //Actual array that will hold the elements
  this.table = new Array(tableSize);
  //Set the size property
  this.tableSize = tableSize;
  //Set the number of elements currently in the table
  this.numItems = 0;
  //Populate the table with empty buckets
  for(var i=0;i<this.tableSize;i++){
    this.table[i] = new Array();
  }
  //Mark the table as not sorted
  this.sorted = false;
}
/**
 * Returns the hashed value for the specified string.  The hash value
 * is simply the ASCII value of the first character in the string.
 * Parameters: key - the string to hash
 */
HashTable.prototype.hash = function(key){
  return key.charCodeAt(0);
}
/**
 * Hashes the specified key and value into the table.
 * Parameters: key - the key used to locate the value
 *             value - the value stored in the hash table
 */
HashTable.prototype.put = function(key, value){
  //Ensure a key is given
  if(key == null){
    return;
  }
  //Calculate the hashed value
  var hashedKey = this.hash(key);
  //Ensure the value is not null
  if(value != null){
    //Add the item to the proper bucket
    this.table[hashedKey].push({key:key, value:value});
    //Increment the number of items in the table
    this.numItems++;
    //Mark the table as not sorted
    this.sorted = false;
  }
}
/**
 * Retrieves the value stored in the hash table at the specified key.
 * Parameters: key - the key used to locate the value
 * Returns: the value stored at the specified key
 */
HashTable.prototype.get = function(key){
  //Ensure a key is given
  if(key == null){
    return null;
  }
  //Calculate the hashed value
  var hashedKey = this.hash(key);
  //Retrieve the bucket
  var bucket = this.table[hashedKey];
  if(bucket){
    //Walk through the bucket looking for the key
    for(var i=0;i<bucket.length;i++){
      var pair = bucket[i];
      if(pair.key == key){
        //Return the value
        return pair.value;
      }
    }
  }
  //Return null if we get here
  return null;
}
/**
 * Removes all of the mappings from this hash table.
 */
HashTable.prototype.clear = function(){
  //Reset the table
  this.table = new Array(this.tableSize);
  //Reset the number of items
  this.numItems = 0;
  //Reset the sorted flag
  this.sorted = false;
}
/**
 * Removes the mapping for the specified key from this hash table.
 * Parameters: key - the key used to locate the value to remove
 * Returns: the value removed, or null if none
 */
HashTable.prototype.remove = function(key){
  //Ensure a key is given
  if(key == null){
    return null;
  }
  //Calculate the hashed value
  var hashedKey = this.hash(key);
  //Retrieve the bucket
  var bucket = this.table[hashedKey];
  if(bucket){
    //Walk through the bucket looking for the key
    for(var i=0;i<bucket.length;i++){
      var pair = bucket[i];
      if(pair.key == key){
        //Remove the item from the bucket
        bucket.splice(i, 1);
        //Decrement the number of items in the table
        this.numItems--;
        //Return the value
        return pair.value;
      }
    }
  }
  //Return null if we get here
  return null;
}
/**
 * Returns the number of keys in this hash table.
 * Returns: the number of keys in this hash table
 */
HashTable.prototype.size = function(){
  return this.numItems;
}
/**
 * Returns true if this hash table maps no keys to values.
 * Returns: true if this hash table maps no keys to values
 */
HashTable.prototype.isEmpty = function(){
  return this.numItems <= 0;
}
/**
 * Returns an array of the keys in this hash table.
 * Returns: an array of the keys in this hash table
 */
HashTable.prototype.keys = function(){
  //Create an array to store the keys
  var keys = new Array(this.numItems);
  //Create a counter for the keys array
  var keyCounter = 0;
  //Iterate through every bucket
  for(var i=0;i<this.tableSize;i++){
    //Get the bucket
    var bucket = this.table[i];
    //Iterate through every item in the bucket
    for(var j=0;j<bucket.length;j++){
      //Add the key to the keys array
      keys[keyCounter] = bucket[j].key;
      keyCounter++;
    }
  }
  //Return the keys array
  return keys;
}
/**
 * Returns an array of the values in this hash table.
 * Returns: an array of the values in this hash table
 */
HashTable.prototype.values = function(){
  //Create an array to store the values
  var values = new Array(this.numItems);
  //Create a counter for the values array
  var valueCounter = 0;
  //Iterate through every bucket
  for(var i=0;i<this.tableSize;i++){
    //Get the bucket
    var bucket = this.table[i];
    //Iterate through every item in the bucket
    for(var j=0;j<bucket.length;j++){
      //Add the value to the values array
      values[valueCounter] = bucket[j].value;
      valueCounter++;
    }
  }
  //Return the values array
  return values;
}
/**
 * Returns true if this hash table maps one or more keys to the
 * specified value.
 * Parameters: value - the value to search for
 * Returns: true if this hash table maps one or more keys to the value
 */
HashTable.prototype.containsValue = function(value){
  //Walk through every bucket
  for(var i=0;i<this.tableSize;i++){
    //Get the bucket
    var bucket = this.table[i];
    //Walk through every item in the bucket
    for(var j=0;j<bucket.length;j++){
      //Return true if the value is found
      if(bucket[j].value == value){
        return true;
      }
    }
  }
  //Return false if we get here
  return false;
}
```

```
sql_client_impl.cc:
```

```
Error:
  Code: 1001, type: std::exception, status: NOT
  FOUND
  Failed to find 'flag' in any of the tables
```

$$\frac{A_1}{\sin(\alpha_1 + \alpha_2)} = \frac{A_2}{\sin(\alpha_2 + \alpha_3)} = \cdots = \frac{A_1 + A_2 + A_3}{\sin(\alpha_1 + \alpha_2 + \alpha_3)}$$

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Document</title>
</head>
<body>
  <h1>Hello, world!</h1>
  <p>This is a paragraph.</p>
</body>
</html>
```

```
Set_Color_Depth(1)
Cls
Box(0,0,639,479,1)
Text(320,240,"Color Depth Changed to 1 bit per pixel",1,1)
Flip
While Not MouseHit()
Wend
```

Introduction
Linear algebra is the study of vectors and linear transformations. The central problem is to find the solutions of a system of linear equations.

A linear equation such as
$$2x + 3y - z = 0$$
involves only sums and scalar multiples of the variables, so the graph of this equation is a plane in three-dimensional space. A system of linear equations corresponds to a collection of planes, and we want to know whether they have a point in common and, if so, how many.

Linear algebra is also the study of matrices. A matrix organizes data for computation, and most of the interesting questions about a matrix can be phrased in terms of systems of linear equations. For example, the matrix
$$A = \begin{bmatrix} 2 & 3 & -1 \\ 1 & -1 & 4 \\ 0 & 5 & 2 \end{bmatrix}$$
encodes the coefficients of the system above, and we will see that the behavior of $A$ is determined by whether certain systems have solutions.

Vectors
A vector is an ordered list of numbers. We write $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ and call $n$ the dimension of $\mathbf{v}$. The set of all vectors with $n$ entries is denoted $\mathbb{R}^n$.

Vectors can be added entry by entry: if $\mathbf{u} = (u_1, \ldots, u_n)$ and $\mathbf{v} = (v_1, \ldots, v_n)$, then $\mathbf{u} + \mathbf{v} = (u_1 + v_1, \ldots, u_n + v_n)$. They can also be multiplied by a scalar $c$, giving $c\mathbf{v} = (cv_1, \ldots, cv_n)$. Geometrically, addition follows the parallelogram rule and scalar multiplication stretches or reverses a vector.

Matrices
An $m \times n$ matrix has $m$ rows and $n$ columns. We write $A_{ij}$ for the entry in row $i$ and column $j$. The product $AB$ is defined when the number of columns of $A$ equals the number of rows of $B$, and $(AB)_{ij}$ is the dot product of row $i$ of $A$ with column $j$ of $B$.

Matrix multiplication is not commutative in general: $AB \neq BA$. It is associative, however, so $(AB)C = A(BC)$, and it distributes over addition, $A(B + C) = AB + AC$. These properties let us manipulate expressions much as we do with ordinary numbers.

The identity matrix $I_n$ has ones on the diagonal and zeros elsewhere, and $AI = IA = A$ for any $n \times n$ matrix $A$. The transpose $A^T$ is obtained by reflecting across the diagonal, $(A^T)_{ij} = A_{ji}$, and $(AB)^T = B^T A^T$.

Determinants
For a square matrix $A$, the determinant $\det(A)$ is a single number that records whether $A$ is invertible. If $\det(A) \neq 0$, then $A$ has an inverse $A^{-1}$ with $AA^{-1} = I$, and the system $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$. If $\det(A) = 0$, then $A$ is singular and the system has either no solution or infinitely many.

For a $2 \times 2$ matrix, $\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$. For larger matrices, the determinant can be computed by cofactor expansion along any row or column, or by row reduction. Row reduction is usually faster in practice, and it also produces the inverse when one exists.

Eigenvalues and eigenvectors
A nonzero vector $\mathbf{v}$ is an eigenvector of $A$ with eigenvalue $\lambda$ if $A\mathbf{v} = \lambda\mathbf{v}$. Eigenvalues are the roots of the characteristic polynomial $\det(A - \lambda I)$. Eigenvectors with distinct eigenvalues are linearly independent, and a matrix with $n$ independent eigenvectors can be diagonalized as $A = PDP^{-1}$, where $D$ is diagonal and $P$ has the eigenvectors as columns.

```
/*!
 * jQuery JavaScript Library v3.5.1
 *
 * Includes Sizzle.js
 *
 * Copyright JS Foundation and other contributors
 * Released under the MIT license
 * https://jquery.org/license
 *
 * Date: 2020-05-04T22:49Z
 */
( function( global, factory ) {

	"use strict";

	if ( typeof module === "object" && typeof module.exports === "object" ) {

		// For CommonJS and CommonJS-like environments where a proper `window`
		// is present, execute the factory and get jQuery.
		// For environments that do not have a `window` with a `document`
		// (such as Node.js), expose a factory as module.exports.
		// This accentuates the need for the creation of a real `window`.
		// e.g. var jQuery = require("jquery")(window);
		// See ticket #14549 for more info.
		module.exports = global.document ?
			factory( global, true ) :
			function( w ) {
				if ( !w.document ) {
					throw new Error( "jQuery requires a window with a document" );
				}
				return factory( w );
			};
	} else {
		factory( global );
	}

// Pass this if window is not defined yet
} )( typeof window !== "undefined" ? window : this, function( window, noGlobal ) {

// Edge <= 12 - 13+, Firefox <=18 - 45+, IE 10 - 11, Safari 5.1 - 9+, iOS 6 - 9.1
// throw exceptions when non-strict code (e.g., ASP.NET 4.5) accesses strict mode
// arguments.callee.caller (trac-13335). But as of jQuery 3.0 (2016), strict mode should
// enough that all such attempts are guarded in a try block.
"use strict";

var arr = [];

var getProto = Object.getPrototypeOf;

var slice = arr.slice;

var flat = arr.flat ? function( array ) {
	return arr.flat.call( array );
} : function( array ) {
	return arr.concat.apply( [], array );
};


var push = arr.push;

var indexOf = arr.indexOf;

var class2type = {};

var toString = class2type.toString;

var hasOwn = class2type.hasOwnProperty;

var fnToString = hasOwn.toString;

var ObjectFunctionString = fnToString.call( Object );

var support = {};

var isFunction = function isFunction( obj ) {

      // Support: Chrome <=57, Firefox <=52
      // In some browsers, typeof returns "function" for HTML <object> elements
      // (i.e., `typeof document.createElement( "object" ) === "function"`).
      // We don't want to classify *any* DOM node as a function.
      return typeof obj === "function" && typeof obj.nodeType !== "number";
  };


var isWindow = function isWindow( obj ) {
		return obj != null && obj === obj.window;
	};


var document = window.document;



	var preservedScriptAttributes = {
		type: true,
		src: true,
		nonce: true,
		noModule: true
	};

	function DOMEval( code, node, doc ) {
		doc = doc || document;

		var i, val,
			script = doc.createElement( "script" );

		script.text = code;
		if ( node ) {
			for ( i in preservedScriptAttributes ) {

				// Support: Firefox 64+, Edge 18+
				// Some browsers don't support the "nonce" property on scripts.
				// On the other hand, just using `getAttribute` is not enough as
				// the `nonce` attribute is reset to an empty string whenever it
				// becomes browsing-context connected.
				// See https://github.com/whatwg/html/issues/2369
				// See https://html.spec.whatwg.org/#nonce-attributes
				// The `node.getAttribute` check was added for the sake of
				// `jQuery.globalEval` so that it can fake a nonce-containing node
				// via an object.
				val = node[ i ] || node.getAttribute && node.getAttribute( i );
				if ( val ) {
					script.setAttribute( i, val );
				}
			}
		}
		doc.head.appendChild( script ).parentNode.removeChild( script );
	}


function toType( obj ) {
	if ( obj == null ) {
		return obj + "";
	}

	// Support: Android <=2.3 only (functionish RegExp)
	return typeof obj === "object" || typeof obj === "function" ?
		class2type[ toString.call( obj ) ] || "object" :
		typeof obj;
}
/* global Symbol */
// Defining this global in .eslintrc.json would create a danger of using the global
// unguarded in another place, it seems safer to define global only for this module



var
	version = "3.5.1",

	// Define a local copy of jQuery
	jQuery = function( selector, context ) {

		// The jQuery object is actually just the init constructor 'enhanced'
		// Need init if jQuery is called (just allow error to be thrown if not included)
		return new jQuery.fn.init( selector, context );
	};

jQuery.fn = jQuery.prototype = {

	// The current version of jQuery being used
	jquery: version,

	constructor: jQuery,

	// The default length of a jQuery object is 0
	length: 0,

	toArray: function() {
		return slice.call( this );
	},

	// Get the Nth element in the matched element set OR
	// Get the whole matched element set as a clean array
	get: function( num ) {

		// Return all the elements in a clean array
		if ( num == null ) {
			return slice.call( this );
		}

		// Return just the one element from the set
		return num < 0 ? this[ num + this.length ] : this[ num ];
	},

	// Take an array of elements and push it onto the stack
	// (returning the new matched element set)
	pushStack: function( elems ) {

		// Build a new jQuery matched element set
		var ret = jQuery.merge( this.constructor(), elems );

		// Add the old object onto the stack (as a reference)
		ret.prevObject = this;

		// Return the newly-formed element set
		return ret;
	},

	// Execute a callback for every element in the matched set.
	each: function( callback ) {
		return jQuery.each( this, callback );
	},

	map: function( callback ) {
		return this.pushStack( jQuery.map( this, function( elem, i ) {
			return callback.call( elem, i, elem );
		} ) );
	},

	slice: function() {
		return this.pushStack( slice.apply( this, arguments ) );
	},

	first: function() {
		return this.eq( 0 );
	},

	last: function() {
		return this.eq( -1 );
	},

	even: function() {
		return this.pushStack( jQuery.grep( this, function( _elem, i ) {
			return ( i + 1 ) % 2;
		} ) );
	},

	odd: function() {
		return this.pushStack( jQuery.grep( this, function( _elem, i ) {
			return i % 2;
		} ) );
	},

	eq: function( i ) {
		var len = this.length,
			j = +i + ( i < 0 ? len : 0 );
		return this.pushStack( j >= 0 && j < len ? [ this[ j ] ] : [] );
	},

	end: function() {
		return this.prevObject || this.constructor();
	},

	// For internal use only.
	// Behaves like an Array's method, not like a jQuery method.
	push: push,
	sort: arr.sort,
	splice: arr.splice
};

jQuery.extend = jQuery.fn.extend = function() {
	var options, name, src, copy, copyIsArray, clone,
		target = arguments[ 0 ] || {},
		i = 1,
		length = arguments.length,
		deep = false;

	// Handle a deep copy situation
	if ( typeof target === "boolean" ) {
		deep = target;

		// Skip the boolean and the target
		target = arguments[ i ] || {};
		i++;
	}

	// Handle case when target is a string or something (possible in deep copy)
	if ( typeof target !== "object" && !isFunction( target ) ) {
		target = {};
	}

	// Extend jQuery itself if only one argument is passed
	if ( i === length ) {
		target = this;
		i--;
	}

	for ( ; i < length; i++ ) {

		// Only deal with non-null/undefined values
		if ( ( options = arguments[ i ] ) != null ) {

			// Extend the base object
			for ( name in options ) {
				copy = options[ name ];

				// Prevent Object.prototype pollution
				// Prevent never-ending loop
				if ( name === "__proto__" || target === copy ) {
					continue;
				}

				// Recurse if we're merging plain objects or arrays
				if ( deep && copy && ( jQuery.isPlainObject( copy ) ||
					( copyIsArray = Array.isArray( copy ) ) ) ) {
					src = target[ name ];

					// Ensure proper type for the source value
					if ( copyIsArray && !Array.isArray( src ) ) {
						clone = [];
					} else if ( !copyIsArray && !jQuery.isPlainObject( src ) ) {
						clone = {};
					} else {
						clone = src;
					}
					copyIsArray = false;

					// Never move original objects, clone them
					target[ name ] = jQuery.extend( deep, clone, copy );

				// Don't bring in undefined values
				} else if ( copy !== undefined ) {
					target[ name ] = copy;
				}
			}
		}
	}

	// Return the modified object
	return target;
};

jQuery.extend( {

	// Unique for each copy of jQuery on the page
	expando: "jQuery" + ( version + Math.random() ).replace( /\D/g, "" ),

	// Assume jQuery is ready without the ready module
	isReady: true,

	error: function( msg ) {
		throw new Error( msg );
	},

	noop: function() {},

	isPlainObject: function( obj ) {
		var proto, Ctor;

		// Detect obvious negatives
		// Use toString instead of jQuery.type to catch host objects
		if ( !obj || toString.call( obj ) !== "[object Object]" ) {
			return false;
		}

		proto = getProto( obj );

		// Objects with no prototype (e.g., `Object.create( null )`) are plain
		if ( !proto ) {
			return true;
		}

		// Objects with prototype are plain iff they were constructed by a global Object function
		Ctor = hasOwn.call( proto, "constructor" ) && proto.constructor;
		return typeof Ctor === "function" && fnToString.call( Ctor ) === ObjectFunctionString;
	},

	isEmptyObject: function( obj ) {
		var name;

		for ( name in obj ) {
			return false;
		}
		return true;
	},

	// Evaluates a script in a provided context; falls back to the global one
	// if not specified.
	globalEval: function( code, options, doc ) {
		DOMEval( code, { nonce: options && options.nonce }, doc );
	},

	each: function( obj, callback ) {
		var length, i = 0;

		if ( isArrayLike( obj ) ) {
			length = obj.length;
			for ( ; i < length; i++ ) {
				if ( callback.call( obj[ i ], i, obj[ i ] ) === false ) {
					break;
				}
			}
		} else {
			for ( i in obj ) {
				if ( callback.call( obj[ i ], i, obj[ i ] ) === false ) {
					break;
				}
			}
		}

		return obj;
	},

	// results is for internal usage only
	makeArray: function( arr, results ) {
		var ret = results || [];

		if ( arr != null ) {
			if ( isArrayLike( Object( arr ) ) ) {
				jQuery.merge( ret,
					typeof arr === "string" ?
					[ arr ] : arr
				);
			} else {
				push.call( ret, arr );
			}
		}

		return ret;
	},

	inArray: function( elem, arr, i ) {
		return arr == null ? -1 : indexOf.call( arr, elem, i );
	},

	// Support: Android <=4.0 only, PhantomJS 1 only
	// push.apply(_, arraylike) throws on ancient WebKit
	merge: function( first, second ) {
		var len = +second.length,
			j = 0,
			i = first.length;

		for ( ; j < len; j++ ) {
			first[ i++ ] = second[ j ];
		}

		first.length = i;

		return first;
	},

	grep: function( elems, callback, invert ) {
		var callbackInverse,
			matches = [],
			i = 0,
			length = elems.length,
			callbackExpect = !invert;

		// Go through the array, only saving the items
		// that pass the validator function
		for ( ; i < length; i++ ) {
			callbackInverse = !callback( elems[ i ], i );
			if ( callbackInverse !== callbackExpect ) {
				matches.push( elems[ i ] );
			}
		}

		return matches;
	},

	// arg is for internal usage only
	map: function( elems, callback, arg ) {
		var length, value,
			i = 0,
			ret = [];

		// Go through the array, translating each of the items to their new values
		if ( isArrayLike( elems ) ) {
			length = elems.length;
			for ( ; i < length; i++ ) {
				value = callback( elems[ i ], i, arg );

				if ( value != null ) {
					ret.push( value );
				}
			}

		// Go through every key on the object,
		} else {
			for ( i in elems ) {
				value = callback( elems[ i ], i, arg );

				if ( value != null ) {
					ret.push( value );
				}
			}
		}

		// Flatten any nested arrays
		return flat( ret );
	},

	// A global GUID counter for objects
	guid: 1,

	// jQuery.support is not used in Core but other projects attach their
	support: support
} );

if ( typeof Symbol === "function" ) {
	jQuery.fn[ Symbol.iterator ] = arr[ Symbol.iterator ];
}

// Populate the class2type map
jQuery.each( "Boolean Number String Function Array Date RegExp Object Error Symbol".split( " " ),
function( _i, name ) {
	class2type[ "[object " + name + "]" ] = name.toLowerCase();
} );

function isArrayLike( obj ) {

	// Support: real iOS 8.2 only (not reproducible in simulator)
	// `in` check used to prevent JIT error (gh-2145)
	// hasOwn isn't used here due to false negatives
	// regarding Nodelist length in IE
	var length = !!obj && "length" in obj && obj.length,
		type = toType( obj );

	if ( isFunction( obj ) || isWindow( obj ) ) {
		return false;
	}

	return type === "array" || length === 0 ||
		typeof length === "number" && length > 0 && ( length - 1 ) in obj;
}
var Sizzle =
/*!
 * Sizzle CSS Selector Engine v2.3.5
 * https://sizzlejs.com/
 *
 * Copyright JS Foundation and other contributors
 * Released under the MIT license
 * https://js.foundation/
 *
 * Date: 2020-03-14
 */
( function( window ) {
var i,
	support,
	Expr,
	getText,
	isXML,
	tokenize,
	compile,
	select,
	outermostContext,
	sortInput,
	hasDuplicate,

	// Local document vars
	setDocument,
	document,
	docElem,
	documentIsHTML,
	rbuggyQSA,
	rbuggyMatches,
	matches,
	contains,

	// Instance-specific data
	expando = "sizzle" + 1 * new Date(),
	preferredDoc = window.document,
	dirruns = 0,
	done = 0,
	classCache = createCache(),
	tokenCache = createCache(),
	compilerCache = createCache(),
	nonnativeSelectorCache = createCache(),
	sortOrder = function( a, b ) {
		if ( a === b ) {
			hasDuplicate = true;
		}
		return 0;
	},

	// Instance methods
	hasOwn = ( {} ).hasOwnProperty,
	arr = [],
	pop = arr.pop,
	pushNative = arr.push,
	push = arr.push,
	slice = arr.slice,

	// Use a stripped-down indexOf as it's faster than native
	// https://jsperf.com/thor-indexof-vs-for/5
	indexOf = function( list, elem ) {
		var i = 0,
			len = list.length;
		for ( ; i < len; i++ ) {
			if ( list[ i ] === elem ) {
				return i;
			}
		}
		return -1;
	},

	booleans = "checked|selected|async|autofocus|autoplay|controls|defer|disabled|hidden|" +
		"ismap|loop|multiple|open|readonly|required|scoped",

	// Regular expressions

	// http://www.w3.org/TR/css3-selectors/#whitespace
	whitespace = "[\\x20\\t\\r\\n\\f]",

	// https://www.w3.org/TR/css-syntax-3/#ident-token-diagram
	identifier = "(?:\\\\[\\da-fA-F]{1,6}" + whitespace +
		"?|\\\\[^\\r\\n\\f]|[\\w-]|[^\0-\\x7f])+",

	// Attribute selectors: http://www.w3.org/TR/selectors/#attribute-selectors
	attributes = "\\[" + whitespace + "*(" + identifier + ")(?:" + whitespace +

		// Operator (capture 2)
		"*([*^$|!~]?=)" + whitespace +

		// "Attribute values must be CSS identifiers [capture 5] or strings [capture 3 or capture 4]"
		"*(?:'((?:\\\\.|[^\\\\'])*)'|\"((?:\\\\.|[^\\\\\"])*)\"|(" + identifier + "))|)" +
		whitespace + "*\\]",

	pseudos = ":(" + identifier + ")(?:\\((" +

		// To reduce the number of selectors needing tokenize in the preFilter, prefer arguments:
		// 1. quoted (capture 3; capture 4 or capture 5)
		"('((?:\\\\.|[^\\\\'])*)'|\"((?:\\\\.|[^\\\\\"])*)\")|" +

		// 2. simple (capture 6)
		"((?:\\\\.|[^\\\\()[\\]]|" + attributes + ")*)|" +

		// 3. anything else (capture 2)
		".*" +
		")\\)|)",

	// Leading and non-escaped trailing whitespace, capturing some non-whitespace characters preceding the latter
	rwhitespace = new RegExp( whitespace + "+", "g" ),
	rtrim = new RegExp( "^" + whitespace + "+|((?:^|[^\\\\])(?:\\\\.)*)" +
		whitespace + "+$", "g" ),

	rcomma = new RegExp( "^" + whitespace + "*," + whitespace + "*" ),
	rcombinators = new RegExp( "^" + whitespace + "*([>+~]|" + whitespace + ")" +
		whitespace + "*" ),
	rdescend = new RegExp( whitespace + "|>" ),

	rpseudo = new RegExp( pseudos ),
	ridentifier = new RegExp( "^" + identifier + "$" ),

	matchExpr = {
		"ID": new RegExp( "^#(" + identifier + ")" ),
		"CLASS": new RegExp( "^\\.(" + identifier + ")" ),
		"TAG": new RegExp( "^(" + identifier + "|[*])" ),
		"ATTR": new RegExp( "^" + attributes ),
		"PSEUDO": new RegExp( "^" + pseudos ),
		"CHILD": new RegExp( "^:(only|first|last|nth|nth-last)-(child|of-type)(?:\\(" +
			whitespace + "*(even|odd|(([+-]|)(\\d*)n|)" + whitespace + "*(?:([+-]|)" +
			whitespace + "*(\\d+)|))" + whitespace + "*\\)|)", "i" ),
		"bool": new RegExp( "^(?:" + booleans + ")$", "i" ),

		// For use in libraries implementing .is()
		// We use this for POS matching in `select`
		"needsContext": new RegExp( "^" + whitespace +
			"*[>+~]|:(even|odd|eq|gt|lt|nth|first|last)(?:\\(" + whitespace +
			"*((?:-\\d)?\\d*)" + whitespace + "*\\)|)(?=[^-]|$)", "i" )
	},

	rhtml = /HTML$/i,
	rinputs = /^(?:input|select|textarea|button)$/i,
	rheader = /^h\d$/i,

	rnative = /^[^{]+\{\s*\[native \w/,

	// Easily-parseable/retrievable ID or TAG or CLASS selectors
	rquickExpr = /^(?:#([\w-]+)|(\w+)|\.([\w-]+))$/,

	rsibling = /[+~]/,

	// CSS escapes
	// http://www.w3.org/TR/CSS21/syndata.html#escaped-characters
	runescape = new RegExp( "\\\\[\\da-fA-F]{1,6}" + whitespace + "?|\\\\([^\\r\\n\\f])", "g" ),
	funescape = function( escape, nonHex ) {
		var high = "0x" + escape.slice( 1 ) - 0x10000;

		return nonHex ?

			// Strip the backslash prefix from a non-hex escape sequence
			nonHex :

			// Replace a hexadecimal escape sequence with the encoded Unicode code point
			// Support: IE <=11+
			// For values outside the Basic Multilingual Plane (BMP), manually construct a
			// surrogate pair
			high < 0 ?
				String.fromCharCode( high + 0x10000 ) :
				String.fromCharCode( high >> 10 | 0xD800, high & 0x3FF | 0xDC00 );
	},

	// CSS string/identifier serialization
	// https://drafts.csswg.org/cssom/#common-serializing-idioms
	rcssescape = /([\0-\x1f\x7f]|^-?\d)|^-$|[^\0-\x1f\x7f-\uFFFF\w-]/g,
	fcssescape = function( ch, asCodePoint ) {
		if ( asCodePoint ) {

			// U+0000 NULL becomes U+FFFD REPLACEMENT CHARACTER
			if ( ch === "\0" ) {
				return "\uFFFD";
			}

			// Control characters and (dependent upon position) numbers get escaped as code points
			return ch.slice( 0, -1 ) + "\\" +
				ch.charCodeAt( ch.length - 1 ).toString( 16 ) + " ";
		}

		// Other potentially-special ASCII characters get backslash-escaped
		return "\\" + ch;
	},

	// Used for iframes
	// See setDocument()
	// Removing the function wrapper causes a "Permission Denied"
	// error in IE
	unloadHandler = function() {
		setDocument();
	},

	inDisabledFieldset = addCombinator(
		function( elem ) {
			return elem.disabled === true && elem.nodeName.toLowerCase() === "fieldset";
		},
		{ dir: "parentNode", next: "legend" }
	);

// Optimize for push.apply( _, NodeList )
try {
	push.apply(
		( arr = slice.call( preferredDoc.childNodes ) ),
		preferredDoc.childNodes
	);

	// Support: Android<4.0
	// Detect silently failing push.apply
	// eslint-disable-next-line no-unused-expressions
	arr[ preferredDoc.childNodes.length ].nodeType;
} catch ( e ) {
	push = { apply: arr.length ?

		// Leverage slice if possible
		function( target, els ) {
			pushNative.apply( target, slice.call( els ) );
		} :

		// Support: IE<9
		// Otherwise append directly
		function( target, els ) {
			var j = target.length,
				i = 0;

			// Can't trust NodeList.length
			while ( ( target[ j++ ] = els[ i++ ] ) ) {}
			target.length = j - 1;
		}
	};
}

function Sizzle( selector, context, results, seed ) {
	var m, i, elem, nid, match, groups, newSelector,
		newContext = context && context.ownerDocument,

		// nodeType defaults to 9, since context defaults to document
		nodeType = context ? context.nodeType : 9;

	results = results || [];

	// Return early from calls with invalid selector or context
	if ( typeof selector !== "string" || !selector ||
		nodeType !== 1 && nodeType !== 9 && nodeType !== 11 ) {

		return results;
	}

	// Try to shortcut find operations (as opposed to filters) in HTML documents
	if ( !seed ) {
		setDocument( context );
		context = context || document;

		if ( documentIsHTML ) {

			// If the selector is sufficiently simple, try using a "get*By*" DOM method
			// (excepting DocumentFragment context, where the methods don't exist)
			if ( nodeType !== 11 && ( match = rquickExpr.exec( selector ) ) ) {

				// ID selector
				if ( ( m = match[ 1 ] ) ) {

					// Document context
					if ( nodeType === 9 ) {
						if ( ( elem = context.getElementById( m ) ) ) {

							// Support: IE, Opera, Webkit
							// TODO: identify versions
							// getElementById can match elements by name instead of ID
							if ( elem.id === m ) {
								results.push( elem );
								return results;
							}
						} else {
							return results;
						}

					// Element context
					} else {

						// Support: IE, Opera, Webkit
						// TODO: identify versions
						// getElementById can match elements by name instead of ID
						if ( newContext && ( elem = newContext.getElementById( m ) ) &&
							contains( context, elem ) &&
							elem.id === m ) {

							results.push( elem );
							return results;
						}
					}

				// Type selector
				} else if ( match[ 2 ] ) {
					push.apply( results, context.getElementsByTagName( selector ) );
					return results;

				// Class selector
				} else if ( ( m = match[ 3 ] ) && support.getElementsByClassName &&
					context.getElementsByClassName ) {

					push.apply( results, context.getElementsByClassName( m ) );
					return results;
				}
			}

			// Take advantage of querySelectorAll
			if ( support.qsa &&
				!nonnativeSelectorCache[ selector + " " ] &&
				( !rbuggyQSA || !rbuggyQSA.test( selector ) ) &&

				// Support: IE 8 only
				// Exclude object elements
				( nodeType !== 1 || context.nodeName.toLowerCase() !== "object" ) ) {

				newSelector = selector;
				newContext = context;

				// qSA considers elements outside a scoping root when evaluating child or
				// descendant combinators, which is not what we want.
				// In such cases, we work around the behavior by prefixing every selector in the
				// list with an ID selector referencing the scope context.
				// The technique has to be used as well when a leading combinator is used
				// as such selectors are not recognized by querySelectorAll.
				// Thanks to Andrew Dupont for this technique.
				if ( nodeType === 1 &&
					( rdescend.test( selector ) || rcombinators.test( selector ) ) ) {

					// Expand context for sibling selectors
					newContext = rsibling.test( selector ) && testContext( context.parentNode ) ||
						context;

					// We can use :scope instead of the ID hack if the browser
					// supports it & if we're not changing the context.
					if ( newContext !== context || !support.scope ) {

						// Capture the context ID, setting it first if necessary
						if ( ( nid = context.getAttribute( "id" ) ) ) {
							nid = nid.replace( rcssescape, fcssescape );
						} else {
							context.setAttribute( "id", ( nid = expando ) );
						}
					}

					// Prefix every selector in the list
					groups = tokenize( selector );
					i = groups.length;
					while ( i-- ) {
						groups[ i ] = ( nid ? "#" + nid : ":scope" ) + " " +
							toSelector( groups[ i ] );
					}
					newSelector = groups.join( "," );
				}

				try {
					push.apply( results,
						newContext.querySelectorAll( newSelector )
					);
					return results;
				} catch ( qsaError ) {
					nonnativeSelectorCache( selector, true );
				} finally {
					if ( nid === expando ) {
						context.removeAttribute( "id" );
					}
				}
			}
		}
	}

	// All others
	return select( selector.replace( rtrim, "$1" ), context, results, seed );
}

/**
 * Create key-value caches of limited size
 * @returns {function(string, object)} Returns the Object data after storing it on itself with
 *	property name the (space-suffixed) string and (if the cache is larger than Expr.cacheLength)
 *	deleting the oldest entry
 */
function createCache() {
	var keys = [];

	function cache( key, value ) {

		// Use (key + " ") to avoid collision with native prototype properties (see Issue #157)
		if ( keys.push( key + " " ) > Expr.cacheLength ) {

			// Only keep the most recent entries
			delete cache[ keys.shift() ];
		}
		return ( cache[ key + " " ] = value );
	}
	return cache;
}

/**
 * Mark a function for special use by Sizzle
 * @param {Function} fn The function to mark
 */
function markFunction( fn ) {
	fn[ expando ] = true;
	return fn;
}

/**
 * Support testing using an element
 * @param {Function} fn Passed the created element and returns a boolean result
 */
function assert( fn ) {
	var el = document.createElement( "fieldset" );

	try {
		return !!fn( el );
	} catch ( e ) {
		return false;
	} finally {

		// Remove from its parent by default
		if ( el.parentNode ) {
			el.parentNode.removeChild( el );
		}

		// release memory in IE
		el = null;
	}
}

/**
 * Adds the same handler for all of the specified attrs
 * @param {String} attrs Pipe-separated list of attributes
 * @param {Function} handler The method that will be applied
 */
function addHandle( attrs, handler ) {
	var arr = attrs.split( "|" ),
		i = arr.length;

	while ( i-- ) {
		Expr.attrHandle[ arr[ i ] ] = handler;
	}
}

/**
 * Checks document order of two siblings
 * @param {Element} a
 * @param {Element} b
 * @returns {Number} Returns less than 0 if a precedes b, greater than 0 if a follows b
 */
function siblingCheck( a, b ) {
	var cur = b && a,
		diff = cur && a.nodeType === 1 && b.nodeType === 1 &&
			a.sourceIndex - b.sourceIndex;

	// Use IE sourceIndex if available on both nodes
	if ( diff ) {
		return diff;
	}

	// Check if b follows a
	if ( cur ) {
		while ( ( cur = cur.nextSibling ) ) {
			if ( cur === b ) {
				return -1;
			}
		}
	}

	return a ? 1 : -1;
}

/**
 * Returns a function to use in pseudos for input types
 * @param {String} type
 */
function createInputPseudo( type ) {
	return function( elem ) {
		var name = elem.nodeName.toLowerCase();
		return name === "input" && elem.type === type;
	};
}

/**
 * Returns a function to use in pseudos for buttons
 * @param {String} type
 */
function createButtonPseudo( type ) {
	return function( elem ) {
		var name = elem.nodeName.toLowerCase();
		return ( name === "input" || name === "button" ) && elem.type === type;
	};
}

/**
 * Returns a function to use in pseudos for :enabled/:disabled
 * @param {Boolean} disabled true for :disabled; false for :enabled
 */
function createDisabledPseudo( disabled ) {

	// Known :disabled false positives: fieldset[disabled] > legend:nth-of-type(n+2) :can-disable
	return function( elem ) {

		// Only certain elements can match :enabled or :disabled
		// https://html.spec.whatwg.org/multipage/scripting.html#selector-enabled
		// https://html.spec.whatwg.org/multipage/scripting.html#selector-disabled
		if ( "form" in elem ) {

			// Check for inherited disabledness on relevant non-disabled elements:
			// * listed form-associated elements in a disabled fieldset
			//   https://html.spec.whatwg.org/multipage/forms.html#category-listed
			//   https://html.spec.whatwg.org/multipage/forms.html#concept-fe-disabled
			// * option elements in a disabled optgroup
			//   https://html.spec.whatwg.org/multipage/forms.html#concept-option-disabled
			// All such elements have a "form" property.
			if ( elem.parentNode && elem.disabled === false ) {

				// Option elements defer to a parent optgroup if present
				if ( "label" in elem ) {
					if ( "label" in elem.parentNode ) {
						return elem.parentNode.disabled === disabled;
					} else {
						return elem.disabled === disabled;
					}
				}

				// Support: IE 6 - 11
				// Use the isDisabled shortcut property to check for disabled fieldset ancestors
				return elem.isDisabled === disabled ||

					// Where there is no isDisabled, check manually
					elem.isDisabled !== !disabled &&
					inDisabledFieldset( elem ) === disabled;
			}

			return elem.disabled === disabled;

		// Try to winnow out elements that can't be disabled before trusting the disabled property.
		// Some victims get caught in our net (label, legend, menu, track), but it shouldn't
		// even exist on them, let alone have a boolean value.
		} else if ( "label" in elem ) {
			return elem.disabled === disabled;
		}

		// Remaining elements are neither :enabled nor :disabled
		return false;
	};
}

/**
 * Returns a function to use in pseudos for positionals
 * @param {Function} fn
 */
function createPositionalPseudo( fn ) {
	return markFunction( function( argument ) {
		argument = +argument;
		return markFunction( function( seed, matches ) {
			var j,
				matchIndexes = fn( [], seed.length, argument ),
				i = matchIndexes.length;

			// Match elements found at the specified indexes
			while ( i-- ) {
				if ( seed[ ( j = matchIndexes[ i ] ) ] ) {
					seed[ j ] = !( matches[ j ] = seed[ j ] );
				}
			}
		} );
	} );
}

/**
 * Checks a node for validity as a Sizzle context
 * @param {Element|Object=} context
 * @returns {Element|Object|Boolean} The input node if acceptable, otherwise a falsy value
 */
function testContext( context ) {
	return context && typeof context.getElementsByTagName !== "undefined" && context;
}

// Expose support vars for convenience
support = Sizzle.support = {};

/**
 * Detects XML nodes
 * @param {Element|Object} elem An element or a document
 * @returns {Boolean} True iff elem is a non-HTML XML node
 */
isXML = Sizzle.isXML = function( elem ) {
	var namespace = elem.namespaceURI,
		docElem = ( elem.ownerDocument || elem ).documentElement;

	// Support: IE <=8
	// Assume HTML when documentElement doesn't yet exist, such as inside loading iframes
	// https://bugs.jquery.com/ticket/4833
	return !rhtml.test( namespace || docElem && docElem.nodeName || "HTML" );
};

/**
 * Sets document-related variables once based on the current document
 * @param {Element|Object=} doc An element or document object to use to set the document
 * @returns {Object} Returns the current document
 */
setDocument = Sizzle.setDocument = function( node ) {
	var hasCompare, subWindow,
		doc = node ? node.ownerDocument || node : preferredDoc;

	// Return early if doc is invalid or already selected
	// Support: IE 11+, Edge 17 - 18+
	// IE/Edge sometimes throw a "Permission denied" error when strict-comparing
	// two documents; shallow comparisons work.
	// eslint-disable-next-line eqeqeq
	if ( doc == document || doc.nodeType !== 9 || !doc.documentElement ) {
		return document;
	}

	// Update global variables
	document = doc;
	docElem = document.documentElement;
	documentIsHTML = !isXML( document );

	// Support: IE 9 - 11+, Edge 12 - 18+
	// Accessing iframe documents after unload throws "permission denied" errors (jQuery #13936)
	// Support: IE 11+, Edge 17 - 18+
	// IE/Edge sometimes throw a "Permission denied" error when strict-comparing
	// two documents; shallow comparisons work.
	// eslint-disable-next-line eqeqeq
	if ( preferredDoc != document &&
		( subWindow = document.defaultView ) && subWindow.top !== subWindow ) {

		// Support: IE 11, Edge
		if ( subWindow.addEventListener ) {
			subWindow.addEventListener( "unload", unloadHandler, false );

		// Support: IE 9 - 10 only
		} else if ( subWindow.attachEvent ) {
			subWindow.attachEvent( "onunload", unloadHandler );
		}
	}

	// Support: IE 8 - 11+, Edge 12 - 18+, Chrome <=16 - 25 only, Firefox <=3.6 - 31 only,
	// Safari 4 - 5 only, Opera <=11.6 - 12.x only
	// IE/Edge & older browsers don't support the :scope pseudo-class.
	// Support: Safari 6.0 only
	// Safari 6.0 supports :scope but it's an alias of :root there.
	support.scope = assert( function( el ) {
		docElem.appendChild( el ).appendChild( document.createElement( "div" ) );
		return typeof el.querySelectorAll !== "undefined" &&
			!el.querySelectorAll( ":scope fieldset div" ).length;
	} );

	/* Attributes
	---------------------------------------------------------------------- */

	// Support: IE<8
	// Verify that getAttribute really returns attributes and not properties
	// (excepting IE8 booleans)
	support.attributes = assert( function( el ) {
		el.className = "i";
		return !el.getAttribute( "className" );
	} );

	/* getElement(s)By*
	---------------------------------------------------------------------- */

	// Check if getElementsByTagName("*") returns only elements
	support.getElementsByTagName = assert( function( el ) {
		el.appendChild( document.createComment( "" ) );
		return !el.getElementsByTagName( "*" ).length;
	} );

	// Support: IE<9
	support.getElementsByClassName = rnative.test( document.getElementsByClassName );

	// Support: IE<10
	// Check if getElementById returns elements by name
	// The broken getElementById methods don't pick up programmatically-set names,
	// so use a roundabout getElementsByName test
	support.getById = assert( function( el ) {
		docElem.appendChild( el ).id = expando;
		return !document.getElementsByName || !document.getElementsByName( expando ).length;
	} );

	// ID filter and find
	if ( support.getById ) {
		Expr.filter[ "ID" ] = function( id ) {
			var attrId = id.replace( runescape, funescape );
			return function( elem ) {
				return elem.getAttribute( "id" ) === attrId;
			};
		};
		Expr.find[ "ID" ] = function( id, context ) {
			if ( typeof context.getElementById !== "undefined" && documentIsHTML ) {
				var elem = context.getElementById( id );
				return elem ? [ elem ] : [];
			}
		};
	} else {
		Expr.filter[ "ID" ] = function( id ) {
			var attrId = id.replace( runescape, funescape );
			return function( elem ) {
				var node = typeof elem.getAttributeNode !== "undefined" &&
					elem.getAttributeNode( "id" );
				return node && node.value === attrId;
			};
		};

		// Support: IE 6 - 7 only
		// getElementById is not reliable as a find shortcut
		Expr.find[ "ID" ] = function( id, context ) {
			if ( typeof context.getElementById !== "undefined" && documentIsHTML ) {
				var node, i, elems,
					elem = context.getElementById( id );

				if ( elem ) {

					// Verify the id attribute
					node = elem.getAttributeNode( "id" );
					if ( node && node.value === id ) {
						return [ elem ];
					}

					// Fall back on getElementsByName
					elems = context.getElementsByName( id );
					i = 0;
					while ( ( elem = elems[ i++ ] ) ) {
						node = elem.getAttributeNode( "id" );
						if ( node && node.value === id ) {
							return [ elem ];
						}
					}
				}

				return [];
			}
		};
	}

	// Tag
	Expr.find[ "TAG" ] = support.getElementsByTagName ?
		function( tag, context ) {
			if ( typeof context.getElementsByTagName !== "undefined" ) {
				return context.getElementsByTagName( tag );

			// DocumentFragment nodes don't have gEBTN
			} else if ( support.qsa ) {
				return context.querySelectorAll( tag );
			}
		} :

		function( tag, context ) {
			var elem,
				tmp = [],
				i = 0,

				// By happy coincidence, a (broken) gEBTN appears on DocumentFragment nodes too
				results = context.getElementsByTagName( tag );

			// Filter out possible comments
			if ( tag === "*" ) {
				while ( ( elem = results[ i++ ] ) ) {
					if ( elem.nodeType === 1 ) {
						tmp.push( elem );
					}
				}

				return tmp;
			}
			return results;
		};

	// Class
	Expr.find[ "CLASS" ] = support.getElementsByClassName && function( className, context ) {
		if ( typeof context.getElementsByClassName !== "undefined" && documentIsHTML ) {
			return context.getElementsByClassName( className );
		}
	};

	/* QSA/matchesSelector
	---------------------------------------------------------------------- */

	// QSA and matchesSelector support

	// matchesSelector(:active) reports false when true (IE9/Opera 11.5)
	rbuggyMatches = [];

	// qSa(:focus) reports false when true (Chrome 21)
	// We allow this because of a bug in IE8/9 that throws an error
	// whenever `document.activeElement` is accessed on an iframe
	// So, we allow :focus to pass through QSA all the time to avoid the IE error
	// See https://bugs.jquery.com/ticket/13378
	rbuggyQSA = [];

	if ( ( support.qsa = rnative.test( document.querySelectorAll ) ) ) {

		// Build QSA regex
		// Regex strategy adopted from Diego Perini
		assert( function( el ) {

			var input;

			// Select is set to empty string on purpose
			// This is to test IE's treatment of not explicitly
			// setting a boolean content attribute,
			// since its presence should be enough
			// https://bugs.jquery.com/ticket/12359
			docElem.appendChild( el ).innerHTML = "<a id='" + expando + "'></a>" +
				"<select id='" + expando + "-\r\\' msallowcapture=''>" +
				"<option selected=''></option></select>";

			// Support: IE8, Opera 11-12.16
			// Nothing should be selected when empty strings follow ^= or $= or *=
			// The test attribute must be unknown in Opera but "safe" for WinRT
			// https://msdn.microsoft.com/en-us/library/ie/hh465388.aspx#attribute_section
			if ( el.querySelectorAll( "[msallowcapture^='']" ).length ) {
				rbuggyQSA.push( "[*^$]=" + whitespace + "*(?:''|\"\")" );
			}

			// Support: IE8
			// Boolean attributes and "value" are not treated correctly
			if ( !el.querySelectorAll( "[selected]" ).length ) {
				rbuggyQSA.push( "\\[" + whitespace + "*(?:value|" + booleans + ")" );
			}

			// Support: Chrome<29, Android<4.4, Safari<7.0+, iOS<7.0+, PhantomJS<1.9.8+
			if ( !el.querySelectorAll( "[id~=" + expando + "-]" ).length ) {
				rbuggyQSA.push( "~=" );
			}

			// Support: IE 11+, Edge 15 - 18+
			// IE 11/Edge
```

```
Set_Color_Depth(1)
Cls
Box(0,0,639,479,1)
Text(320,240,"Color Depth Example",1,1,0,1)
Wait_Key()
End
```

1 day

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

At vero eos et accusamus et iusto odio dignissimos ducimus qui blanditiis praesentium voluptatum deleniti atque corrupti quos dolores et quas molestias excepturi sint occaecati cupiditate non provident, similique sunt in culpa qui officia deserunt mollitia animi, id est laborum et dolorum fuga. Et harum quidem rerum facilis est et expedita distinctio. Nam libero tempore, cum soluta nobis est eligendi optio cumque nihil impedit quo minus id quod maxime placeat facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat.

$$\sum_{i=1}^{n} \frac{1}{i^2} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots$$

1 day

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

$$\theta = \theta_0 - \alpha \cdot \nabla_\theta \cdot \ell(\cdot, \cdot) \cdot \cdot \cdot

```
plot(ggplot_frame)
```

```
## Warning: Removed 1 rows containing missing values (geom_point).
```



Ch 1 Ch 2 Ch 3 Ch 4 Ch 5 Ch 6 Ch 7 Ch 8 Ch 9 Ch 10 Ch 11 Ch 12 Ch 13 Ch 14 Ch 15 Ch 16 Ch 17 Ch 18 Ch 19 Ch 20 Ch

# Lecture 1: Introduction to Linear Algebra

Prof. Gilbert Strang, MIT

## 1.1 Vectors and Linear Combinations

A vector $\mathbf{v}$ in $\mathbb{R}^n$ is an ordered list of $n$ numbers. Vectors can be added and scaled: $\mathbf{v} + \mathbf{w}$ and $c\mathbf{v}$ are also in $\mathbb{R}^n$.

A linear combination of $\mathbf{v}$ and $\mathbf{w}$ is $c\mathbf{v} + d\mathbf{w}$ for any scalars $c$ and $d$. The set of all linear combinations fills a plane through the origin (unless $\mathbf{v}$ and $\mathbf{w}$ point in the same direction).

## 1.2 Lengths and Dot Products

The length of $\mathbf{v}$ is $\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$. The dot product $\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + \cdots + v_n w_n$.

The angle $\theta$ between $\mathbf{v}$ and $\mathbf{w}$ satisfies $\cos\theta = \dfrac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \, \|\mathbf{w}\|}$.

$$\cos\theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \, \|\mathbf{w}\|} \qquad (1.1)$$

## 1.3 Matrices

A matrix $A$ with $m$ rows and $n$ columns is an $m \times n$ array. If $A$ has columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$, then $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$.

```
# Multiply A by x: combine the columns of A
def matvec(A, x):
    m, n = len(A), len(A[0])
    return [sum(A[i][j] * x[j] for j in range(n)) for i in range(m)]
```

## 1.4 Solving $A\mathbf{x} = \mathbf{b}$

Elimination reduces $A$ to an upper triangular $U$. The pivots are the diagonal entries of $U$. Back substitution then solves $U\mathbf{x} = \mathbf{c}$.

Name: Ayush Agrawal Roll No: 2019101049

Q. 1. Explain the concept of Language Model. Discuss how collocations help in building natural sounding language models?

Ans. A language model is a probability distribution over sequences of words. This means that, according to a given language model, we are able to determine the probability of occurrence of a number of words in a particular sequence. Also, we can predict the probability of occurrence of a word by looking at the previous few words. Language models are used in natural language processing applications such as machine translation and speech recognition.

Collocations are expressions of multiple words which commonly co-occur.

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Recurse -Include *.ps1 |
Unblock-File
```

$x_1$
$x_2$
$x_3$

Two equations, two unknowns

$$\begin{aligned} 2x - y &= 0 \\ -x + 2y &= 3 \end{aligned}$$

The first equation $2x - y = 0$ produces a straight line. Solving for $y$ gives $y = 2x$. Plotting the points $(x, y) = (0, 0)$ and $(1, 2)$ and drawing the line through them produces the first graph.

The second equation $-x + 2y = 3$ produces another straight line. Solving for $y$ gives $y = \frac{1}{2}x + \frac{3}{2}$. Plotting the points $(x, y) = (3, 3)$ and $(1, 2)$ and drawing the line through them produces the second graph.

The point where the two lines intersect is the solution to both equations. This is the only point that lies on both lines. In this case, the solution is $(x, y) = (1, 2)$.

Three equations, three unknowns

$$\begin{aligned} 2x - y &= 0 \\ -x + 2y - z &= -2 \\ -3y + 4z &= 4 \end{aligned}$$

The first equation $2x - y = 0$ produces a plane. The second equation $-x + 2y - z = -2$ produces another plane. The third equation $-3y + 4z = 4$ produces a third plane.

The solution to all three equations is the point where all three planes intersect. In this case, the solution is $(x, y, z) = (0, 0, 1)$.

Four equations, four unknowns

$$\begin{aligned} 2x - y &= 0 \\ -x + 2y - z &= -2 \\ -3y + 4z - t &= 4 \\ -z + 2t &= 4 \end{aligned}$$

The first equation $2x - y = 0$ produces a plane in four-dimensional space. The second equation $-x + 2y - z = -2$ produces another plane. The third equation $-3y + 4z - t = 4$ produces a third plane. The fourth equation $-z + 2t = 4$ produces a fourth plane.

The solution to all four equations is the point where all four planes intersect. In this case, the solution is $(x, y, z, t) = (0, 0, 1, 2)$.

Row picture: I drew this picture in two dimensions, with two equations and two unknowns. The equations produced two lines, and the solution was the point where the two lines intersected. This is called the row picture because each equation corresponds to a row of the matrix.

Column picture: Now I'm drawing the same problem in four dimensions, with four equations and four unknowns. The equations produce four planes, and the solution is the point where all four planes intersect. This is called the column picture because each column of the matrix corresponds to a vector.

Matrix form: The equations can be written in matrix form as $Ax = b$, where $A$ is the matrix of coefficients, $x$ is the vector of unknowns, and $b$ is the vector of constants. In this case, the matrix is

$$A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -3 & 4 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

the vector of unknowns is

$$x = \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix}$$

and the vector of constants is

$$b = \begin{bmatrix} 0 \\ -2 \\ 4 \\ 4 \end{bmatrix}$$

Solving the matrix equation $Ax = b$ will give the solution to the four equations.

Singular case: In the singular case, the four planes do not intersect at a point. Instead, they intersect along a line, or they are parallel and do not intersect at all. This is called the singular case because the matrix of coefficients is singular, which means that it does not have an inverse.

Solving $Ax = b$: To solve the equation $Ax = b$, we can use Gaussian elimination. This is a method for solving systems of linear equations by transforming the matrix of coefficients into an upper triangular matrix. Then we can solve for the unknowns by back substitution.

Elimination: Elimination is the process of transforming the matrix of coefficients into an upper triangular matrix. This is done by adding multiples of one row to another row. The goal is to create zeros below the diagonal of the matrix.

Back substitution: Back substitution is the process of solving for the unknowns in an upper triangular matrix. We start with the last row and solve for the last unknown. Then we substitute this value into the second-to-last row and solve for the second-to-last unknown. We continue this process until we have solved for all of the unknowns.

Inverse matrices: The inverse of a matrix $A$ is a matrix $A^{-1}$ such that $AA^{-1} = A^{-1}A = I$, where $I$ is the identity matrix. The identity matrix is a square matrix with ones on the diagonal and zeros everywhere else. If a matrix has an inverse, then it is called invertible or nonsingular. If a matrix does not have an inverse, then it is called singular or noninvertible.

Finding $A^{-1}$: To find the inverse of a matrix $A$, we can use the following formula:

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

where $\det(A)$ is the determinant of $A$ and $\operatorname{adj}(A)$ is the adjugate of $A$. The determinant of a matrix is a scalar value that can be used to determine whether the matrix is invertible. The adjugate of a matrix is a matrix that is obtained by taking the transpose of the cofactor matrix.

Solving $Ax = b$ using $A^{-1}$: If we have the inverse of $A$, then we can solve the equation $Ax = b$ by multiplying both sides by $A^{-1}$:

$$\begin{aligned} Ax &= b \\ A^{-1}Ax &= A^{-1}b \\ Ix &= A^{-1}b \\ x &= A^{-1}b \end{aligned}$$

Therefore, the solution to $Ax = b$ is $x = A^{-1}b$.

Determinants: The determinant of a matrix is a scalar value that can be used to determine whether the matrix is invertible. It can also be used to find the area of a parallelogram or the volume of a parallelepiped. The determinant of a matrix $A$ is denoted by $\det(A)$ or $|A|$.

Cofactor expansion: The cofactor expansion is a method for finding the determinant of a matrix. It involves expanding the determinant along a row or column. The cofactor of an element $a_{ij}$ is the determinant of the matrix that is obtained by deleting the $i$th row and $j$th column of $A$, multiplied by $(-1)^{i+j}$. The determinant of $A$ is then the sum of the products of the elements of any row or column and their cofactors.

Properties of determinants: The determinant of a matrix has several important properties. For example, the determinant of a product of two matrices is the product of their determinants. The determinant of a matrix is also equal to the determinant of its transpose. If a matrix has a row or column of zeros, then its determinant is zero. If a matrix has two identical rows or columns, then its determinant is zero.

Applications of determinants: Determinants have many applications in linear algebra and other fields. They can be used to find the area of a parallelogram or the volume of a parallelepiped. They can also be used to find the eigenvalues of a matrix, which are important in many applications, such as stability analysis and quantum mechanics.

Eigenvalues and eigenvectors: Eigenvalues and eigenvectors are important concepts in linear algebra. An eigenvector of a matrix $A$ is a nonzero vector $v$ such that $Av = \lambda v$ for some scalar $\lambda$. The scalar $\lambda$ is called the eigenvalue corresponding to the eigenvector $v$. Eigenvalues and eigenvectors can be used to find the solutions to systems of linear differential equations, to analyze the stability of dynamical systems, and to find the principal components of a data set.

Finding eigenvalues and eigenvectors: To find the eigenvalues of a matrix $A$, we can solve the characteristic equation $\det(A - \lambda I) = 0$. The solutions to this equation are the eigenvalues of $A$. To find the eigenvector corresponding to an eigenvalue $\lambda$, we can solve the equation $(A - \lambda I)v = 0$.

Diagonalization: A matrix $A$ is diagonalizable if there exists a matrix $P$ such that $P^{-1}AP = D$, where $D$ is a diagonal matrix. The matrix $P$ is called the matrix of eigenvectors of $A$, and the diagonal entries of $D$ are the eigenvalues of $A$. Diagonalization can be used to simplify many calculations involving matrices, such as finding powers of matrices and solving systems of linear differential equations.

Singular value decomposition: The singular value decomposition (SVD) is a factorization of a matrix $A$ into the product of three matrices: $A = U\Sigma V^T$, where $U$ and $V$ are orthogonal matrices and $\Sigma$ is a diagonal matrix. The diagonal entries of $\Sigma$ are the singular values of $A$, which are the square roots of the eigenvalues of $A^TA$. The SVD can be used to find the rank of a matrix, to find the pseudoinverse of a matrix, and to solve least squares problems.

Applications of singular value decomposition: The singular value decomposition has many applications in data analysis, image processing, and machine learning. For example, it can be used to reduce the dimensionality of a data set, to compress images, and to find the principal components of a data set.

Conclusion: In this lecture, we have introduced the basic concepts of linear algebra, including systems of linear equations, matrices, determinants, eigenvalues, and eigenvectors. We have also discussed some of the applications of linear algebra in various fields. In the next lecture, we will continue our study of linear algebra by exploring more advanced topics, such as vector spaces, linear transformations, and inner product spaces.

```
<!--
  ~ Copyright 2018 Google LLC
  ~
  ~ Licensed under the Apache License, Version 2.0 (the "License");
  ~ you may not use this file except in compliance with the License.
  ~ You may obtain a copy of the License at
  ~
  ~      https://www.apache.org/licenses/LICENSE-2.0
  ~
  ~ Unless required by applicable law or agreed to in writing, software
  ~ distributed under the License is distributed on an "AS IS" BASIS,
  ~ WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  ~ See the License for the specific language governing permissions and
  ~ limitations under the License.
  -->

<resources>
    <string name="app_name">Android Things Blinky Sample</string>
</resources>
```

```
<resources>
    <string name="app_name">Android Things Blinky Sample</string>
</resources>
```

```
<resources>
    <string name="app_name">Android Things Blinky Sample</string>
</resources>
```

```
<resources>
    <string name="app_name">Android Things Blinky Sample</string>
</resources>
```

```
<resources>
    <string name="app_name">Android Things Blinky Sample</string>
</resources>
```

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Recurse -Include *.ps1 |
    Unblock-File
```

1.1 Motivation

Let $\mathcal{M}$ be a smooth manifold of dimension $n$, and let $\omega \in \Omega^2(\mathcal{M})$ be a closed, non-degenerate 2-form. The pair $(\mathcal{M}, \omega)$ is called a symplectic manifold.

1.2 Hamiltonian Mechanics

Let $H : \mathcal{M} \to \mathbb{R}$ be a smooth function. The Hamiltonian vector field $X_H$ is defined by the equation $\iota_{X_H}\omega = dH$. In local coordinates $(q^1, \dots, q^n, p_1, \dots, p_n)$ where $\omega = \sum_{i=1}^{n} dq^i \wedge dp_i$, Hamilton's equations read

$$\dot{q}^i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q^i}.$$

1.3 Poisson Brackets

The Poisson bracket of two functions $f, g \in C^\infty(\mathcal{M})$ is $\{f, g\} = \omega(X_f, X_g)$. It satisfies the Jacobi identity $\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0$, which makes $(C^\infty(\mathcal{M}), \{\cdot,\cdot\})$ a Lie algebra.

1.4 Liouville's Theorem

Liouville's theorem states that the flow of $X_H$ preserves the symplectic volume form $\omega^n / n!$. Consequently, the phase space density is constant along trajectories.

$$\int_{\mathcal{M}} \frac{\omega^n}{n!} = \int_{\mathcal{M}} \frac{\omega^n}{n!} \circ \varphi_t^{-1}$$

1.5 Integrable Systems

A system with $n$ degrees of freedom is integrable if it admits $n$ independent first integrals $F_1 = H, F_2, \dots, F_n$ in involution, $\{F_i, F_j\} = 0$. The Liouville–Arnold theorem then guarantees that the level sets $\{F_1 = c_1, \dots, F_n = c_n\}$ are invariant tori on which the flow is linear.

1.6 Action–Angle Variables

Introduce the action variables $I_i = \frac{1}{2\pi} \oint_{\gamma_i} p \, dq$ and the conjugate angle variables $\theta^i$. In these coordinates the Hamiltonian depends only on the actions, $H = H(I_1, \dots, I_n)$, and the equations of motion become $\dot{I}_i = 0$, $\dot{\theta}^i = \omega_i(I)$.

1.7 Perturbation Theory

Consider $H = H_0 + \varepsilon H_1$ with $H_0$ integrable and $\varepsilon \ll 1$. Resonant tori with $k \cdot \omega(I) = 0$ for some $k \in \mathbb{Z}^n \setminus \{0\}$ are destroyed, while non-resonant tori survive by the KAM theorem.

1.8 Summary

This chapter has introduced the basic objects of symplectic geometry and Hamiltonian mechanics. The next chapter develops the theory of Lagrangian submanifolds.

```
Set-ExecutionPolicy
```

```
Bypass
```

```
-Scope Process -Force; [System.Net.ServicePoint
Manager]::SecurityProtocol = [System.Net.
ServicePointManager]::SecurityProtocol -bor 3072;
iex ((New-Object System.Net.WebClient).
DownloadString('https://community.chocolatey.org/
install.ps1'))
```

```
\begin{table}
\caption{Comparison of error categories between two models.}
\label{tab:error_comparison}
\begin{tabular}{lrrr}
\toprule
Error Category & Model A & Model B & $\Delta$ \\
\midrule
Factual inconsistency & 23.4\% & 15.1\% & $-8.3$ \\
Omission of key information & 18.7\% & 12.9\% & $-5.8$ \\
Grammatical errors & 9.2\% & 8.8\% & $-0.4$ \\
Hallucination & 14.6\% & 7.3\% & $-7.3$ \\
\bottomrule
\end{tabular}
\end{table}
```

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Recurse -Include *.ps1 |
    Unblock-File
```

b
a
c

# Chapter 2.1: Ohm's Law

## Learning Objectives

By the end of this section, you should be able to:

- State Ohm's law and identify the quantities it relates
- Distinguish between resistance and conductance
- Calculate power dissipated in a resistor using multiple forms of the power equation
- Apply Ohm's law to simple circuits with one resistor

## Introduction

Georg Simon Ohm published his findings in 1827 after years of experimental work. His discovery is deceptively simple: for many materials, the current flowing through a conductor is directly proportional to the voltage across it. This relationship, now known as Ohm's law, is the foundation upon which nearly all of circuit analysis is built.

Not every material obeys Ohm's law. Diodes, transistors, and incandescent light bulbs all exhibit nonlinear behavior. But resistors, wires, and many other components do follow it closely over a wide range of conditions, and we will spend the next several chapters working exclusively with such linear elements.

## Ohm's Law

For a resistor with resistance $R$ carrying current $i$ from its positive terminal to its negative terminal, the voltage across the resistor is

$$v = iR$$

where $v$ is measured in volts (V), $i$ in amperes (A), and $R$ in ohms ($\Omega$). Solving for current gives the alternative form

$$i = \frac{v}{R}$$

The reciprocal $G = 1/R$ is called the conductance, measured in siemens (S). Some engineers prefer working with conductance because it makes the equations look more symmetric, but we will use resistance throughout this course.

## Power Dissipated in a Resistor

Whenever current flows through a resistor, electrical energy is converted into heat. The rate at which this conversion occurs is the power dissipated by the resistor. Using Ohm's law, we can write power in three equivalent forms:

$$p = vi = i^2 R = \frac{v^2}{R}$$

Which form is most convenient depends on what you already know. If you have already found $i$, the middle form is quickest. If you have the voltage directly, the last form avoids an unnecessary multiplication. The first form is rarely the most convenient, but it is the most general, since it applies to any two-terminal element, not just resistors.

## Worked Example 2.1.1

A $470\ \Omega$ resistor is connected across a $12\ \text{V}$ battery. Find the current through the resistor and the power it dissipates.

**Solution.** Using $i = v/R$:

$$i = \frac{12\ \text{V}}{470\ \Omega} \approx 25.5\ \text{mA}$$

Using $p = vi$:

$$p = (12\ \text{V})(25.5\ \text{mA}) \approx 306\ \text{mW}$$

As a check, $p = i^2 R = (25.5\ \text{mA})^2 (470\ \Omega) \approx 306\ \text{mW}$, which agrees.

## Worked Example 2.1.2

A resistor dissipates $0.5\ \text{W}$ when $10\ \text{V}$ appears across it. Find its resistance.

**Solution.** Using $R = v^2/p$:

$$R = \frac{(10\ \text{V})^2}{0.5\ \text{W}} = 200\ \Omega$$

## Practice Problems

1. A $1\ \text{k}\Omega$ resistor has $5\ \text{V}$ across it. Find the current and the power dissipated.
2. A $100\ \Omega$ resistor carries $200\ \text{mA}$. Find the voltage across it and the power dissipated.
3. A $2.2\ \text{k}\Omega$ resistor dissipates $1\ \text{W}$. Find the maximum current it can safely carry.
4. Two resistors, $R_1 = 100\ \Omega$ and $R_2 = 300\ \Omega$, are connected in series with a $20\ \text{V}$ source. Find the current through each resistor and the power dissipated by each.
5. Derive the conductance form of Ohm's law, $i = Gv$, starting from $v = iR$.

## Summary

- Ohm's law: $v = iR$, or equivalently $i = v/R$
- Conductance $G = 1/R$, measured in siemens (S)
- Power dissipated in a resistor: $p = vi = i^2 R = v^2/R$
- Ohm's law is a model that holds well for many materials over a wide range of conditions

## Exercises

1. Explain in your own words why the power dissipated in a resistor is always positive, regardless of the direction of the current.
2. A resistor is rated at $1/4\ \text{W}$. What is the largest voltage you can safely apply across it? Show your reasoning.
3. Two students measure the same resistor and get slightly different values. Suggest two reasons why this might happen.
4. Sketch a graph of $v$ versus $i$ for a $50\ \Omega$ resistor. What is the slope of the line?
5. Why do you think Ohm's law is so useful in practice, even though it is only an approximation?

Ref. 4 with Patch - Install Playstation™...

Ref. 4 with Patch - Install Playstation™...

Ref. 4 with Patch - Install Playstation™...

Ref.
Ref. Ref. Ref.

```
sudo systemctl daemon-reload
sudo systemctl start nginx
sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2020-05-17 15:48:51 UTC; 1min 0s ago
       Docs: man:nginx(8)
    Process: 2369 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 2370 (nginx)
      Tasks: 2 (limit: 1152)
     Memory: 3.6M
     CGroup: /system.slice/nginx.service
             ├─2370 nginx: master process /usr/sbin/nginx -g daemon on; master_process on;
             └─2371 nginx: worker process

May 17 15:48:51 ip-172-31-31-160 systemd[1]: Starting A high performance web server and a reverse proxy server...
May 17 15:48:51 ip-172-31-31-160 systemd[1]: Started A high performance web server and a reverse proxy server.
```

1
0
−1
x

## Vectors as Arrows in Space

Let's start with the most basic idea in linear algebra: a vector is an arrow in space. It has a direction and a length, but it doesn't have a fixed position. You can slide it around anywhere, as long as you don't change its direction or length.

### 2D Vectors

In two dimensions, a vector is an arrow on a flat plane. We can describe it using two numbers, like $\vec{v} = (3, 2)$. This means: from wherever you are, move 3 units to the right and 2 units up. That's the arrow.




### 3D Vectors

Add a third dimension and you get a vector like $\vec{w} = (1, 4, -2)$. Now you move 1 unit right, 4 units up, and 2 units backward (the negative sign). It's the same idea, just in more room.

### Why Arrows Matter in AI

Vectors are the fundamental way we represent data in machine learning. A single image, a sentence, or a user's preferences can all be turned into a vector. The arrow's direction and length then tell us how similar or different two pieces of data are.

```
Set_Color_Depth(1)
Cls
Box(0,0,639,479,1)
Text(320,240,"Hello World!",1,1,0,0)
Flip
WaitKey()
End
```

Vectors in 2D and 3D: Lines in a Coordinate Plane and Solid Geometry

Introduction

Vectors are fundamental objects in mathematics, physics, and engineering that help us describe quantities with both magnitude and direction. In this section, we explore vectors in two and three dimensional spaces, focusing on how they can be used to represent lines in a coordinate plane and in solid geometry. Understanding vectors in these contexts is crucial for solving problems in navigation, computer graphics, and many other fields.

Vectors in 2D and 3D Space

A vector is a mathematical object that has both a magnitude (or length) and a direction. Vectors can be represented geometrically as arrows in space, where the length of the arrow corresponds to the magnitude of the vector, and the direction of the arrow indicates the direction of the vector. Vectors are usually denoted by boldface letters, such as $\mathbf{v}$, or by letters with an arrow above them, such as $\vec{v}$.

In two dimensional space, a vector can be represented by an ordered pair of real numbers, $\mathbf{v} = \langle v_1, v_2 \rangle$, where $v_1$ and $v_2$ are the components of the vector along the $x$ and $y$ axes, respectively. Similarly, in three dimensional space, a vector can be represented by an ordered triple of real numbers, $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$, where $v_1$, $v_2$, and $v_3$ are the components of the vector along the $x$, $y$, and $z$ axes, respectively.

Vectors can be added together and multiplied by scalars (real numbers) to produce new vectors. The sum of two vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ is defined as $\mathbf{u} + \mathbf{v} = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle$. The scalar multiple of a vector $\mathbf{v}$ by a scalar $c$ is defined as $c\mathbf{v} = \langle cv_1, cv_2, cv_3 \rangle$. These operations have important geometric interpretations, which we will explore in more detail later.

Lines in a Coordinate Plane

One of the most basic applications of vectors is in describing lines in a coordinate plane. A line can be uniquely determined by a point on the line and a vector that is parallel to the line. This vector is called the **direction vector** of the line. Given a point $P_0 = (x_0, y_0)$ on a line and a direction vector $\mathbf{d} = \langle a, b \rangle$, we can write the parametric equations of the line as

$$x = x_0 + at, \quad y = y_0 + bt,$$

where $t$ is a real number called the parameter. As $t$ varies over all real numbers, the point $(x, y)$ traces out the entire line.

Example 1.1.1

Find the parametric equations of the line passing through the point $P_0 = (2, -1)$ and parallel to the vector $\mathbf{d} = \langle 3, 4 \rangle$.

Solution

Using the formula above, we have $x = 2 + 3t$ and $y = -1 + 4t$. Therefore, the parametric equations of the line are $x = 2 + 3t$, $y = -1 + 4t$.

Lines in Solid Geometry

In three dimensional space, a line can also be uniquely determined by a point on the line and a direction vector. The parametric equations of a line in 3D are similar to those in 2D, with an additional equation for the $z$ coordinate. Given a point $P_0 = (x_0, y_0, z_0)$ on a line and a direction vector $\mathbf{d} = \langle a, b, c \rangle$, the parametric equations of the line are

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct,$$

where $t$ is a real number called the parameter. As $t$ varies over all real numbers, the point $(x, y, z)$ traces out the entire line.

Example 1.1.2

Find the parametric equations of the line passing through the point $P_0 = (1, 0, -2)$ and parallel to the vector $\mathbf{d} = \langle 2, -3, 5 \rangle$.

Solution

Using the formula above, we have $x = 1 + 2t$, $y = -3t$, and $z = -2 + 5t$. Therefore, the parametric equations of the line are $x = 1 + 2t$, $y = -3t$, $z = -2 + 5t$.

Conclusion

In this section, we have explored vectors in two and three dimensional spaces, focusing on how they can be used to represent lines in a coordinate plane and in solid geometry. We have seen that vectors can be added together and multiplied by scalars, and that these operations have important geometric interpretations. We have also seen that lines in 2D and 3D can be uniquely determined by a point on the line and a direction vector, and that the parametric equations of a line can be easily obtained from this information. In the next section, we will explore the dot product and the cross product, two important operations on vectors that have many applications in physics and engineering.

```
\\
\\
\\
```

```
.gitignore Modified A M ??

M  .gitignore
M  .vscode/settings.json
?? .vscode/launch.json
?? .vscode/tasks.json
?? Untitled-1.ipynb
?? Untitled-2.ipynb
?? Untitled-3.ipynb
?? Untitled-4.ipynb
?? Untitled-5.ipynb
?? Untitled-6.ipynb
?? Untitled-7.ipynb
?? Untitled-8.ipynb
?? Untitled-9.ipynb
?? Untitled-10.ipynb
?? Untitled-11.ipynb
?? Untitled-12.ipynb
?? Untitled-13.ipynb
?? Untitled-14.ipynb
?? Untitled-15.ipynb
?? Untitled-16.ipynb
?? Untitled-17.ipynb
?? Untitled-18.ipynb
?? Untitled-19.ipynb
?? Untitled-20.ipynb
??
```

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Recurse -Include *.ps1 | Unblock-File
```

```
.\Install-ExchangeCertificate.ps1
```

Resistor
Capacitor
Inductor
Transistor
Voltmeter
Bulb
Battery
Battery in Cell
DC source
Diode
Switch
Light dependent resistor
Variable resistor

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Document</title>
</head>
<body>
  <h1>Hello, world!</h1>
  <p>Welcome to your first HTML document.</p>
</body>
</html>
```




Ok, I don't think you can't

Alright, I'll just continue with the next one then. Ok, next one:




Alright



Alright



Ok


Hi, I'm Ivan - a human - not a bot!

```
# Author: Przemyslaw Kaminski <cgmb@slackware.com>
# Modified by: Eric Hameleers <alien@slackware.com>
# Modified by: Willy Sudiarto Raharjo <willysr@slackware.com>
# Modified by: Matteo Rossini <matteorossini@gmail.com>
# Modified by: Robby Workman <rworkman@slackware.com>
# Modified by: Niels Horn <niels.horn@slackware.com>
# Modified by: Christoph Willing <chris.willing@linux.com>
# Modified by: Mauro Giachero <mauro.giachero@gmail.com>
# Modified by: Alan Hicks <alan@lizella.net>
# Modified by: Patrick J Volkerding <volkerdi@slackware.com>
# Modified by: Vincent Batts <vbatts@hashbangbash.com>
# Modified by: Daniel de Kok <sonny@xs4all.nl>
# Modified by: David Somero <dsomero@gmail.com>
# Modified by: Heinz Wiesinger <pprkut@slackware.com>
# Modified by: Fernando Lopez-Lezcano <nando@ccrma.stanford.edu>
# Modified by: Robby Workman <rworkman@slackware
```

Apr 1 [user] Audrey enrolled at ?

Apr [user] Audrey's ?

Apr 1 [user] Audrey's [user] ?

Apr 1 [user] Audrey's [user] 9781984817624 Physics : Concise Psychological ?

Apr 1 [user] ?

Apr 1 [user] ?

Apr 1 [user] ?

Apr 1 [user] Audrey's [user] ?

Apr 1 [user] Audrey's [user] ?

Apr 1 [user] ?

Apr 1 [user] Audrey's [user] ?

Apr 1 [user] Audrey's [user] ?

Apr 1 [user] Audrey's

Title: Jeremy Bearer's Bear Lore

Author: Jeremy Bearer

Release Date: [EBook #11111]
[Last updated: February 11, 2007]
Language: English
Character set encoding: ISO-8859-1
Produced by: Jeremy Bearer, Jeremy Bearer's Bear Lore Proofreaders, and Project Gutenberg Proofreading Team

*** START OF THIS PROJECT GUTENBERG EBOOK JEREMY BEARER'S BEAR LORE ***

[Illustration: Jeremy Bearer's Bear Lore]

Jeremy Bearer's Bear Lore

Jeremy Bearer's Bear Lore

Jeremy Bearer's Bear Lore

```
<div class="page">

<p>Lorem ipsum dolor sit amet, <b>consectetur adipiscing elit</b>. Fusce a pretium ut leo gravida. Etiam faucibus tellus, <i>aliquet</i> <u>aliquam</u> risus <b><i>fermentum</i></b> tempus. Nullam <b>ac</b> <i>ligula</i> <u>ut</u> <b><i><u>nisi</u></i></b> <u><b>ullamcorper</b></u> <u><i>ullamcorper</i></u> <b><u><i>ut</i></u></b> <i><u><b>ligula</b></u></i> <b><i><u>ut</u></i></b> <u><b><i>nisi</i></b></u> <u><b><i>ullamcorper</i>
```

```
<div id="_" class="layout">

<div id="_" class="header">

</div>
```

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Path "C:\Program Files\Java\jdk-17" -Recurse -Filter "java.exe"
```

```
Test-NetConnection -ComputerName 192.168.1.100 -Port 8080
```

y axis
x axis
y axis
x axis

Author block

Prof. Dr. Ahmed Ali Abdalla
Department of Mathematics, Faculty of Science, Cairo University, Giza, Egypt

Abstract

We develop the determinant function axiomatically, prove its existence and uniqueness, derive the cofactor expansion along any row or column, and establish the multiplicative property $\det(AB) = \det(A)\det(B)$. We then use these results to obtain Cramer's rule for square systems with nonzero determinant and to characterize invertibility of a matrix in terms of its determinant. Worked examples and exercises are included throughout.

Publication info

Typeset in LaTeX, December 2024.

Rule

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. If $ad - bc \neq 0$, then $A$ is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

If $ad - bc = 0$, then $A$ is not invertible.

Proof

Multiply $A$ by the proposed inverse. The $(1,1)$ entry of the product is $\frac{ad - bc}{ad - bc} = 1$, and the $(1,2)$ entry is $\frac{-ab + ba}{ad - bc} = 0$. The $(2,1)$ and $(2,2)$ entries are computed in the same way and also equal $1$ and $0$. Hence the product is the identity matrix. Conversely, if $ad - bc = 0$, then $ad = bc$, so the rows of $A$ are proportional: $(c, d) = \frac{c}{a}(a, b)$ when $a \neq 0$, and similarly in the other cases. A nontrivial linear combination of the rows therefore vanishes, so $A$ is not invertible. $\square$

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```
int i=0;char buf[] = {"abcdefghijklmnopqrstuvwxyz"};
while(i<26){printf("%c",buf[i]);i++;}
```

```
int i=0;while(i<26){printf("%c",'a'+i
```

EXHIBIT A

IN THE SUPREME COURT
STATE OF NEW YORK
COUNTY OF ERIE

Index No. 809000/2019

Plaintiff designates Erie County as the place of trial.

The basis of venue is Plaintiff's residence

SUMMONS

To the above named defendant:

YOU are hereby summoned to answer the complaint in this action and to serve a copy of your answer, or, if the complaint is not served with this summons, to serve a notice of appearance, on the Plaintiff's attorneys within 20 days after the service of this summons, exclusive of the day of service (or within 30 days after the service is complete if this summons is not personally delivered to you within the State of New York); and in case of your failure to appear or answer, judgment will be taken against you by default for the relief demanded in the complaint.

Dated: March 29, 2019

Buffalo, New York

Yours, etc.

**THE LAW OFFICE OF JAMES P. KEANE, ESQ.**

Attorneys for Plaintiff

**JAMES P. KEANE, ESQ.**

Of Counsel

1250 Scottsville Road, Suite 225

Rochester, New York 14624

(585) 436-7518

TO:

**Defendant**

**Address unknown**

**Defendant**

**Address unknown**

**Defendant**

**Address unknown**

IN THE SUPREME COURT
STATE OF NEW YORK
COUNTY OF ERIE

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**COMPLAINT**

Plaintiff, by and through his attorneys, THE LAW OFFICE OF JAMES P. KEANE, ESQ., as and for his Complaint, alleges upon information and belief as follows:

1. That at all times hereinafter mentioned, Plaintiff was a resident of the County of Erie, State of New York.

2. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was a foreign corporation duly organized and existing under and by virtue of the laws of the State of Ohio.

3. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was authorized to do business in the State of New York.

4. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was doing business in the State of New York.

5. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was transacting business in the State of New York.

6. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was engaged in the business of insurance.

7. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was licensed to write insurance in the State of New York.

8. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was licensed to write insurance in the State of New York by the New York State Department of Financial Services.

9. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was authorized to write insurance in the State of New York by the New York State Department of Financial Services.

10. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was authorized to write insurance in the State of New York by the New York State Department of Financial Services pursuant to the Insurance Law of the State of New York.

11. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was authorized to write insurance in the State of New York by the New York State Department of Financial Services pursuant to the Insurance Law of the State of New York and the regulations promulgated thereunder.

12. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was authorized to write insurance in the State of New York by the New York State Department of Financial Services pursuant to the Insurance Law of the State of New York and the regulations promulgated thereunder and was in good standing.

13. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was authorized to write insurance in the State of New York by the New York State Department of Financial Services pursuant to the Insurance Law of the State of New York and the regulations promulgated thereunder and was in good standing and was not insolvent.

14. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was authorized to write insurance in the State of New York by the New York State Department of Financial Services pursuant to the Insurance Law of the State of New York and the regulations promulgated thereunder and was in good standing and was not insolvent and was not in receivership.

15. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was authorized to write insurance in the State of New York by the New York State Department of Financial Services pursuant to the Insurance Law of the State of New York and the regulations promulgated thereunder and was in good standing and was not insolvent and was not in receivership and was not in conservatorship.

16. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was authorized to write insurance in the State of New York by the New York State Department of Financial Services pursuant to the Insurance Law of the State of New York and the regulations promulgated thereunder and was in good standing and was not insolvent and was not in receivership and was not in conservatorship and was not in liquidation.

17. That at all times hereinafter mentioned, Defendant, **Progressive Casualty Insurance Company**, was authorized to write insurance in the State of New York by the New York State Department of Financial Services pursuant to the Insurance Law of the State of New York and the regulations promulgated thereunder and was in good standing and was not insolvent and was not in receivership and was not in conservatorship and was not in liquidation and was not in rehabilitation.

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```
<br />
<br />
<br />
```

Ein Sprichwort sagt, dass Geschichte sich wiederholt. Leider trifft es zu, dass es nur begrenzten Platz gibt, um Menschen in Gräber zu legen.
Stanisław Jerzy Lec

```
<?xml version="1.0"?>

<!-- ================================================================== -->
<!-- ANTIPATTERN: Every element is wrapped in a <div> with no semantic meaning -->
<!-- ================================================================== -->
<div>
  <div type="message">
    <div type="sender">
      <div type="username">Mr. Bones</div>
    </div>
    <div type="content">Mr. Bones is not registered for special events Team Trivia Halloween.</div>
  </div>
</div>

<!-- ================================================================== -->
<!-- ANTIPATTERN: Machine data in CDATA -->
<!-- Should be machine-readable but not in the text flow. Use key/value pairs, proper elements, or machine-readable attributes. -->
<!-- ================================================================== -->
<log_entry><![CDATA[2024-10-31 23:59:59 ERROR: Connection timeout after 30000ms
host=db-primary-01.internal port=5432 retry_count=3 pool_size=20 active=20 idle=0 waiting=47
stack_trace=at com.example.db.ConnectionPool.acquire(ConnectionPool.java:142)
at com.example.db.QueryExecutor.execute(QueryExecutor.java:89)
at com.example.app.UserService.findById(UserService.java:56)]]></log_entry>
```

Claim 1

A [redacted] method for the use of a [redacted] to assist through the operation, [redacted] use of the [redacted] [redacted] of [redacted],

the [redacted] [redacted] [redacted]

```
# Author: Krishna Kumar <krkumar@us.ibm.com>
# Copyright 2007 IBM
# Licensed under the Apache License, Version 2.0
# http://www.apache.org/licenses/LICENSE-2.0
# Maintained at www.ibm.com/developerworks/linux/linux390/perf/tuning_rhel.html
# Edited by Marcel Wiesweg <marcel.wiesweg@gmx.de> since 2011-10-20
# Functions to set various tunables. Some inspiration obtained from
# tuned and sysctl.conf man page.
#
# List of functions:
#
# Command line options set:
#
# Set/clear irq affinity for cpu0 (default is any cpu/all)
# cpu_affinity [clear]
# Set/clear transmit packet steering for cpu0 (default is any cpu/all)
# xps_affinity [clear]
# Set/clear receive packet steering for cpu0 (default is any cpu/all)
# rps_affinity [clear]
# Set/clear receive flow steering for cpu0 (default is any cpu/all)
# rfs_affinity [clear]
# Set/clear XPS and RPS/RFS together for cpu0 (default is any cpu/all)
# net_affinity [clear]
#
# Functions called:
#
# Get input parameters
# get_force
# get_if <interface> <ip/hostname>
#
# Logging:
# log_msg
# log_echo
#
# Basic system settings
# set_kernel_settings <sysctl.conf file to load>
# (RHEL4 only)
# set_time_wait <net.ipv4.tcp_tw_recycle> <net.ipv4.tcp_tw_reuse>
# set_ipv4_settings <net.ipv4.ip_local_port_range> <net.ipv4.tcp_max_syn_backlog>
# set_core_settings <net.core.rmem_max> <net.core.wmem_max>
# set_qdisc <interface> <fifo_pfifo_fast|sfq|tbf|htb|prio|cbq|red|etc.>
# set_txqueuelen <interface> <txqueuelen>
#
# Ethtool settings
# set_ring <interface> <rx|tx> <value>
# set_coalesce <interface> <on|off> <value>
# set_offload <interface> <rx|tx|sg|tso|ufo|gso|gro|lro|rxvlan|txvlan> <on|off>
# set_features <interface> <feature> <on|off>
# set_one_offload <interface> <on|off>
# set_all_offload <interface> <on|off>
#
# Set/clear irq affinity for cpu0 (default is any cpu/all)
# set_irq_affinity <interface> [clear]
# Set/clear transmit packet steering for cpu0 (default is any cpu/all)
# set_xps_affinity <interface> [clear]
# Set/clear receive packet steering for cpu0 (default is any cpu/all)
# set_rps_affinity <interface> [clear]
# Set/clear receive flow steering for cpu0 (default is any cpu/all)
# set_rfs_affinity <interface> [clear]
# Set/clear XPS and RPS/RFS together for cpu0 (default is any cpu/all)
# set_net_affinity <interface> [clear]
#
# Bonding settings
# set_bonding_options <bond_interface> <"bonding options">
# set_bonding_affinity <bond_interface> <slave_interface>
#
# Tuned settings
# set_tuned_profile <profile_name>
#
# Main routine
# main <interface> <ip/hostname>
#

# Set default values
force=0
# Set values to default if not set by user
[ -z "$cpu_affinity" ] && cpu_affinity="ffffffff,ffffffff,ffffffff,ffffffff"
[ -z "$xps_affinity" ] && xps_affinity="ffffffff,ffffffff,ffffffff,ffffffff"
[ -z "$rps_affinity" ] && rps_affinity="ffffffff,ffffffff,ffffffff,ffffffff"
[ -z "$rfs_affinity" ] && rfs_affinity="ffffffff,ffffffff,ffffffff,ffffffff"
[ -z "$net_affinity" ] && net_affinity="ffffffff,ffffffff,ffffffff,ffffffff"

# Get input parameters
get_force() {
        force=1
}

get_if() {
        [ $# -lt 2 ] && return
        ifname=$1
        ip=$2
        # Is this a hostname?
        echo $ip | grep "[a-zA-Z]" > /dev/null
        if [ $? -eq 0 ]; then
                ip=`gethostip -d $ip`
        fi
        # Is this a valid ip?
        ifconfig $ifname | grep "inet addr:$ip " > /dev/null
        if [ $? -ne 0 ]; then
                echo "$ip is not assigned to $ifname"
                exit 1
        fi
}

# Logging
log_msg() {
        [ $# -lt 1 ] && return
        [ $force -eq 0 ] && return
        logger -p user.notice -t ethtool_set "$1"
}

log_echo() {
        [ $# -lt 1 ] && return
        [ $force -eq 0 ] && return
        logger -p user.notice -t ethtool_set "$1"
        echo "$1"
}

# Basic system settings
set_kernel_settings() {
        [ $# -lt 1 ] && return
        [ ! -f "$1" ] && return
        log_msg "Setting kernel settings from file $1"
        sysctl -p $1 > /dev/null
}

set_time_wait() {
        [ $# -lt 2 ] && return
        log_msg "Setting net.ipv4.tcp_tw_recycle=$1 net.ipv4.tcp_tw_reuse=$2"
        sysctl -w net.ipv4.tcp_tw_recycle=$1 > /dev/null
        sysctl -w net.ipv4.tcp_tw_reuse=$2 > /dev/null
}

set_ipv4_settings() {
        [ $# -lt 2 ] && return
        log_msg "Setting net.ipv4.ip_local_port_range=$1 net.ipv4.tcp_max_syn_backlog=$2"
        sysctl -w net.ipv4.ip_local_port_range="$1" > /dev/null
        sysctl -w net.ipv4.tcp_max_syn_backlog=$2 > /dev/null
}

set_core_settings() {
        [ $# -lt 2 ] && return
        log_msg "Setting net.core.rmem_max=$1 net.core.wmem_max=$2"
        sysctl -w net.core.rmem_max=$1 > /dev/null
        sysctl -w net.core.wmem_max=$2 > /dev/null
}

set_qdisc() {
        [ $# -lt 2 ] && return
        log_msg "Setting qdisc $2 on interface $1"
        tc qdisc del dev $1 root > /dev/null 2>&1
        tc qdisc add dev $1 root $2 > /dev/null 2>&1
}

set_txqueuelen() {
        [ $# -lt 2 ] && return
        log_msg "Setting txqueuelen $2 on interface $1"
        ifconfig $1 txqueuelen $2
}

# Ethtool settings
set_ring() {
        [ $# -lt 3 ] && return
        log_msg "Setting ring $2 $3 on interface $1"
        ethtool -G $1 $2 $3
}

set_coalesce() {
        [ $# -lt 2 ] && return
        log_msg "Setting coalesce $2 on interface $1"
        ethtool -C $1 $2
}

set_offload() {
        [ $# -lt 3 ] && return
        log_msg "Setting offload $2 $3 on interface $1"
        ethtool -K $1 $2 $3
}

set_features() {
        [ $# -lt 3 ] && return
        log_msg "Setting feature $2 $3 on interface $1"
        ethtool --set-priv-flags $1 $2 $3
}

set_one_offload() {
        [ $# -lt 2 ] && return
        log_msg "Setting offload $2 on interface $1"
        ethtool -K $1 rx $2 tx $2 sg $2 tso $2 ufo $2 gso $2 gro $2 lro $2 rxvlan $2 txvlan $2
}

set_all_offload() {
        [ $# -lt 2 ] && return
        log_msg "Setting all offload $2 on interface $1"
        ethtool -K $1 rx $2 tx $2 sg $2 tso $2 ufo $2 gso $2 gro $2 lro $2 rxvlan $2 txvlan $2
}

# Set/clear irq affinity for cpu0 (default is any cpu/all)
set_irq_affinity() {
        [ $# -lt 1 ] && return
        for file in /proc/irq/*/$1-*; do
                [ "$file" = "/proc/irq/*/$1-*" ] && break
                log_msg "Setting irq affinity $2 for file $file"
                echo $2 > $file/../smp_affinity
        done
}

# Set/clear transmit packet steering for cpu0 (default is any cpu/all)
set_xps_affinity() {
        [ $# -lt 1 ] && return
        for file in /sys/class/net/$1/queues_tx-*; do
                [ "$file" = "/sys/class/net/$1/queues_tx-*" ] && break
                log_msg "Setting xps affinity $2 for file $file"
                echo $2 > $file/xps_cpus
        done
}

# Set/clear receive packet steering for cpu0 (default is any cpu/all)
set_rps_affinity() {
        [ $# -lt 1 ] && return
        for file in /sys/class/net/$1/queues_rx-*; do
                [ "$file" = "/sys/class/net/$1/queues_rx-*" ] && break
                log_msg "Setting rps affinity $2 for file $file"
                echo $2 > $file/rps_cpus
        done
}

# Set/clear receive flow steering for cpu0 (default is any cpu/all)
set_rfs_affinity() {
        [ $# -lt 1 ] && return
        for file in /sys/class/net/$1/queues_rx-*; do
                [ "$file" = "/sys/class/net/$1/queues_rx-*" ] && break
                log_msg "Setting rps affinity $2 for file $file"
                echo $2 > $file/rps_flow_cnt
        done
}

# Set/clear XPS and RPS/RFS together for cpu0 (default is any cpu/all)
set_net_affinity() {
        [ $# -lt 1 ] && return
        set_irq_affinity $1 $2
        set_xps_affinity $1 $2
        set_rps_affinity $1 $2
        set_rfs_affinity $1 $2
}

# Bonding settings
set_bonding_options() {
        [ $# -lt 2 ] && return
        log_msg "Setting bonding options $2 on interface $1"
        echo "$2" > /sys/class/net/$1/bonding/mode
}

set_bonding_affinity() {
        [ $# -lt 2 ] && return
        log_msg "Setting bonding affinity for slave $2 on interface $1"
        for file in /sys/class/net/$1/slave_$2/queues_*/*_cpus; do
                [ "$file" = "/sys/class/net/$1/slave_$2/queues_*/*_cpus" ] && break
                log_msg "Setting bonding affinity $3 for file $file"
                echo $3 > $file
        done
}

# Tuned settings
set_tuned_profile() {
        [ $# -lt 1 ] && return
        log_msg "Setting tuned profile $1"
        tuned-adm profile $1
}

# Main routine
main() {
        [ $# -lt 2 ] && return
        get_if $1 $2
        log_msg "Setting ethtool settings for interface $ifname"
        set_kernel_settings /etc/sysctl.conf
        set_ring $ifname rx 4096
        set_ring $ifname tx 4096
        set_coalesce $ifname adaptive-rx on
        set_coalesce $ifname adaptive-tx on
        set_offload $ifname sg on
        set_offload $ifname tso on
        set_offload $ifname ufo on
        set_offload $ifname gso on
        set_offload $ifname gro on
        set_offload $ifname lro on
        set_offload $ifname rxvlan on
        set_offload $ifname txvlan on
        set_features $ifname flow-director-atr on
        set_one_offload $ifname on
        set_all_offload $ifname on
        set_qdisc $ifname fifo_pfifo_fast
        set_txqueuelen $ifname 1000
        set_net_affinity $ifname
        set_tuned_profile latency-performance
}

# Execute main routine
main $1 $2
```

Apr 1 [user:hudsonmcmahan] *

Apr 1 [user:Hudson] *

Apr 1 [user:Hudson]/[user:
Hudson] Initial [user:Hudson] STUDENT Physics - Honors Term Project
Apr 1
[user:
Hudson]
Apr 1 [user:Hudson]/[user:Hudson] *

Apr 1 [user:Hudson] Initial [user:Hudson] *
Apr 1

Apr 1 [user:Hudson] Record [user:Hudson] on Day 1 Period 3, Building [user:1] - [user:Term Number] - [user:Term Description] Time [user:Student Time] Temperature [user:Temperature] Pressure [user:Pressure]

```
Set-ExecutionPolicy
```

```
Bypass
```

```
Get-ChildItem -Recurse -Include *.exe,*.dll
| Unblock-File
```

A
B
C
D

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UNITED STATES DISTRICT COURT
WESTERN DISTRICT OF NEW YORK

Case No. 1:20-cv-393

Plaintiff demands a jury trial on all issues so triable.

Plaintiff,

vs.

JP Morgan Chase Bank, N.A.

Defendant.

Plaintiff, Laurie Schaffner, and for Complaint against the Defendant, JP Morgan Chase Bank, N.A., alleges as follows:

1. This is an action brought by Plaintiff, Laurie Schaffner, against her former employer, Defendant, JP Morgan Chase Bank, N.A., for violations of the Family and Medical Leave Act, 29 U.S.C. § 2601 et seq. ("FMLA"), and the New York Human Rights Law, New York Executive Law § 290 et seq. ("NYHRL"). Plaintiff seeks damages to redress the injuries she has suffered as a result of Defendant's unlawful employment practices.

2. This Court has jurisdiction over this action pursuant to 28 U.S.C. § 1331 and 29 U.S.C. § 2617(a)(2). Supplemental jurisdiction over Plaintiff's NYHRL claim is conferred by 28 U.S.C. § 1367(a).

3. Venue is proper in this District pursuant to 28 U.S.C. § 1391(b)(2) because a substantial part of the events or omissions giving rise to the claims occurred in this District.

4. Plaintiff is a resident of Erie County, New York.

5. Defendant is a foreign corporation with a principal place of business in New York, New York.

6. Defendant employs more than 50 employees within 75 miles of Plaintiff's worksite.

7. Defendant is an employer as defined by the FMLA, 29 U.S.C. § 2611(4)(A).

8. Plaintiff was employed by Defendant from approximately October 2, 2006 through her termination on or about March 26, 2019.

9. Plaintiff was employed by Defendant as a Senior Vice President.

10. Plaintiff's duties included, but were not limited to, supervising a staff of approximately 10 employees.

11. Plaintiff's annual salary was approximately $150,000.00.

12. Plaintiff's duties required her to work more than 1,250 hours over the 12 months prior to her termination.

13. Plaintiff suffers from a serious health condition, namely, Crohn's disease which often flares up.

14. Plaintiff's Crohn's disease substantially limits her ability to perform major life activities, including, but not limited to, eating, digesting, and working.

15. Plaintiff's Crohn's disease substantially limits the operation of her digestive system.

16. Plaintiff's Crohn's disease substantially limits the operation of her immune system.

17. Plaintiff's Crohn's disease substantially limits the operation of her endocrine system.

18. Plaintiff's Crohn's disease substantially limits the operation of her bowel.

19. Plaintiff's Crohn's disease substantially limits the operation of her bowel.

20. Plaintiff's Crohn's disease substantially limits the operation of her bowel.

Tue
Morning, Recordings, Band Call
Rec: Lead Vocals, Guitar, Bass, Drums
Mix: Lead Vocals, Guitar, Bass, Drums
Mas: Lead Vocals, Guitar, Bass, Drums
Wed
Rec: Lead Vocals, Guitar, Bass, Drums
Mix: Lead Vocals, Guitar, Bass, Drums
Mas: Lead Vocals, Guitar, Bass, Drums
Thu
Rec: Lead Vocals, Guitar, Bass, Drums
Mix: Lead Vocals, Guitar, Bass, Drums
Mas: Lead Vocals, Guitar, Bass, Drums
Fri
Rec: Lead Vocals, Guitar, Bass, Drums
Mix: Lead Vocals, Guitar, Bass, Drums
Mas: Lead Vocals, Guitar, Bass, Drums
Sat
Rec: Lead Vocals, Guitar, Bass, Drums
Mix: Lead Vocals, Guitar, Bass, Drums
Mas: Lead Vocals, Guitar, Bass, Drums
Sun
Rec: Lead Vocals, Guitar, Bass, Drums
Mix: Lead Vocals, Guitar, Bass, Drums
Mas: Lead Vocals, Guitar, Bass, Drums
Mon
Rec: Lead Vocals, Guitar, Bass, Drums
Mix: Lead Vocals, Guitar, Bass, Drums
Mas: Lead Vocals, Guitar, Bass, Drums
Tue

```
#!/bin/bash
# Author: Ali Mirghasemi
# Description: Build and Flash Bootloader for ESP32
# Usage: ./build.sh [clean] [flash] [monitor] [port=/dev/ttyUSB0] [baud=921600]
# Example: ./build.sh flash monitor port=/dev/ttyUSB0 baud=921600
# Example: ./build.sh clean flash monitor port=/dev/ttyUSB0 baud=921600
# Example: ./build.sh flash monitor port=/dev/ttyUSB0 baud=921600 secure=1 key=secure_boot_signing_key.pem
# Example: ./build.sh flash monitor port=/dev/ttyUSB0 baud=921600 secure=1 key=secure_boot_signing_key.pem encrypt=1 key=flash_encryption_key.bin
# Example: ./build.sh flash monitor port=/dev/ttyUSB0 baud=921600 secure=1 key=secure_boot_signing_key.pem encrypt=1 key=flash_encryption_key.bin efuse=1

# Exit on error
set -e

# Check if ESP-IDF is set
if [ -z "$IDF_PATH" ]; then
    echo "Error: IDF_PATH is not set. Please source export.sh from ESP-IDF."
    echo "Example: . $HOME/esp/esp-idf/export.sh"
    exit 1
fi

# Parse arguments
CLEAN=0
FLASH=0
MONITOR=0
PORT="/dev/ttyUSB0"
BAUD="921600"
SECURE=0
KEY=""
ENCRYPT=0
EFUSE=0

for arg in "$@"; do
    case $arg in
        clean) CLEAN=1 ;;
        flash) FLASH=1 ;;
        monitor) MONITOR=1 ;;
        port=*) PORT="${arg#*=}" ;;
        baud=*) BAUD="${arg#*=}" ;;
        secure=1) SECURE=1 ;;
        key=*) KEY="${arg#*=}" ;;
        encrypt=1) ENCRYPT=1 ;;
        efuse=1) EFUSE=1 ;;
        *) echo "Unknown argument: $arg"; exit 1 ;;
    esac
done

# Clean if requested
if [ $CLEAN -eq 1 ]; then
    echo "Cleaning build directory..."
    idf.py fullclean
fi

# Build
echo "Building bootloader..."
idf.py build

# Flash if requested
if [ $FLASH -eq 1 ]; then
    echo "Flashing bootloader to $PORT at $BAUD baud..."
    idf.py -p $PORT -b $BAUD flash
fi

# Monitor if requested
if [ $MONITOR -eq 1 ]; then
    echo "Starting monitor..."
    idf.py -p $PORT monitor
fi

# Secure boot if requested
if [ $SECURE -eq 1 ]; then
    if [ -z "$KEY" ]; then
        echo "Error: Secure boot key not specified. Use key=path/to/key.pem"
        exit 1
    fi
    echo "Enabling secure boot with key: $KEY"
    idf.py secure-boot-v2 --keyfile $KEY
fi

# Encrypt if requested
if [ $ENCRYPT -eq 1 ]; then
    if [ -z "$KEY" ]; then
        echo "Error: Encryption key not specified. Use key=path/to/key.bin"
        exit 1
    fi
    echo "Enabling flash encryption with key: $KEY"
    idf.py flash-encryption --keyfile $KEY
fi

# Burn efuses if requested
if [ $EFUSE -eq 1 ]; then
    echo "Burning efuses..."
    idf.py efuse-burn
fi

# Print usage if no arguments
if [ $# -eq 0 ]; then
    echo "Usage: $0 [clean] [flash] [monitor] [port=/dev/ttyUSB0] [baud=921600] [secure=1] [key=path/to/key.pem] [encrypt=1] [key=path/to/key.bin] [efuse=1]"
    echo "Example: $0 flash monitor port=/dev/ttyUSB0 baud=921600"
    echo "Example: $0 clean flash monitor port=/dev/ttyUSB0 baud=921600 secure=1 key=secure_boot_signing_key.pem"
    echo "Example: $0 flash monitor port=/dev/ttyUSB0 baud=921600 secure=1 key=secure_boot_signing_key.pem encrypt=1 key=flash_encryption_key.bin"
    echo "Example: $0 flash monitor port=/dev/ttyUSB0 baud=921600 secure=1 key=secure_boot_signing_key.pem encrypt=1 key=flash_encryption_key.bin efuse=1"
fi
```

```
# Espressif ESP-IDF Bootloader 2nd stage
# See https://docs.espressif.com/projects/esp-idf/en/latest/api-reference/system/bootloader.html
# See https://github.com/espressif/esp-idf/tree/master/components/bootloader

# Application Level Configuration

# CONFIG_BOOTLOADER_OFFSET_IN_FLASH is not set
CONFIG_BOOTLOADER_FLASH_XMC_SUPPORT=y
# CONFIG_BOOTLOADER_DISABLE_RANDOM_CACHE_WAIT_DELAY is not set
CONFIG_BOOTLOADER_CACHE_32BIT_ADDR_QUAD_FLASH=y
# CONFIG_BOOTLOADER_SKIP_VALIDATE_IN_DEEP_SLEEP is not set
# CONFIG_BOOTLOADER_SKIP_VALIDATE_ALWAYS is not set
# CONFIG_BOOTLOADER_RESERVE_RTC_MEM is not set

#
# Log
#
# CONFIG_BOOTLOADER_LOG_VERSION_1 is not set
CONFIG_BOOTLOADER_LOG_VERSION_2=y
# CONFIG_BOOTLOADER_LOG_LEVEL_NONE is not set
# CONFIG_BOOTLOADER_LOG_LEVEL_ERROR is not set
# CONFIG_BOOTLOADER_LOG_LEVEL_WARN is not set
CONFIG_BOOTLOADER_LOG_LEVEL_INFO=y
# CONFIG_BOOTLOADER_LOG_LEVEL_DEBUG is not set
# CONFIG_BOOTLOADER_LOG_LEVEL_VERBOSE is not set
CONFIG_BOOTLOADER_LOG_LEVEL=3
# CONFIG_BOOTLOADER_LOG_COLORS is not set
CONFIG_BOOTLOADER_LOG_TIMESTAMP_SOURCE_NONE=y
# CONFIG_BOOTLOADER_LOG_TIMESTAMP_SOURCE_RTOS is not set
# CONFIG_BOOTLOADER_LOG_TIMESTAMP_SOURCE_SYSTEM is not set

#
# Serial Flash Configurations
#
CONFIG_BOOTLOADER_FLASH_DC_AWARE=y
CONFIG_BOOTLOADER_FLASH_1MB_SUPPORT=y
CONFIG_BOOTLOADER_FLASH_DUMMY_DUAL_FLASH_SUPPORT=y
CONFIG_BOOTLOADER_FLASH_DUMMY_QUAD_FLASH_SUPPORT=y
CONFIG_BOOTLOADER_FLASH_DUMMY_OCT_FLASH_SUPPORT=y
CONFIG_BOOTLOADER_FLASH_DUMMY_AUTO_DETECT=y
CONFIG_BOOTLOADER_FLASH_DUMMY_AUTO_DETECT_DEFAULT_FALLBACK=y
CONFIG_BOOTLOADER_FLASH_DUMMY_AUTO_DETECT_FALLBACK_STRICT=y
CONFIG_BOOTLOADER_FLASH_DUMMY_AUTO_DETECT_FALLBACK_RELAXED=y
CONFIG_BOOTLOADER_FLASH_DUMMY_AUTO_DETECT_FALLBACK_DISABLE=y
CONFIG_BOOTLOADER_FLASH_DUMMY_AUTO_DETECT_FALLBACK_ERROR=y
CONFIG_BOOTLOADER_FLASH_DUMMY_AUTO_DETECT_FALLBACK_ERROR_RELAXED=y
CONFIG_BOOTLOADER_FLASH_DUMMY_AUTO_DETECT_FALLBACK_ERROR_DISABLE=y
CONFIG_BOOTLOADER_FLASH_DUMMY_AUTO_DETECT_FALLBACK_ERROR_DISABLE_RELAXED=y
CONFIG_BOOTLOADER_FLASH_DUMMY_AUTO_DETECT_FALLBACK_ERROR_DISABLE_STRICT=y
CONFIG_BOOTLOADER_FLASH_DUMMY_AUTO_DETECT_FALLBACK_ERROR_DISABLE_RELAXED_STRICT=y
CONFIG_BOOTLOADER_FLASH_DUMMY_AUTO_DETECT_FALLBACK_ERROR_DISABLE_STRICT_RELAXED=y
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CONFIG_BOOTLOADER_FLASH_DUMMY_AUTO_DETECT_FALLBACK_ERROR_DISABLE_STRICT_RELAXED
```

Apr 1 [user] Audrey's birthday! ?

Apr 1 [user] Audrey's birthday!

Apr 1 [user] Celebrations! #FOMMM Pauline, Connie, Rachel Aggie

Apr 1

Apr 1

Apr 1

Apr 1 [user] Pollyanna Audrey!!

Apr 1

Apr 1 [user] Happy! [user] Challenge! ?

Apr 1 [user]

Apr 1

Apr 1 [user] Birthday! Rachel [user] Audrey! ? [user] Thumbsup! ?

Apr 1 [user] Audrey! [user] Thumbsup! ?

Apr 1

Apr 1 [user] Birthday! Rachel [user] Audrey! ? [user] Thumbsup! ?

# Comparison Study on Physical Properties of Cu-Mn-Ni Alloy Coated Thin Films Produced by Dip Coating Technique

Dr. Ajay Kumar Singh, Dr. Ashok Kumar Singh, Dr. Ram Kishor Singh,

Date: 14th November 2008 Date of Publication: 21st November 2008 NAME OF PUBLICATION: Legal Notice no. 46 of 2008, Supplement Number 2

**PART II**

Notice is hereby given to all persons to whom it may concern that the Minister for Home Affairs and Municipal Affairs and Municipal Affairs will, on or around the date given in the number and names of persons specified in the Schedule hereto, apply on the

**SCHEDULE**

**PART I**

Name of person, Parish or Village in which ordinarily resident Date of his intended Petition to be presented to Parliament, Court of Chief Magistrate, Principal Magistrate, Principal Magistrate, Principal

Tue
Jan 1
Jerusalem, Ancient, Jew, Israelite, Pilgrimage, Church, Pilgrimage

Tue, Jan 1 

Ref:
Ref: Advocate Manoj Dhote Nanded
Ref: Ref: Ref:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Document</title>
</head>
<body>
<h1>Hello World!</h1>
<p>Welcome to my first webpage.</p>
<p>I hope you enjoy it.</p>
</body>
</html>
```

# Minutes: Board of Trustees Public Meeting

**Date:** Wednesday, September 17, 2025
**Time:** 6:00 p.m.
**Location:** Council Chambers, Lakeside Civic Center, 1 Main Plaza, Lakeside Bay City, Crystal Shore Province 5560

**Call to Order**

The meeting was called to order at 6:00 p.m. by Board Chair Dr. Helena Voss.

**Roll Call**

Present: Dr. Helena Voss, Mr. Jonathan Reed, Ms. Priya Desai, Mr. Carlos Mendoza, Ms. Aisha Thompson, Mr. Robert Lin, Ms. Fatima Al-Sayed.

Absent: Mr. Thomas Greene (excused).

Also present: Superintendent Dr. Marcus Whitfield, Chief Financial Officer Linda Park, Clerk of the Board Jessica Morales, and approximately 40 members of the public.

**Approval of Agenda**

Ms. Desai moved to approve the agenda as presented. Mr. Reed seconded. The motion passed unanimously, 7–0.

**Approval of Minutes**

Mr. Lin moved to approve the minutes of the August 20, 2025, regular meeting. Ms. Thompson seconded. The motion passed unanimously, 7–0.

**Public Comment**

Eleven members of the public addressed the board. Topics included the proposed school consolidation plan, the new bus routing system, and the upcoming referendum. Each speaker was limited to three minutes.

**Superintendent's Report**

Dr. Whitfield reported that enrollment for the 2025–2026 school year is 12,847 students, a decrease of 213 from last year. He noted that the decline is smaller than projected. He also announced that the district received a $1.2 million grant from the Crystal Shore Department of Education for literacy coaching.

**Financial Report**

Ms. Park presented the monthly financial report. As of August 31, the district has spent 16.4% of its budget, which is on track for the point in the year. Transportation costs are running 3% over budget due to fuel prices.

**Action Items**

1. **Consolidation Study Committee.** Ms. Desai moved to establish a committee of seven members to study the proposed consolidation of Maple Grove Elementary and Riverside Elementary. The committee will report its findings by January 15, 2026. Mr. Mendoza seconded. Motion passed, 7–0.

2. **Bus Routing Contract.** Mr. Reed moved to award the bus routing contract to Lakeside Transportation Services for the 2025–2026 and 2026–2027 school years, with an option for a third year. Ms. Thompson seconded. Motion passed, 7–0.

3. **Referendum Resolution.** Mr. Lin moved to place a $4.5 million operating referendum on the November 4, 2025, ballot. Ms. Al-Sayed seconded. Motion passed, 7–0.

**Discussion Items**

The board discussed the timeline for the referendum campaign. Dr. Whitfield will present a communications plan at the October meeting.

**Adjournment**

Ms. Desai moved to adjourn. Mr. Reed seconded. The meeting was adjourned at 8:42 p.m.

Respectfully submitted,
Jessica Morales, Clerk of the Board

Name: Bethany Jennings Date: 10/5/2015 Period: 1 Title: **LAB REPORT: Liquid Density Physics Physics Physics Physics Lab**

Aim: To determine which substance has the most density.

Hypothesis: If I pour the same amount of water, oil and corn syrup in a cup and measure the mass of each substance, then the corn syrup will have the most mass, and the oil will have the least mass. Because the corn syrup is the most dense liquid, and the oil is the least dense liquid, which has the most mass.

Materials: 1 beaker, 1 graduated cylinder, 1 scale, 1 cup of water, 1 cup of oil, 1 cup of corn syrup

Procedure:

1. Measure 100 grams of water in a beaker. Pour **Through This Sheet of Paper Or** Objects Into Aluminum **Still** into a graduated cylinder of 100 ml. Be careful not to let the water spill over the 100 ml mark. Record the mass of water. Pour the water back into the beaker. Repeat this step for the oil and corn syrup.
2. Pour the water into the graduated cylinder. Record the volume of water. Repeat this step for the oil and corn syrup.
3. Calculate the density of water by dividing the mass by the volume. Repeat this step for the oil and corn syrup.

Data: Liquid Density Experiment 1 Liquid: **A Wizards Wand Is A Wizard's** Mass: 100 g Volume: 100 ml Density: 1 g/ml Liquid: Oil Mass: 100 g Volume: 100 ml Density: 1 g/ml Liquid: Corn Syrup Mass: 100 g Volume: 100 ml Density: 1 g/ml

Data: Liquid Density Experiment 2 Liquid: **A Wizards Wand Is** Mass: 100 g Volume: 100 ml Density: 1 g/ml Liquid: Oil Mass: 100 g Volume: 100 ml Density: 1 g/ml Liquid: Corn Syrup Mass: 100 g Volume: 100 ml Density: 1 g/ml

Data: Liquid Density Experiment 3 Liquid: **A Wizards Wand Is A Wizard's** Mass: 100 g Volume: 100 ml Density: 1 g/ml Liquid: Oil Mass: 100 g Volume: 100 ml Density: 1 g/ml Liquid: Corn Syrup Mass: 100 g Volume: 100 ml Density: 1 g/ml

Data: Liquid Density Experiment 4 Liquid: **A Wizards Wand Is A Wizard's** Mass: 100 g Volume: 100 ml Density: 1 g/ml Liquid: Oil Mass: 100 g Volume: 100 ml Density: 1 g/ml Liquid: Corn Syrup Mass: 100 g Volume: 100 ml Density: 1 g/ml

Data: Liquid Density Experiment 5 Liquid: **A Wizards Wand Is A Wizard's** Mass: 100 g Volume: 100 ml Density: 1 g/ml Liquid: Oil Mass: 100 g Volume: 100 ml Density: 1 g/ml Liquid: Corn Syrup Mass: 100 g Volume: 100 ml Density: 1 g/ml

Conclusion: My hypothesis was correct. The corn syrup was the most dense liquid, and the oil was the least dense liquid. Because the corn syrup had the most mass, and the oil had the least mass. Which proves that the corn syrup is the most dense liquid, and the oil is the least dense liquid. Because the corn syrup had the most mass, and the oil had the least mass.

Liquid Density
Physics
Physics
Physics

```
#!/bin/bash
# Author: Ali Shitiwi
# Description: Automatically Backup and Restore Home Folder
# Version: 1.0
# Date: 2018-03-27
# Usage: ./backup.sh -b -d /path/to/backup/dir -n backup_name
# Usage: ./backup.sh -r -f /path/to/backup/file.tar.gz -t /path/to/restore/dir
# Usage: ./backup.sh -h
# Usage: ./backup.sh -v
# Usage: ./backup.sh -b -d /path/to/backup/dir -n backup_name -r -f /path/to/backup/file.tar.gz -t /path/to/restore/dir
# Usage: ./backup.sh -b -d /path/to/backup/dir -n backup_name -r -f /path/to/backup/file.tar.gz -t /path/to/restore/dir -h -v
# Usage: ./backup.sh -b -d /path/to/backup/dir -n backup_name -r -f /path/to/backup/file.tar.gz -t /path/to/restore/dir -h -v
```

```
Set-ExecutionPolicy Unrestricted

Set-ExecutionPolicy -

Set-ExecutionPolicy Unrestricted
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\ActiveDirectory\ActiveDirectory.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\GroupPolicy\GroupPolicy.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\ServerManager\ServerManager.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\TroubleshootingPack\TroubleshootingPack.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\TrustedPlatformModule\TrustedPlatformModule.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\WDS\WDS.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\WebAdministration\WebAdministration.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\WindowsSearch\WindowsSearch.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\WindowsUpdate\WindowsUpdate.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\Dism\Dism.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\ADRMS\ADRMS.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\ADRMSAdmin\ADRMSAdmin.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\ADFS\ADFS.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\ADFSAdmin\ADFSAdmin.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\ADLDS\ADLDS.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\ADLDSAdmin\ADLDSAdmin.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\ADRMSAdmin\ADRMSAdmin.psd1
Import-Module C:\Windows\System32\WindowsPowerShell\v1.0\Modules\ADRMS\ADRMS
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```
#!/bin/bash
# Advanced Pipeline Operations

# Create test files
echo -e "apple\nbanana\ncherry\napple\ndate" > fruits.txt
echo -e "1\n2\n3\n4\n5" > numbers.txt

# Basic pipeline
cat fruits.txt | sort | uniq

# Count occurrences
cat fruits.txt | sort | uniq -c | sort -nr

# Multi-stage pipeline
cat numbers.txt | while read num; do
    echo "Number: $num, Square: $((num * num))"
done

# Process substitution
diff <(sort fruits.txt) <(sort -r fruits.txt)

# Named pipe (FIFO)
mkfifo /tmp/mypipe
echo "Hello through pipe" > /tmp/mypipe &
cat < /tmp/mypipe
rm /tmp/mypipe
```

Chapter 1 Introduction
Chapter 2 Background
Chapter 3 Methodology
Chapter 4 Results
Chapter 5 Discussion
Chapter 6 Conclusion
Chapter 7 Future Work
Chapter 8 Appendix
Chapter 9 Bibliography

1:2 Author Name et al.

$$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)$$

Author Name et al.
Author Name et al.

$$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)$$

$$\begin{aligned} \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) \\ \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) \end{aligned}$$

$$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)

```
%%{
  init: {
    'theme': 'forest',
    'themeVariables': {
      'primaryColor': '#BB2528',
      'primaryTextColor': '#fff',
      'primaryBorderColor': '#7C0000',
      'lineColor': '#F8B229',
      'secondaryColor': '#006100',
      'tertiaryColor': '#fff'
    }
  }
}%%
graph LR;
    A--Text-->B{Let me think};
    B--Text-->C{Let me think};
    C--Text-->D{Let me think};
    D--Text-->A;
```

$$\alpha = \frac{1}{2} - \beta + \beta^2$$

```
%%{
  init: {
    'theme': 'base',
    'themeVariables': {
      'primaryColor': '#ffffff',
      'primaryTextColor': '#ffffff',
      'primaryBorderColor': '#ffffff',
      'lineColor': '#ffffff',
      'secondaryColor': '#ffffff',
      'tertiaryColor': '#ffffff'
    }
  }
}%%
graph LR;
    A--Text-->B{Let me think};
    B--Text-->C{Let me think};
    C--Text-->D{Let me think};
    D--Text-->A;
```

$$\alpha = \frac{1}{2} - \beta + \beta^2$$

```
%%{
  init: {
    'theme': 'dark',
    'themeVariables': {
      'primaryColor': '#BB2528',
      'primaryTextColor': '#fff',
      'primaryBorderColor': '#7C0000',
      'lineColor': '#F8B229',
      'secondaryColor': '#006100',
      'tertiaryColor': '#fff'
    }
  }
}%%
graph LR;
    A--Text-->B{Let me think};
    B--Text-->C{Let me think};
    C--Text-->D{Let me think};
    D--Text-->A;
```

$$\alpha = \frac{1}{2} - \beta + \beta^2$$

```
%%{
  init: {
    'theme': 'neutral',
    'themeVariables': {
      'primaryColor': '#BB2528',
      'primaryTextColor': '#fff',
      'primaryBorderColor': '#7C0000',
      'lineColor': '#F8B229',
      'secondaryColor': '#006100',
      'tertiaryColor': '#fff'
    }
  }
}%%
graph LR;
    A--Text-->B{Let me think};
    B--Text-->C{Let me think};
    C--Text-->D{Let me think};
    D--Text-->A;
```

$$\alpha = \frac{1}{2} - \beta + \beta^2$$

Dr. Jonathan Ellery Whitcombe, Department of Early Modern Studies, Hollowbrook College, Dr. Miriam Sol Ortega, Department of Early Modern Studies, Hollowbrook College

Input:
Integrate -3/(4x^2+4x+5) dx, x=-1/2 to 1/2, Riemann sum
Use 4 subintervals, Midpoint Rule, Round to 4 decimal places
Hint: Use midpoints, Midpoint Rule: ∫[a,b] f(x) dx ≈ Δx[f(m1)+f(m2)+...+f(mn)]

Output:
Enter your answer(s) delimited by ;

Input:
Math_Code("Number") Compute(Solve(2x+3=0,x),x)

Output:

$$x = -\frac{3}{2} \quad x = -\frac{3}{2}$$

Input:
Math_Code("Number") Compute(Solve(2x+3=0, x), "Type in the form: fill in(a), fill in(b)", "a,b")

Output:
Type in the form: fill in(a), fill in(b)

Input:
Math_Code("Number") Compute(Solve(2x+3=0,x),"Integer")

Output:

$$2x + 3 = 0 \quad x = -\frac{3}{2}$$

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Output:

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Math_Code("Number") Compute(Solve(2x+3=0,x),"Integer")

Output:

$$2x + 3 = 0 \quad x = -\frac{3}{2}$$